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Beyond All-Digital PLL for RF and Millimeter-Wave Frequency Synthesis

Robert Bogdan Staszewski, with Yizhe Hu and Teerachot Siriburanon

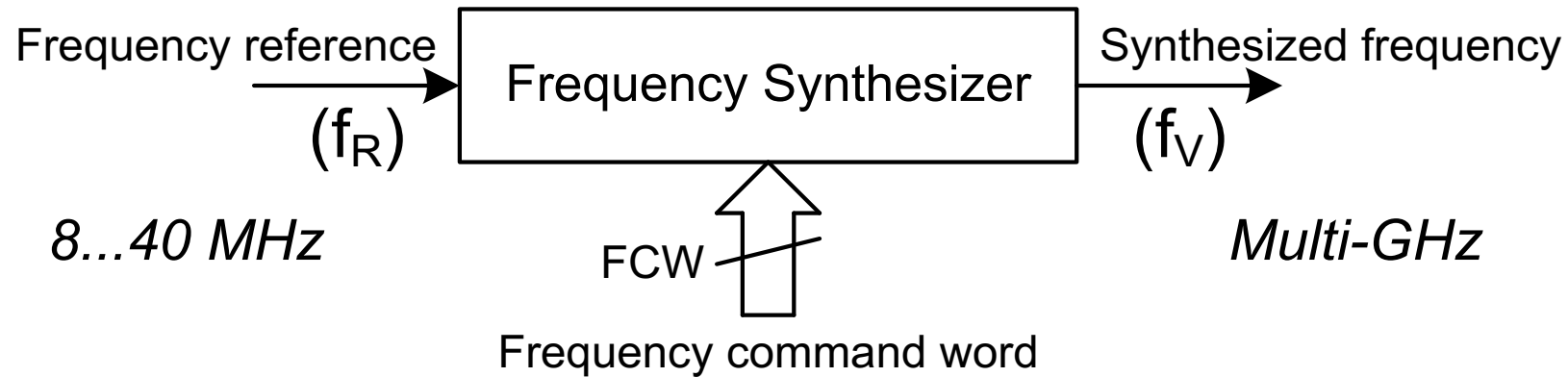
University College Dublin, Ireland

2020-03-22

Outline

- Overview: RF design in nanometer-scale CMOS environment
- All-digital phase-locked loop (ADPLL)
 - Digitally-controlled oscillator (DCO)
 - Time-to-digital converter (TDC)
 - Digital loop filter
- Ultra-fast ADPLL settling
- Direct frequency modulation
- Dithering and rotation of TDC
- Digital assistance of RF
- New type of frequency synthesis: charge-Sharing locking (CSL)
- CSL experimental results

Local Oscillator

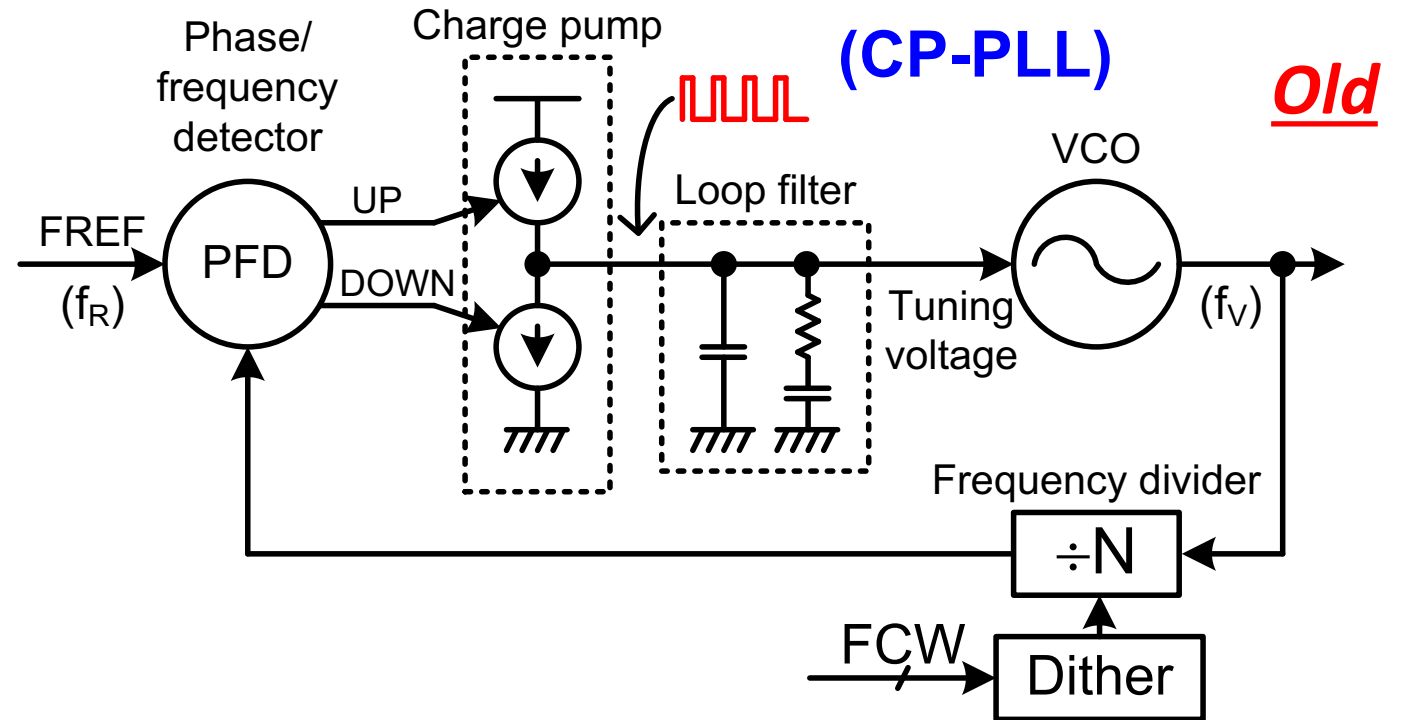


- Frequency reference (FREF) source usually built as tunable crystal oscillator (XO)
 - Voltage-controlled temperature-compensated XO (VCTCXO)
 - External module; expensive (~\$1)
 - Digitally-controlled XO (DCXO)
 - Requires external XTAL (~\$0.2)
 - MEMS resonators, RC-locked resonators, etc...
 - Emerging technology
- Three major frequency synthesis techniques:
 - Direct-analog (error correction process is avoided)
 - Direct-digital
 - Indirect or phase-locked loop

Motivation for Digitally-Intensive RF PLL

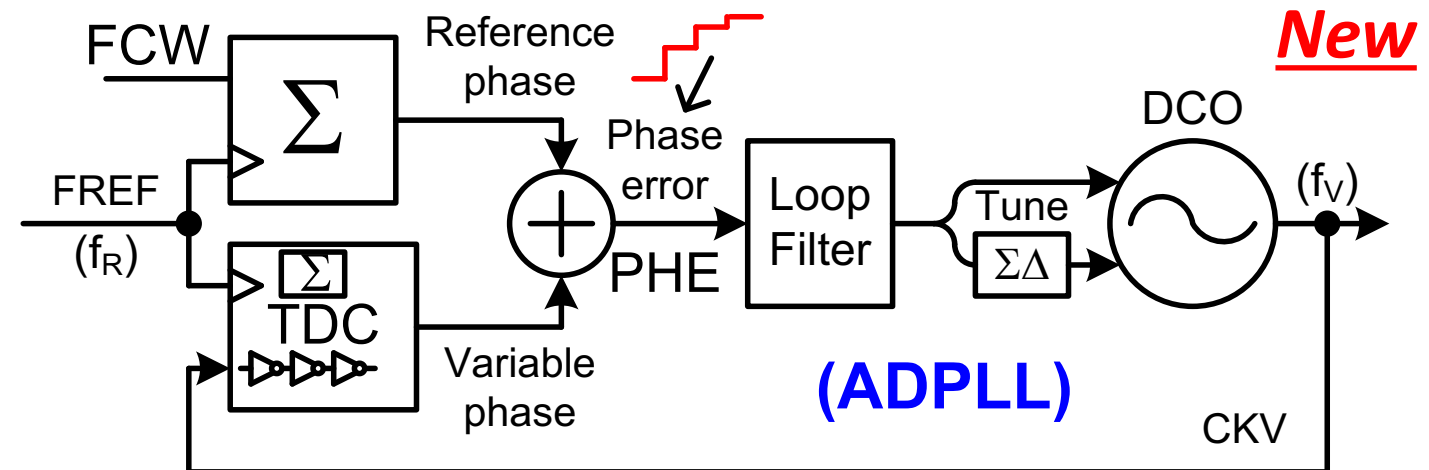
- **Charge-pump PLL:**

- Leaky loop filter caps
- Charge pump distortions
- Suffers from reference spurs
- Tradeoff: bandwidth against spur level



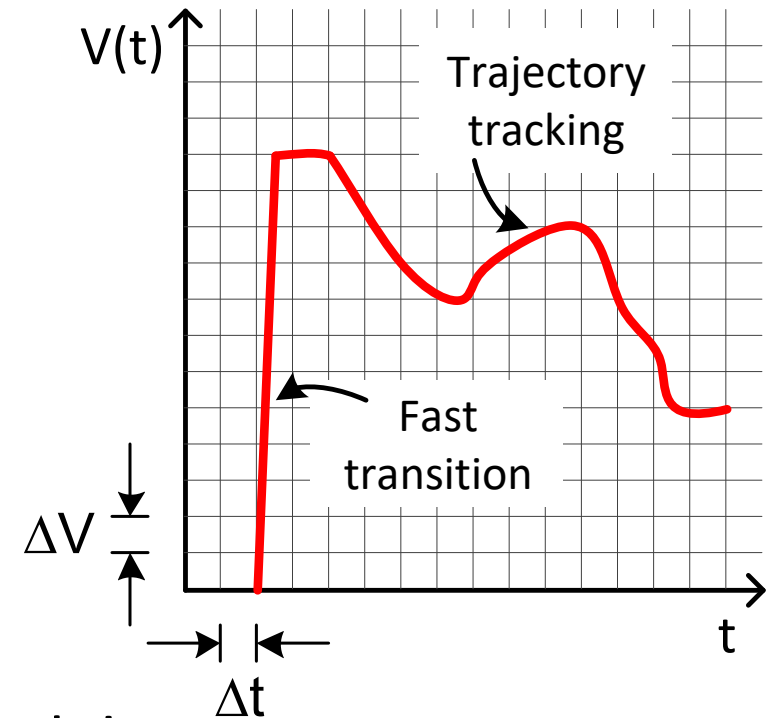
- **All-digital PLL:**

- True phase domain operation
- None of the above problems



RF in Nanoscale CMOS: New Paradigm

- Now the mainstream CMOS technology is extremely fast
 - Good news for the time-domain!
- However, the voltage supply levels are very small: Difficult to resolve voltage-domain signals
 - “Don't care” for the time-domain!
- Seems like the mainstream CMOS technology is now ready!
- It is time to use **time**!
- However, obstacles:
 - New theory of the time-domain signal processing and circuit architectures is not well developed
 - Reluctance to change in scientific and industrial communities



Rules of nm-Scale CMOS

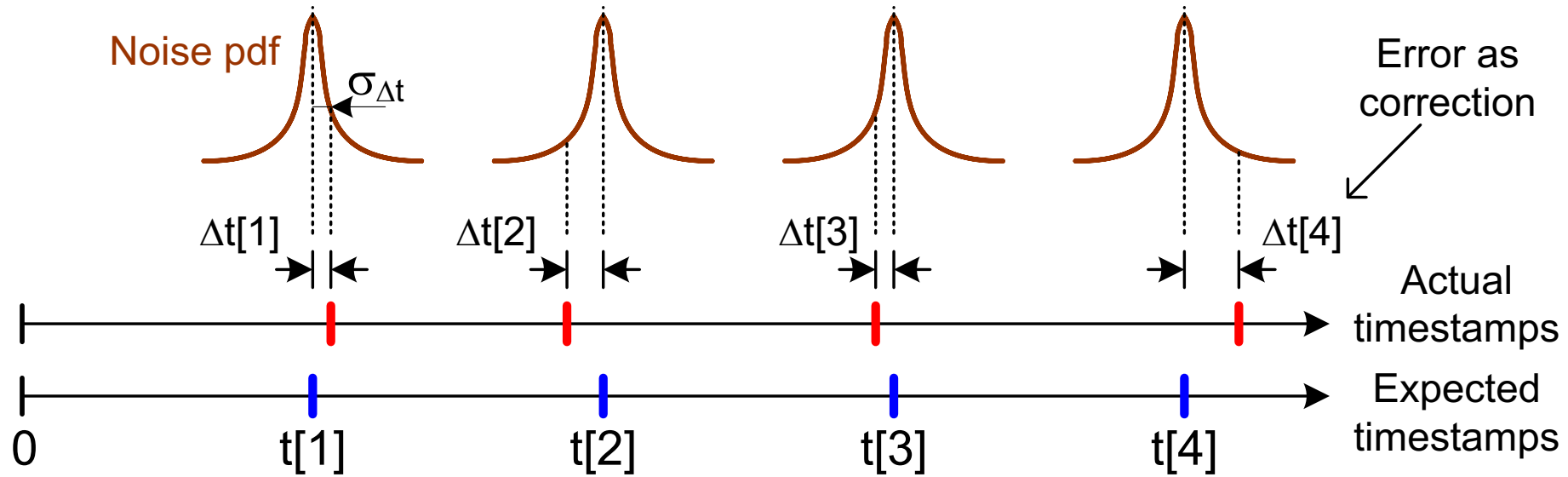
- **Exploit:**

- Fast switching characteristics of MOS transistors
 - 10 ps transition and f_T of 400 GHz in 28-nm CMOS
 - Improves 30% per process node (18—24 months)
- Small device geometries and precise device matching
- High density of digital logic: 2 M gates/mm² in 28-nm CMOS
 - 2x scaling at each process node (18—24 months)
- High density of SRAM memory: 8 Mbits/mm² in 28-nm CMOS

- **Avoid:**

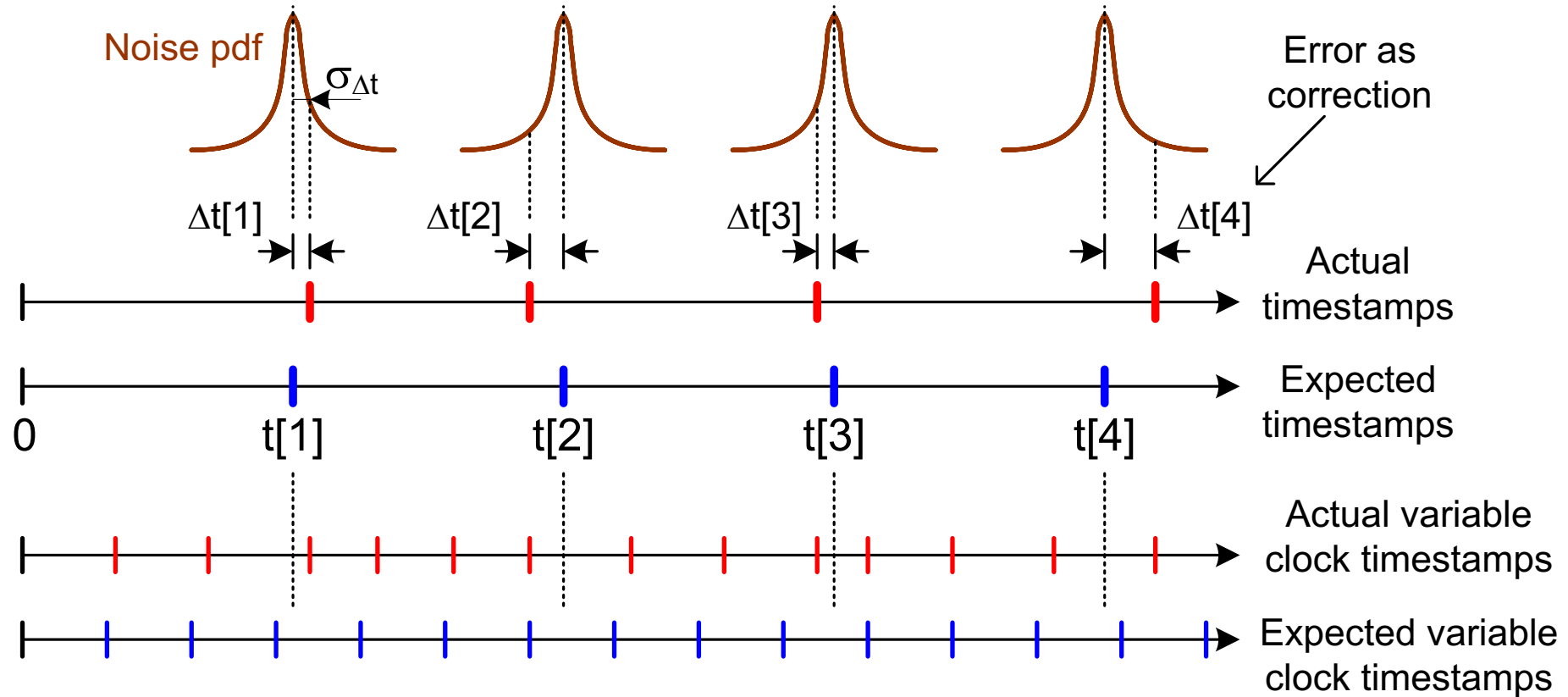
- Biasing currents for analog circuits
- Reliance on voltage resolution
- Nonstandard devices not needed for memory and logic

ADPLL: Simple Idea of Phase Domain



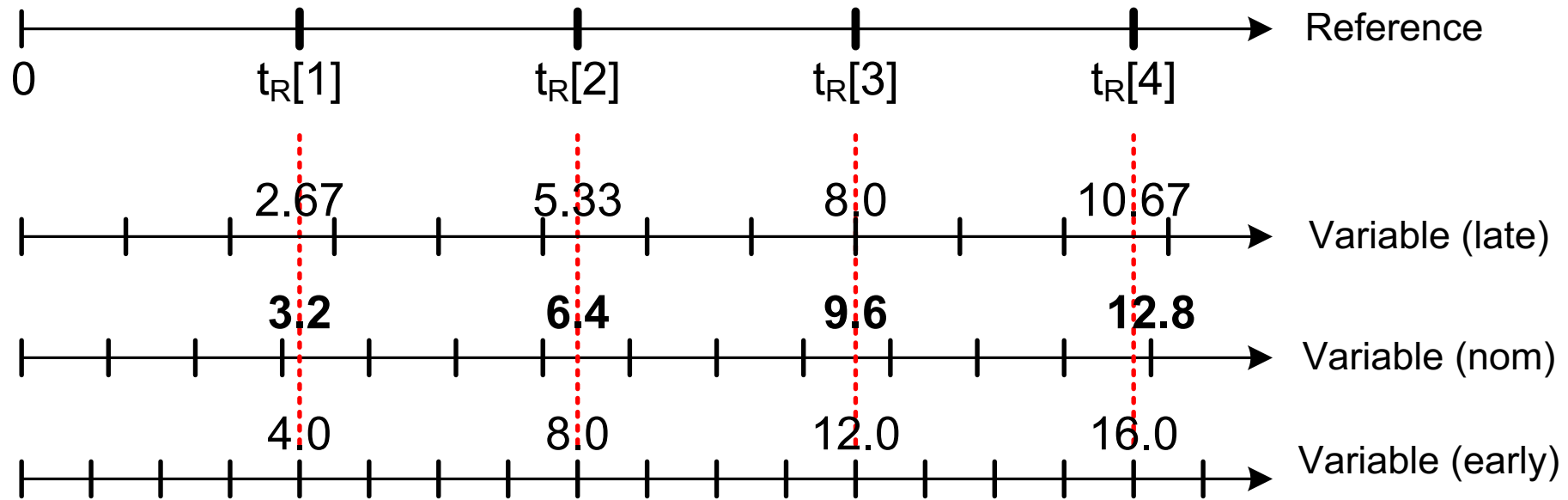
- Construct expected or ideal timestamps
 - “Reference phase”
- Measure the actual timestamps
 - “Variable phase”
- Their difference is the time error
 - Used for future timing corrections as negative feedback

Issues with the Simple Idea



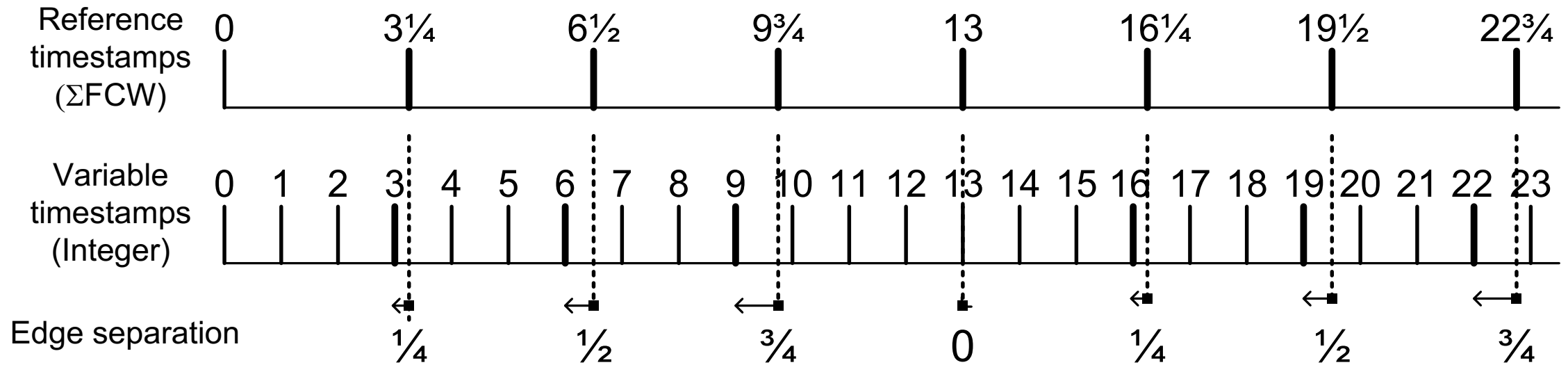
- Synthesized (variable) frequency is typically much higher than the reference frequency
- Non-integer relationship
- How to truly operate in time domain, i.e., “timestamps”?

Proposed Solution



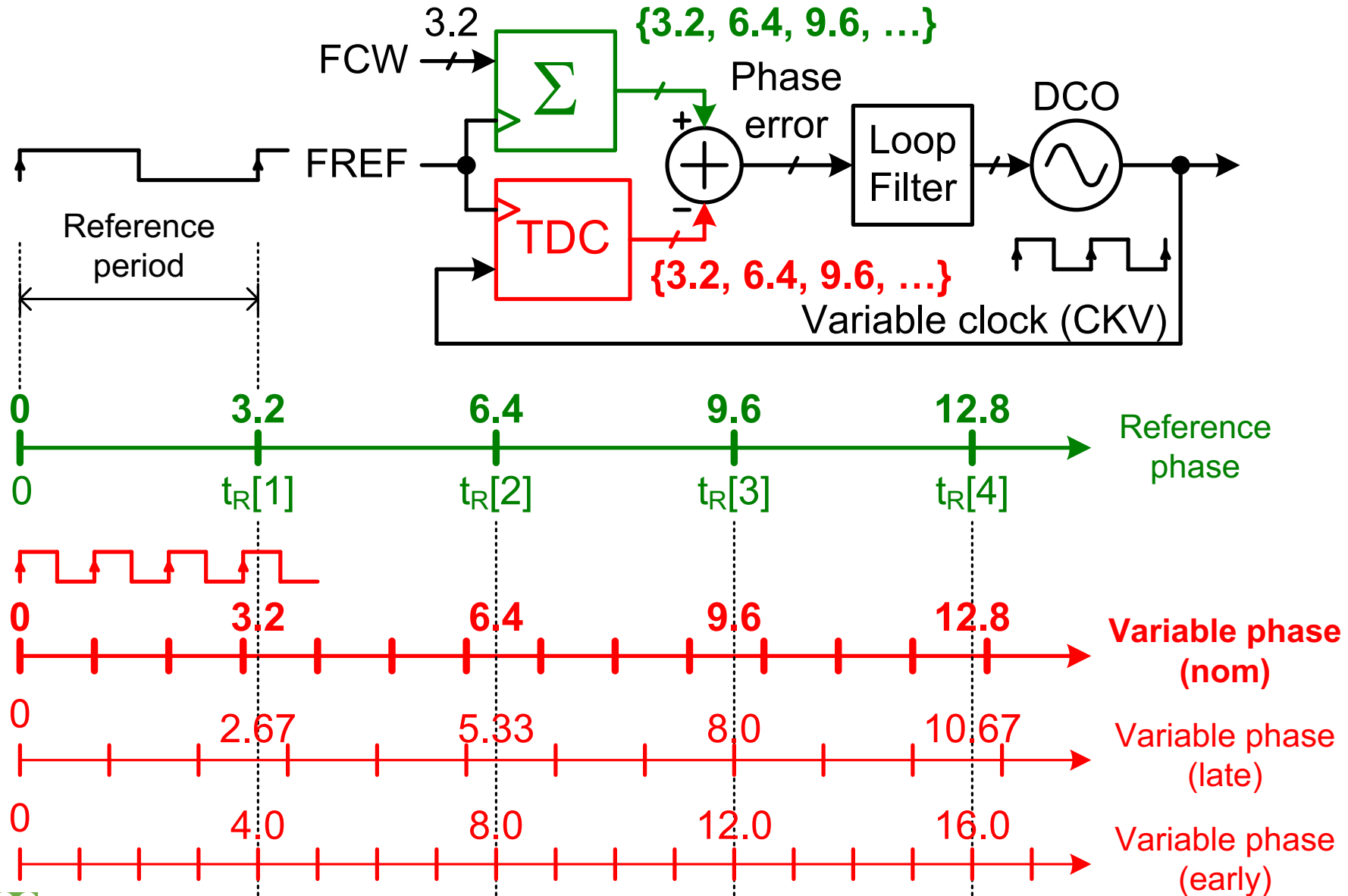
- Reference and variable signals are digital clocks
 - Use only their (rising) edges
- Phase error calculation to be performed on reference edges
- Turn the phase detection problem around
 - Measure the reference timestamps with the variable clock
 - Sample the count of variable clock cycles with each reference edge
 - If variable phase drifts, their sampled count will get affected

Hardware Needed for Digital Operation

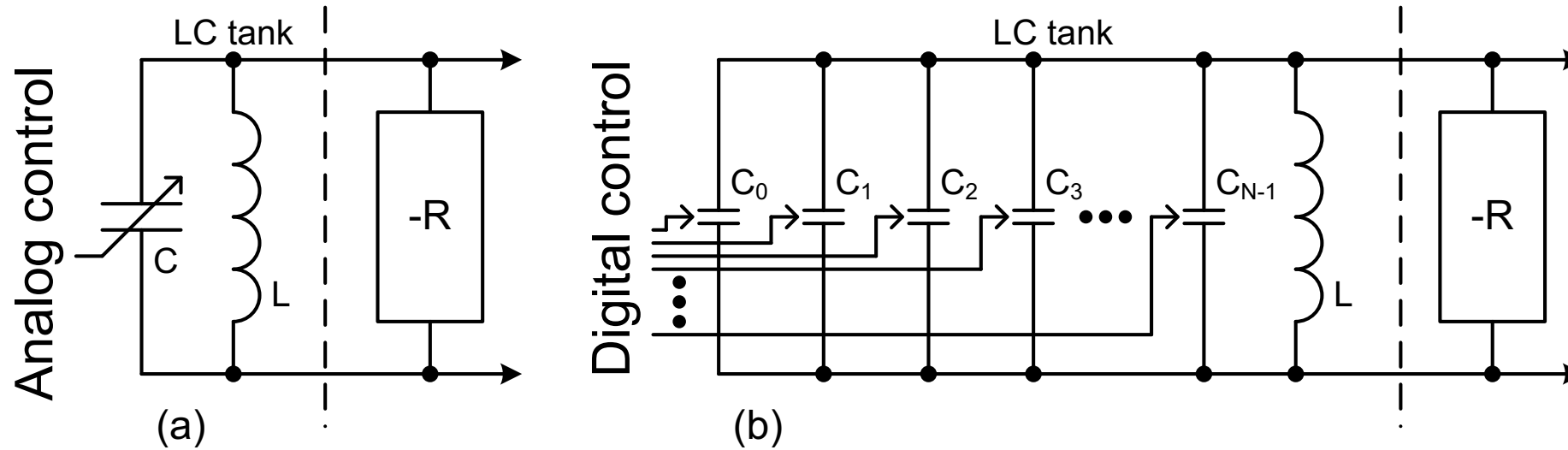


- Accumulator of FCW
- Interpolator or normalize estimator of edge separation
 - Time-to-digital converter (TDC)
- RF oscillator needs to be numerically controlled:
 - Digitally-controlled oscillator (DCO)
- Digital phase detector and loop filter

All-Digital Phase Locked Loop (APLL)



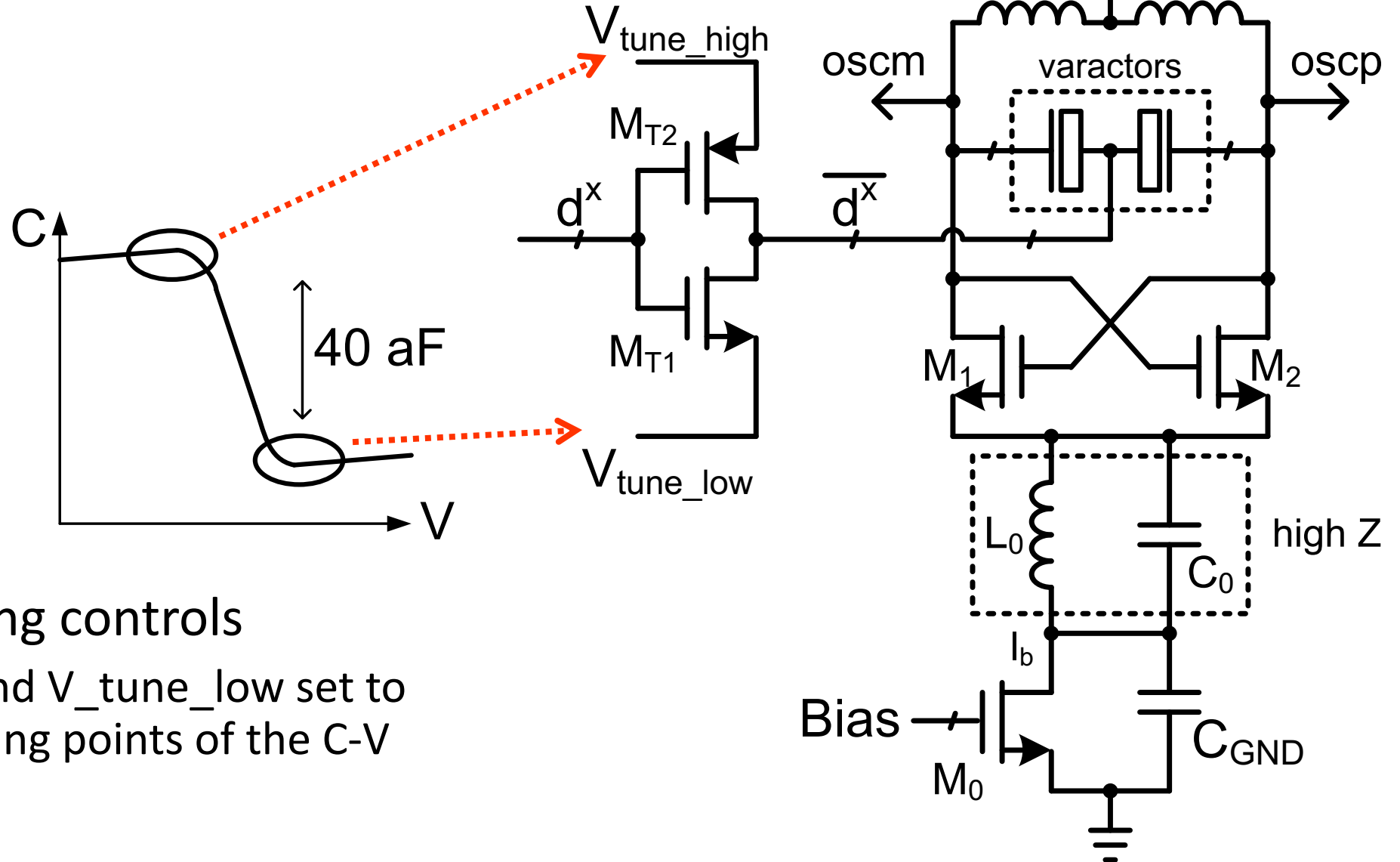
Digital Tuning of DCO



- A large linear varactor in conventional VCO replaced with a large number of tiny binary-controlled varactors in a digitally controlled oscillator (DCO)
 - Smallest varactor size: tens of atto-farad
- Deliberate avoidance of any analog tuning
- The feedback loop could now be fully digital

DCO Core

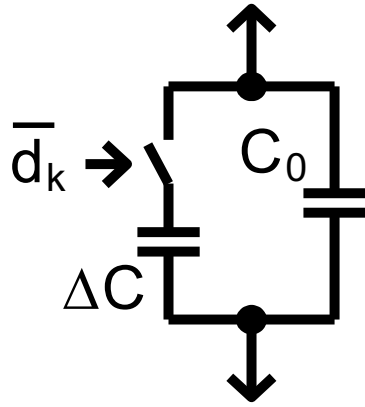
- 3.2—4 GHz range



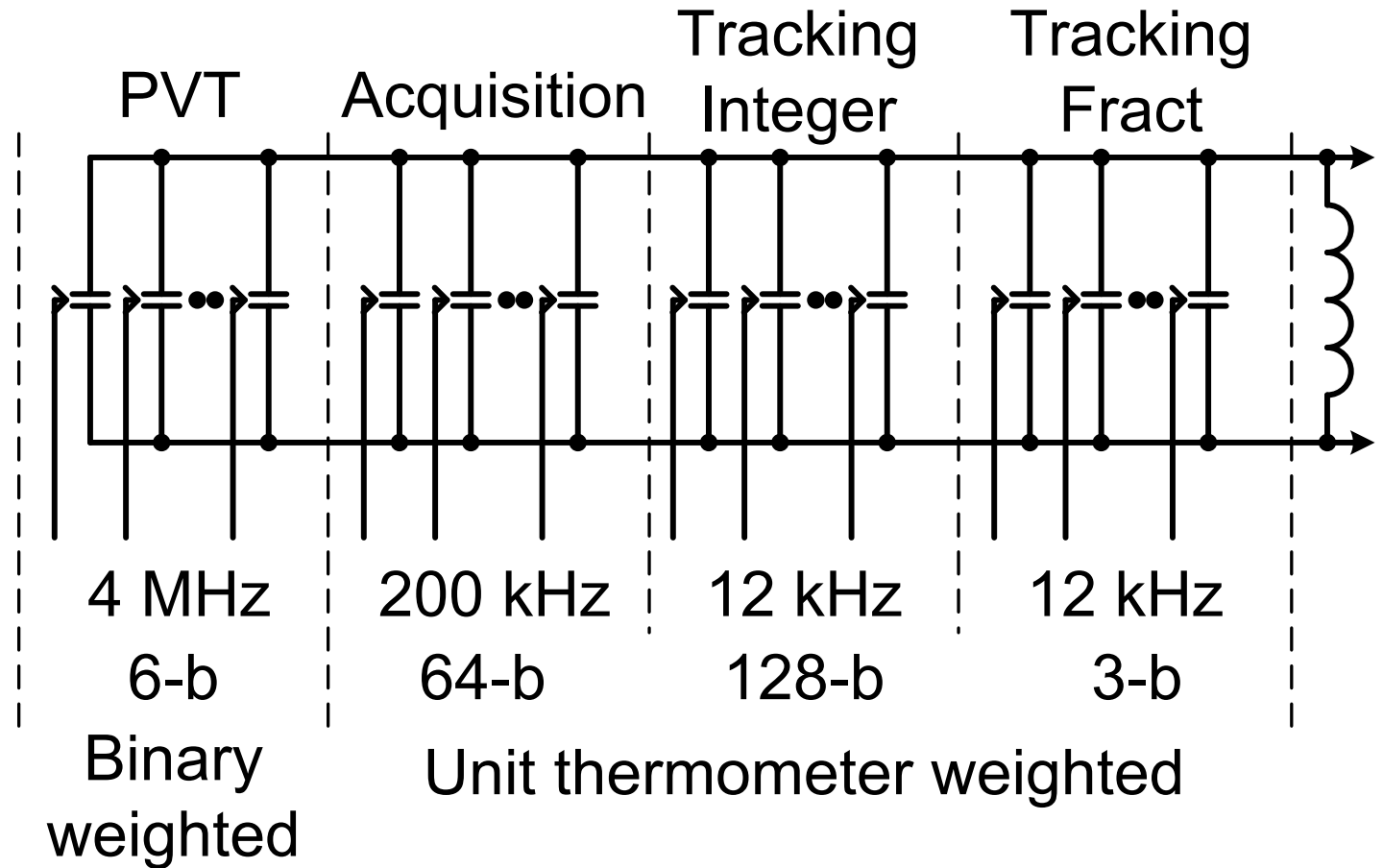
- No analog tuning controls
 - $V_{\text{tune_high}}$ and $V_{\text{tune_low}}$ set to two flat operating points of the C-V curve

DCO Varactor Functional Banks

Varactor model:

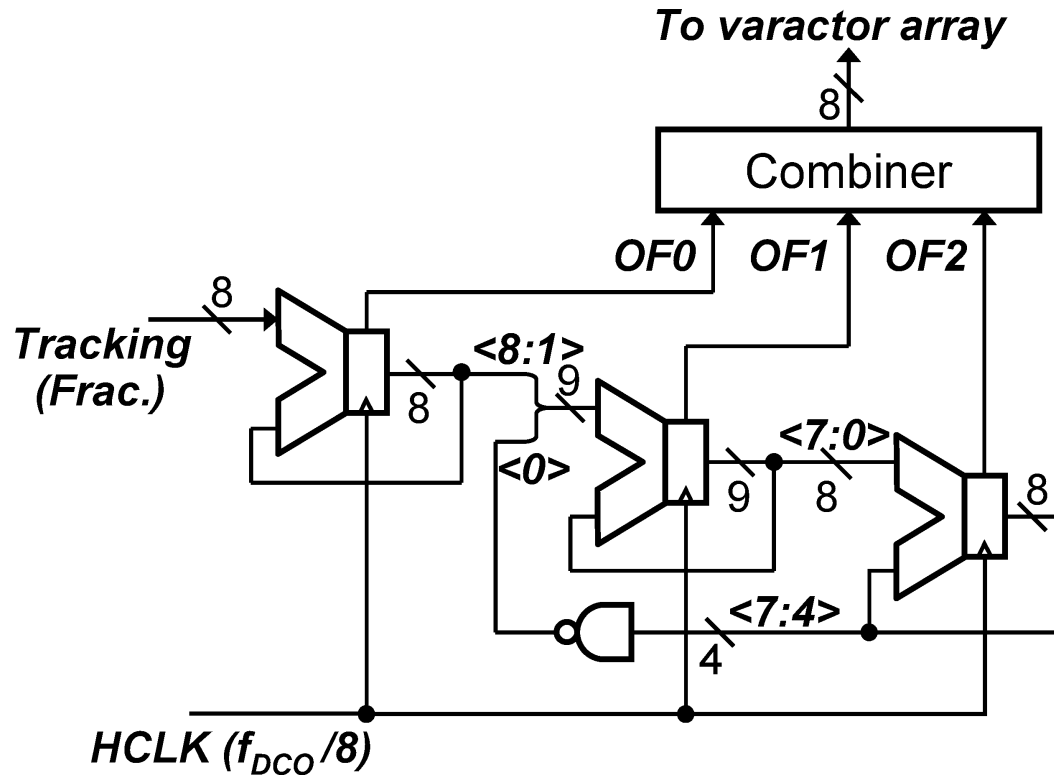


- Process/voltage/temperature (PVT) calibration mode
- Acquisition mode (during channel select)
- Tracking mode (during the actual TX and RX)
 - Dithering to improve resolution

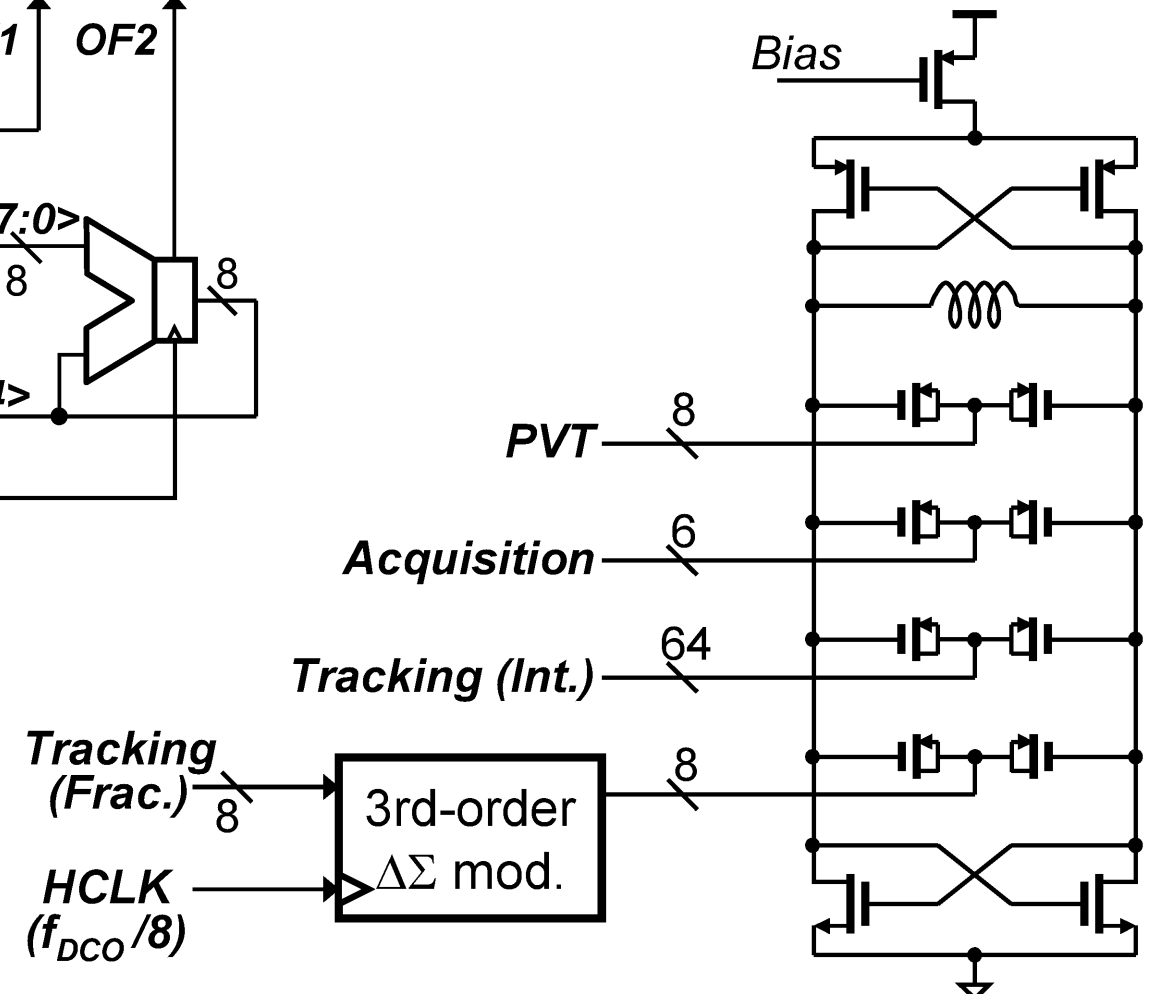


DCO with Sigma-Delta Modulation

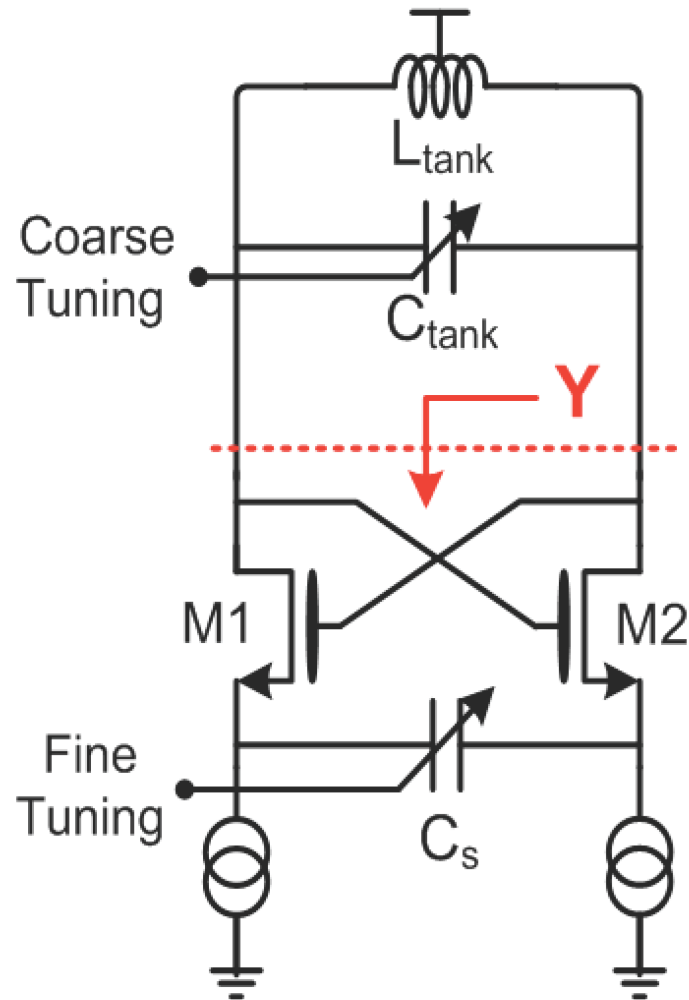
[Tokairin ISSCC 2010]



- Dithering of SDM
- Breaks the periodicity for static inputs



DCO Δf Scaling: Switching Pair Degeneration



- A large C_s between the sources is shrunk in parallel to the tank

For $C_s \gg \bar{g}_m/(2\omega_0)$:

$$Y = -\frac{\bar{g}_m}{2} - j\omega_0 C_s \left(\frac{\bar{g}_m}{2\omega_0 C_s} \right)^2$$

Classic -R

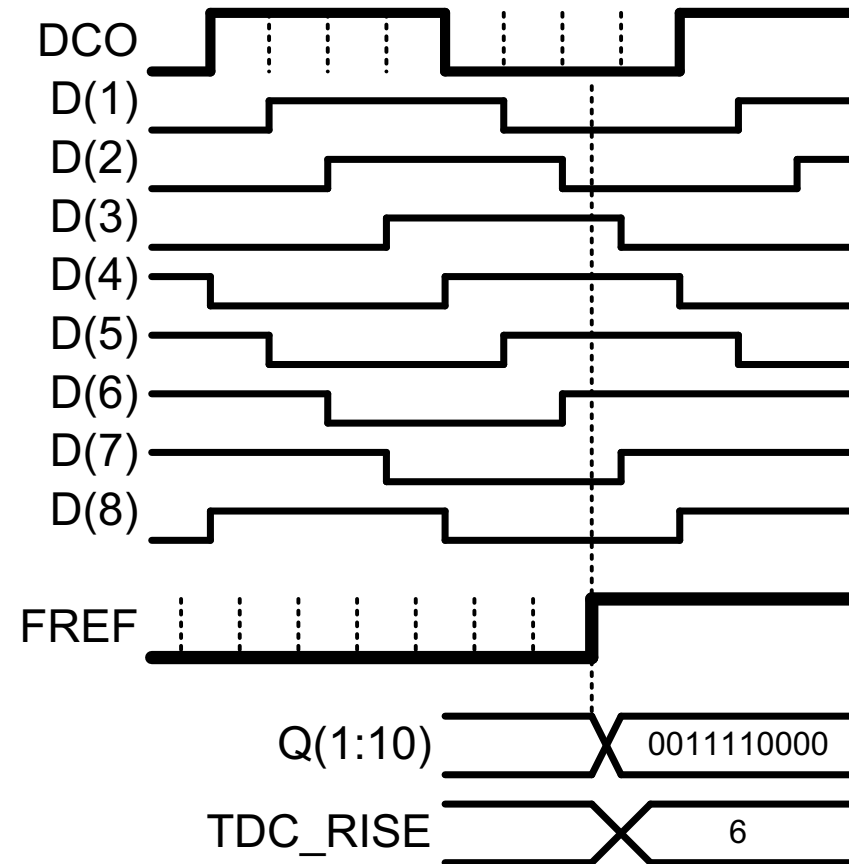
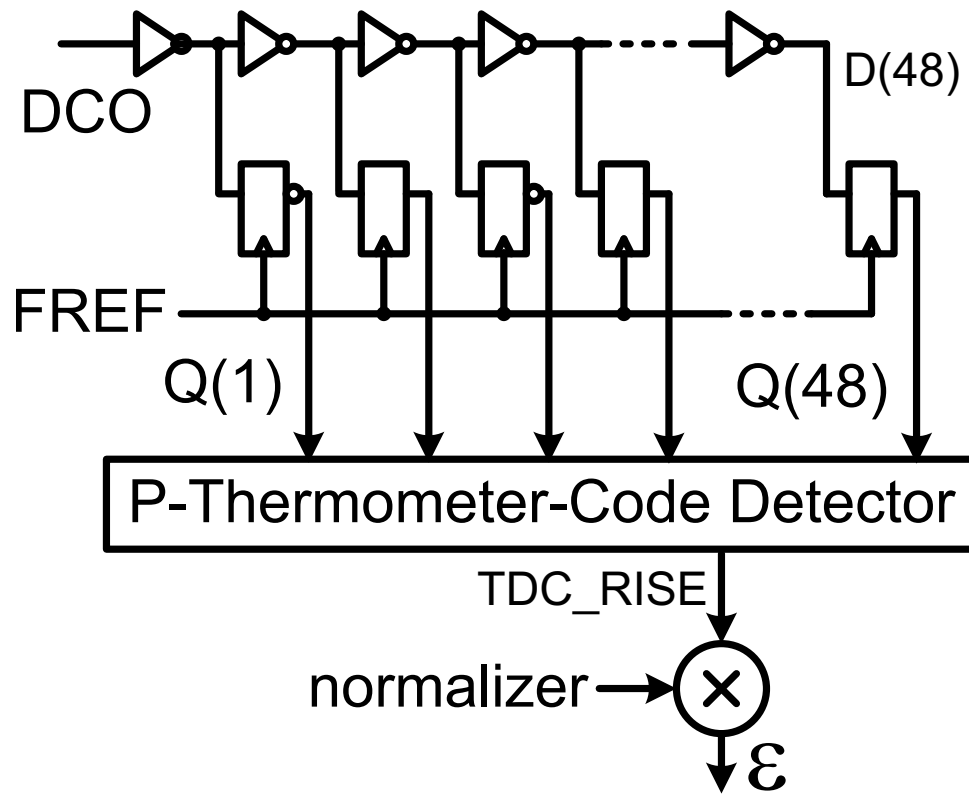
C_s is shrunk

- Minimum capacitive step beyond technology limitations

- Step size can be below 1kHz
 - No need for $\Sigma\Delta$ dithering

[Fanori ISSCC 2010]

Time-to-Digital Converter (TDC)

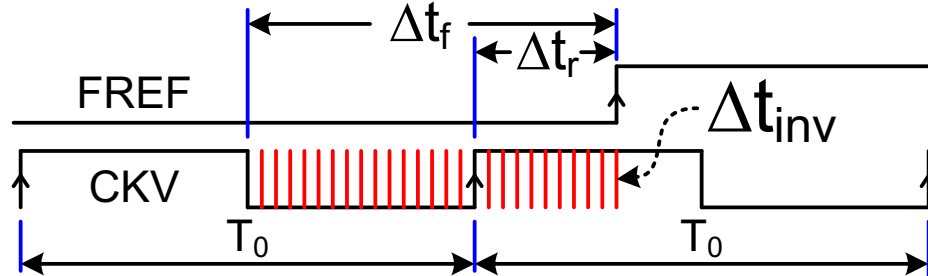


- Quantized phase detector with resolution of about 10 ps
- DCO clock passes through the inverter chain
- Delayed outputs are sampled by FREF

TDC Normalization

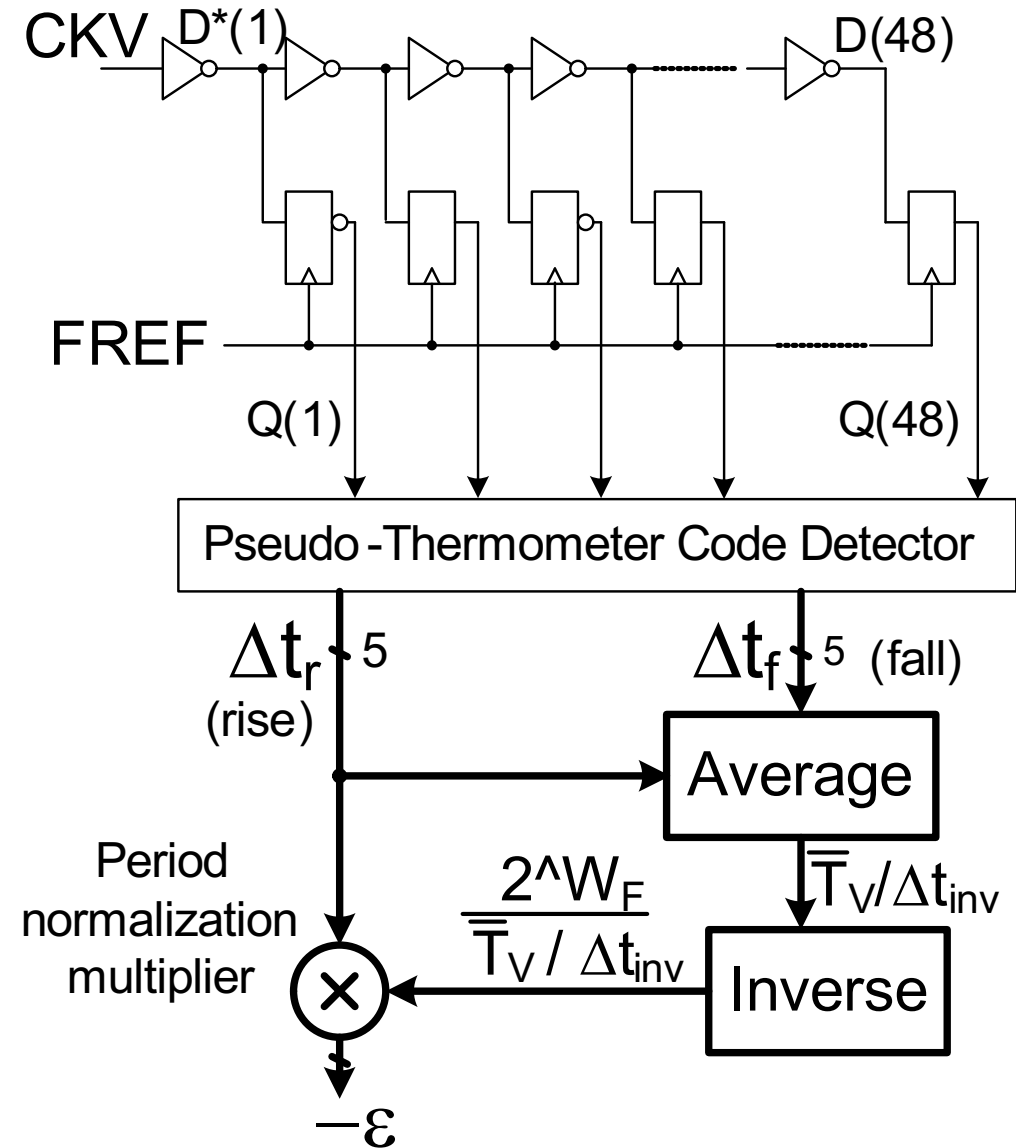
- Expected output between 0.0 – 1.0 UI

$$-\varepsilon = \frac{2^{W_F}}{\bar{T}_V / \Delta t_{inv}}$$



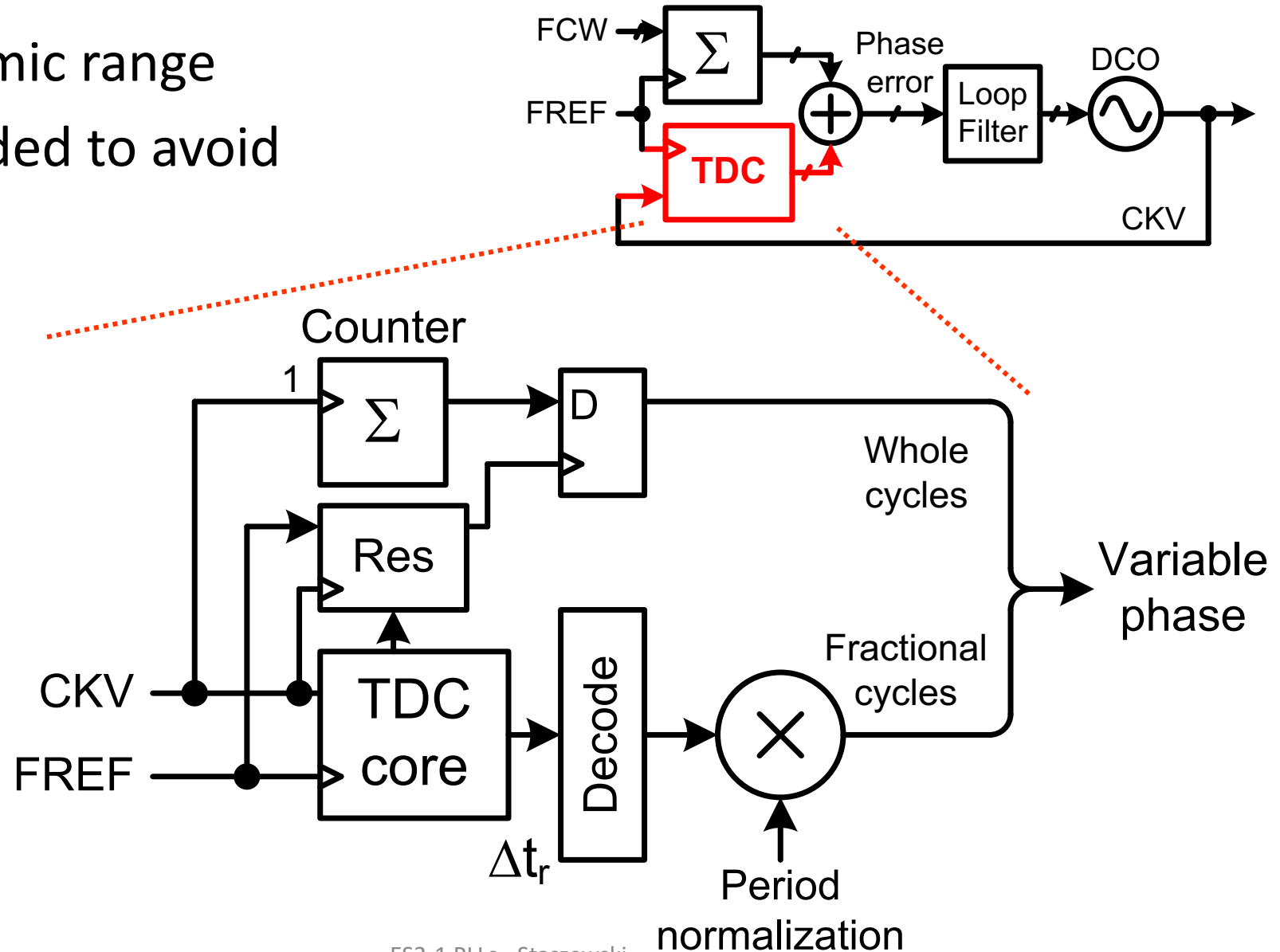
- Accurate calibration of the inverter delay

$$\bar{T}_V = \frac{1}{N_{avg}} \sum_{k=1}^{N_{avg}} T_V[k]$$

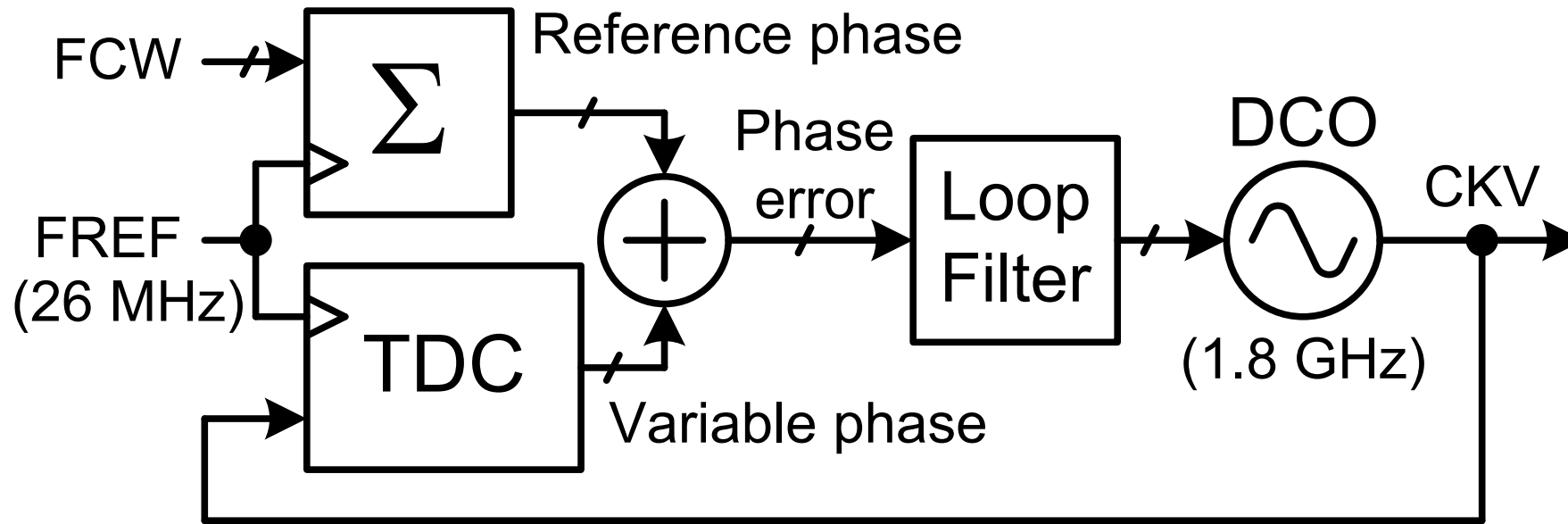


TDC System

- Extended dynamic range
- Resampler needed to avoid metastability



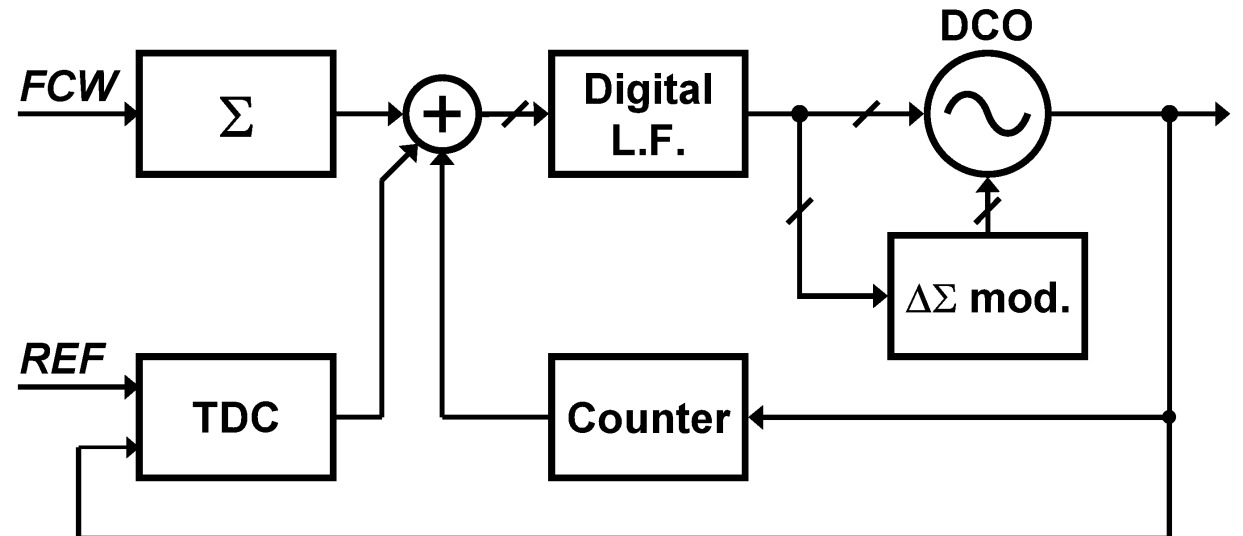
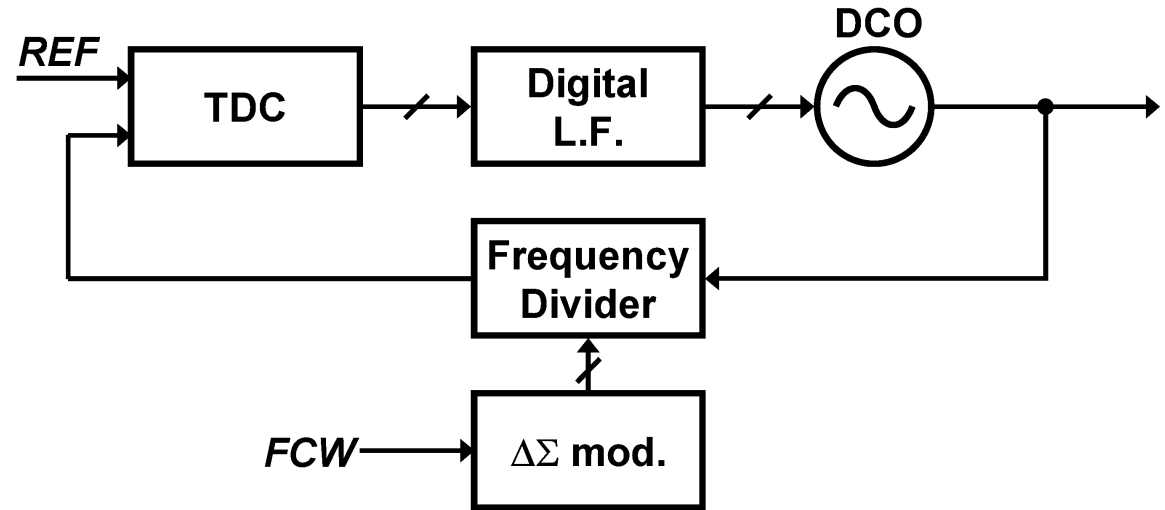
TDC-Based All-Digital PLL



- TDC replaces the conventional PFD and charge pump
- TDC measures the actual F_{REF} timestamps
- FCW is a fixed-point frequency multiplication ratio
- F_{REF} timestamps compared with accumulated FCW

Two Types of ADPLL Architecture

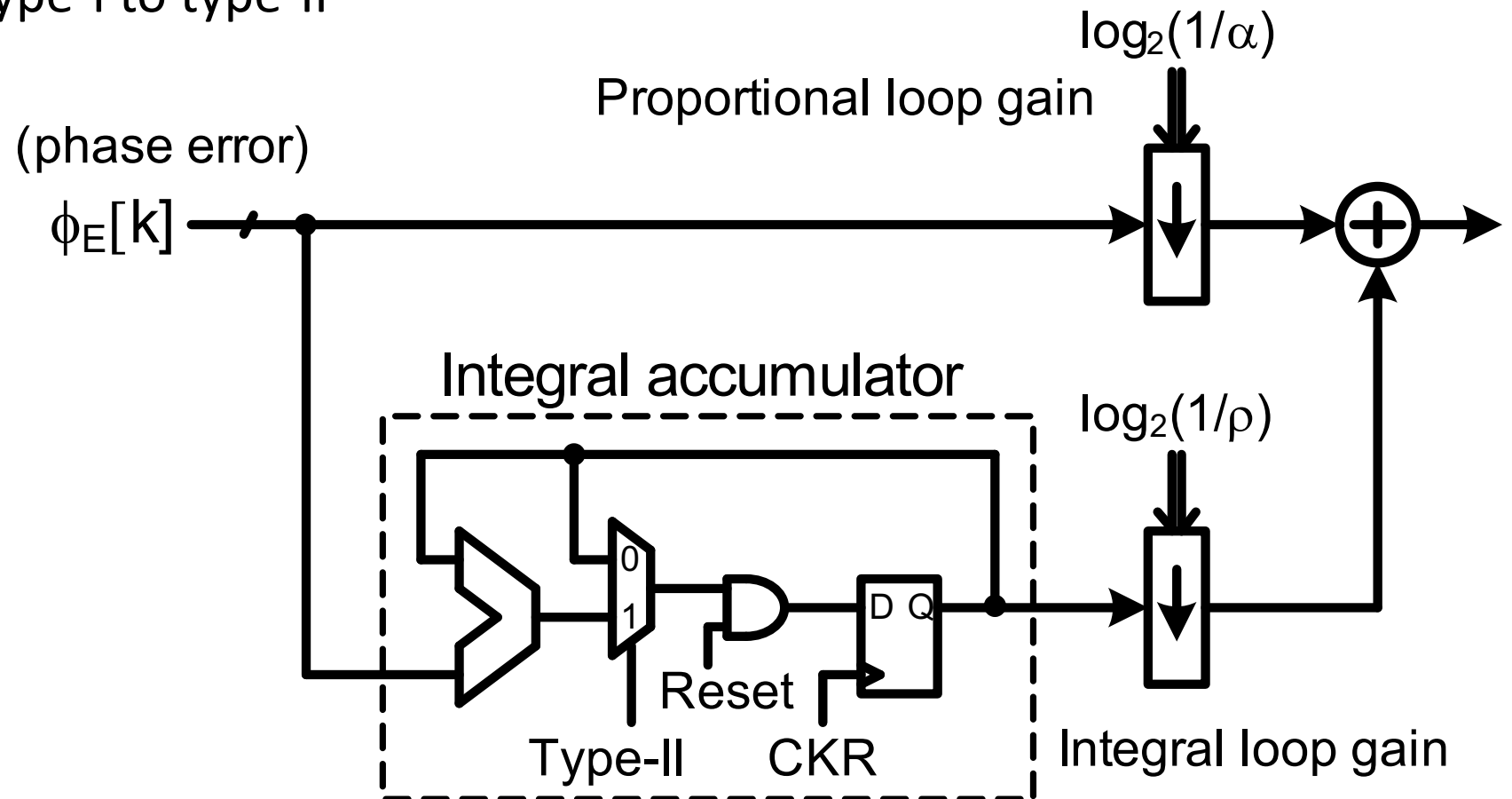
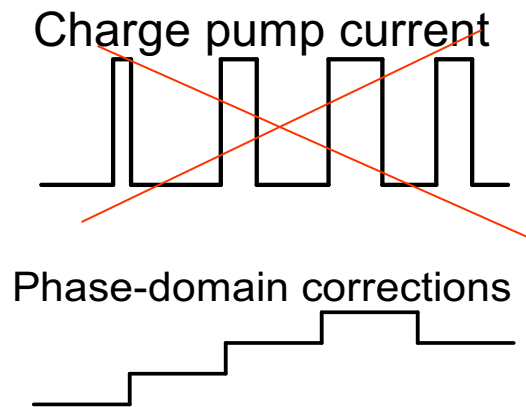
- Fractional frequency divider in the feedback path
 - Excessive quantization noise
- Counter-based
 - Chosen in that paper



[Tokairin ISSCC 2010]

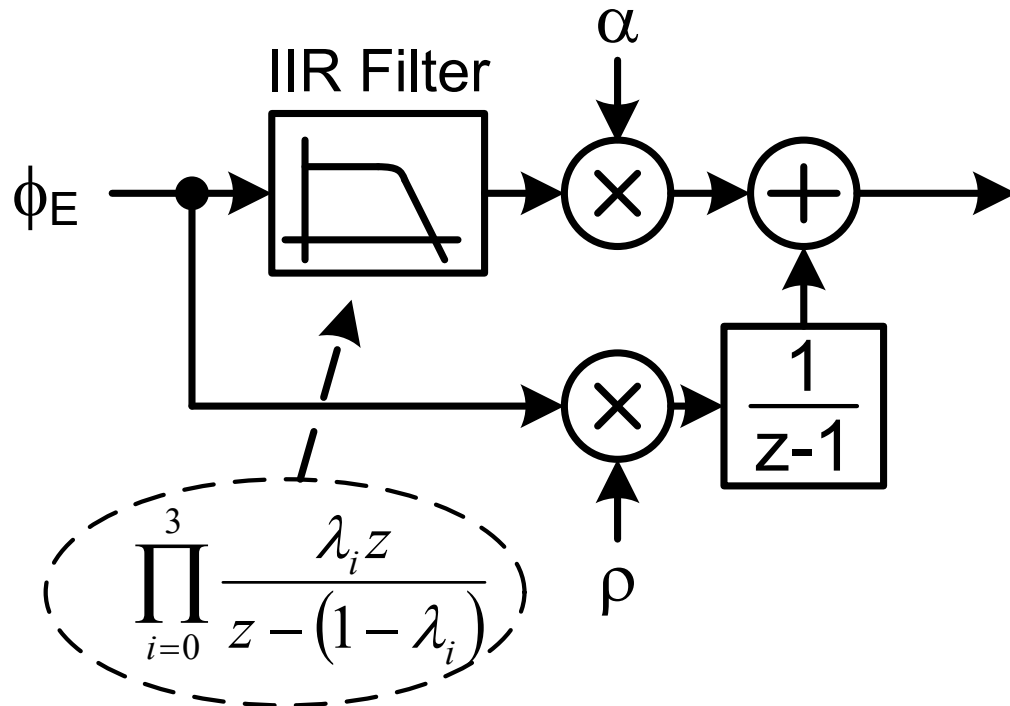
Type-II Loop Filter

- No correlative detection spurs
- Software programmed PLL loop:
 - Gentle transition of type-I to type-II
- Two “knobs” α , ρ



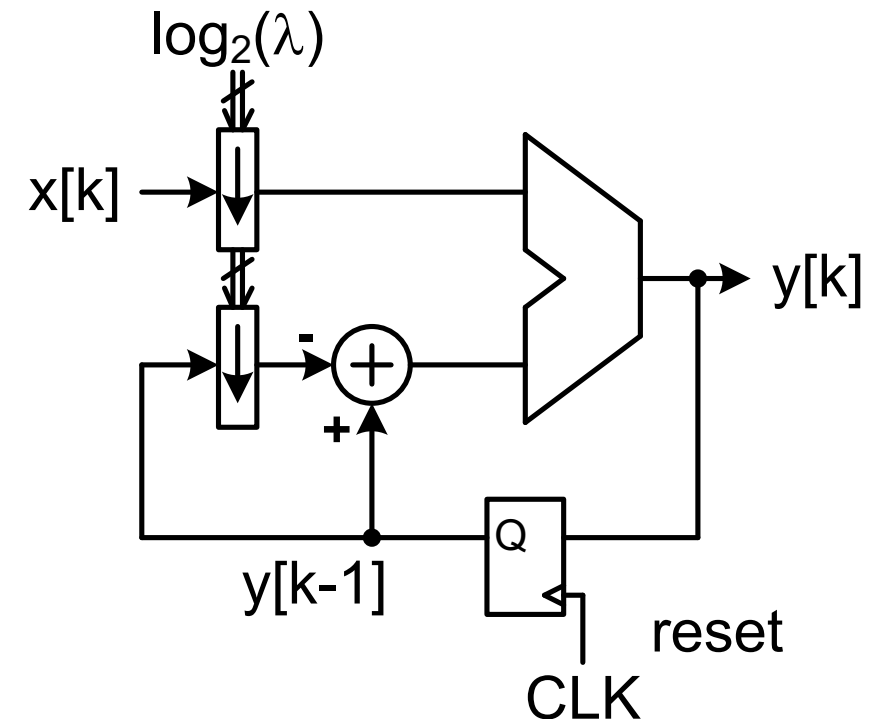
Loop Filter with IIR Filtering

- 4th order digital IIR loop filter to suppress the frequency reference and TDC quantization noise
- Unconditionally stable IIR filter



Single-pole IIR stage:

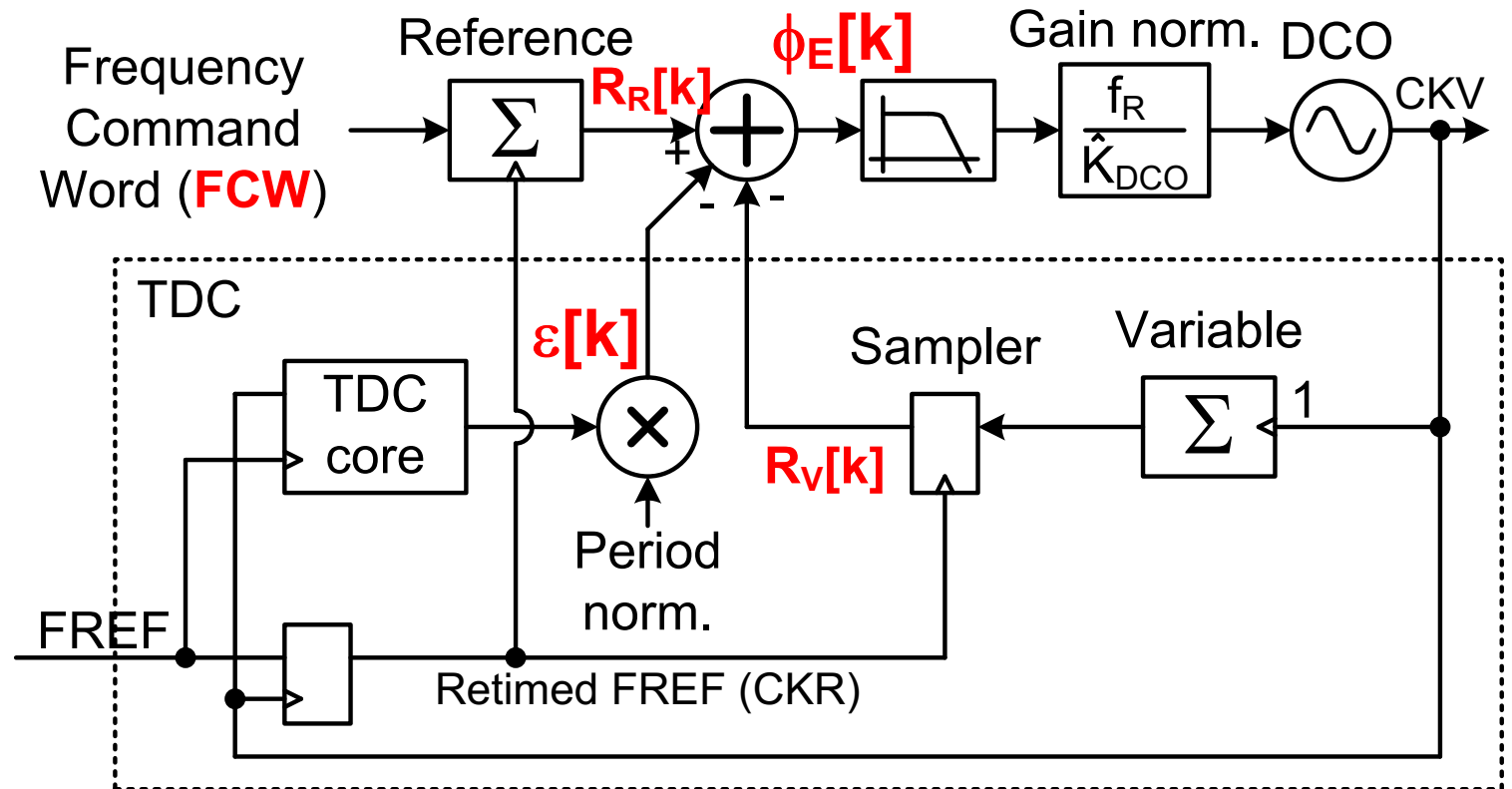
$$y[k] = (1 - \lambda) \cdot y[k - 1] + \lambda \cdot x[k]$$



All-Digital PLL (ADPLL)

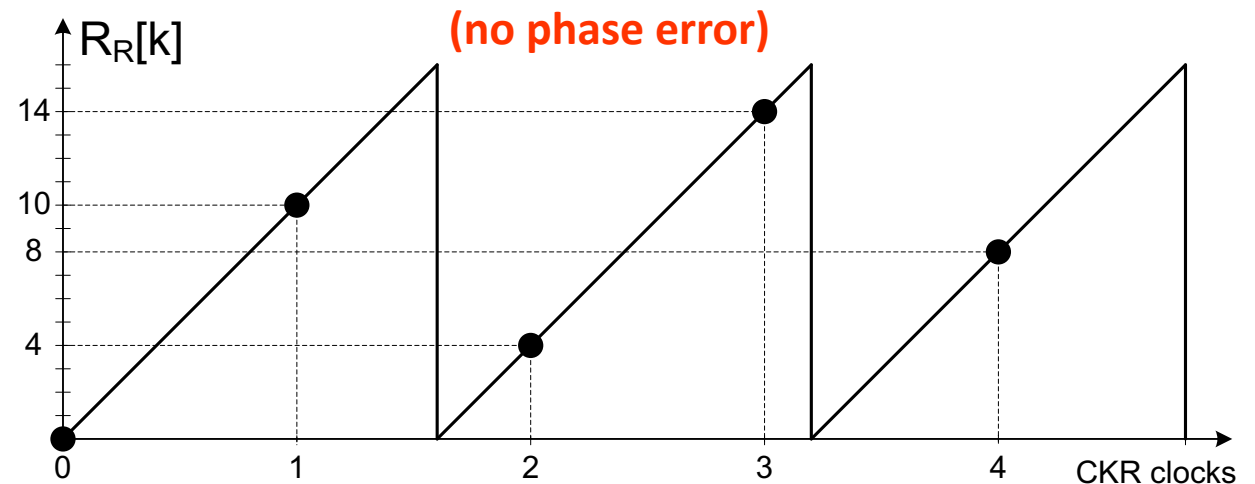
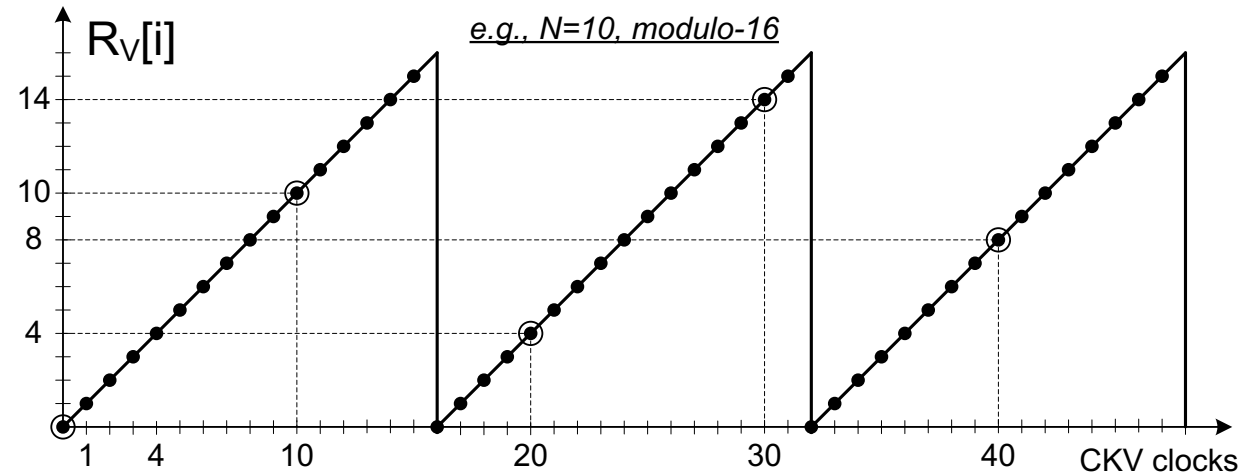
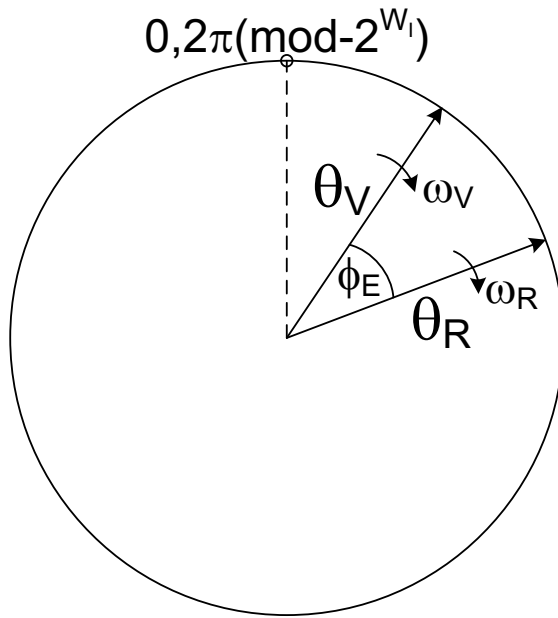
- Phase domain operation
- Digitally synchronous fixed-point arithmetic
- Phase signals cannot be corrupted by noise

$$\varphi_E = R_R[k] - (R_V[k] + \varepsilon[k])$$



Modulo Arithmetic

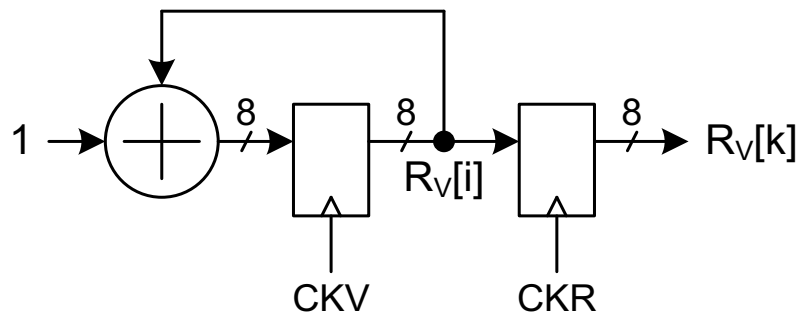
- Theoretically, reference θ_R and variable θ_V phases grow without bound
- θ_R and θ_V implemented in modulo arithmetic to limit wordlength: $W_I = 8$, $W_F = 15$



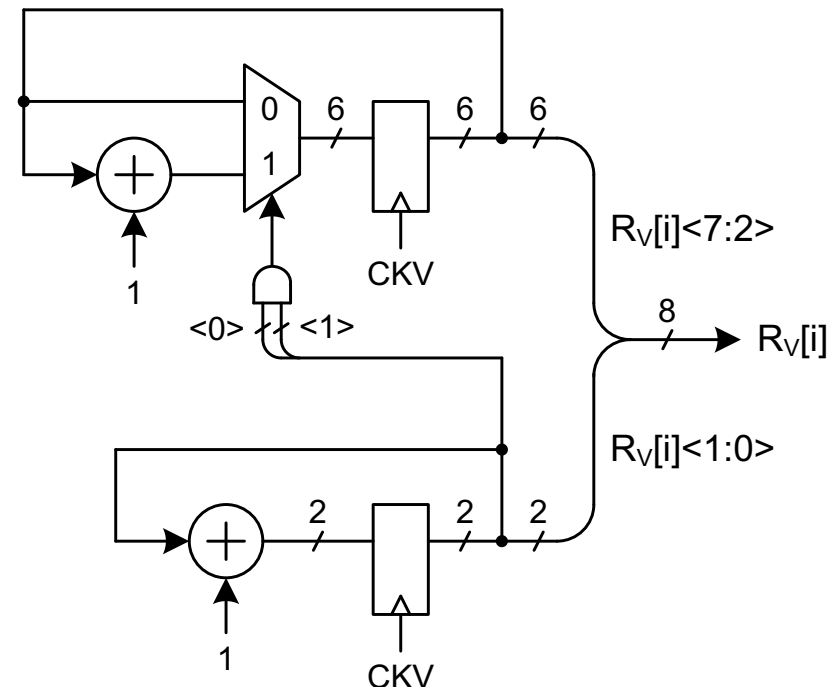
Variable Phase Accumulator

- Implements DCO clock count incrementing
- Deep-submicron process capable of 8-bit incrementing at 2.4 GHz
- Extra timing margin required to guarantee robust operation
- Separated calculation between lower and higher order bits

Function:

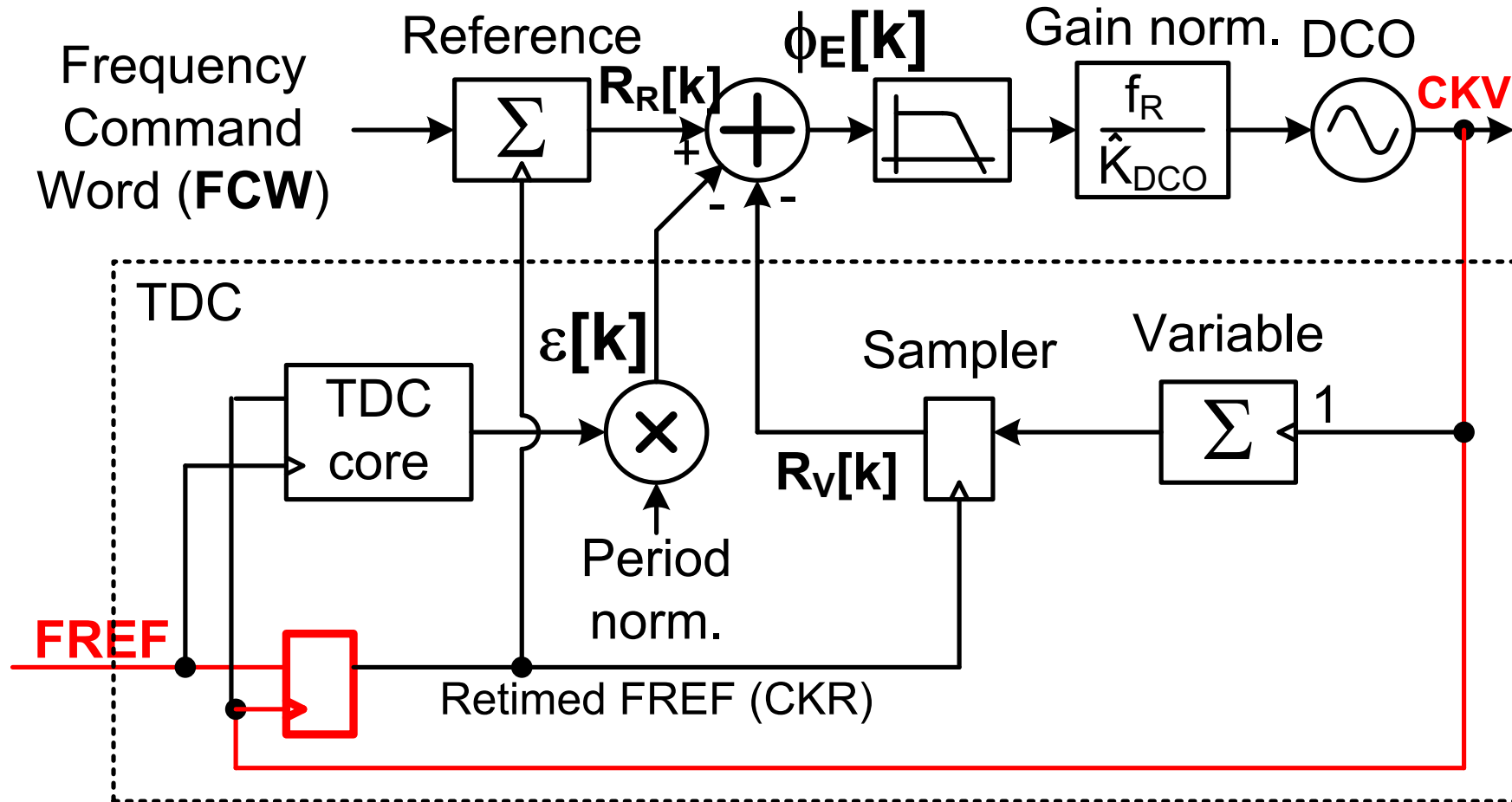


Using pipelining:



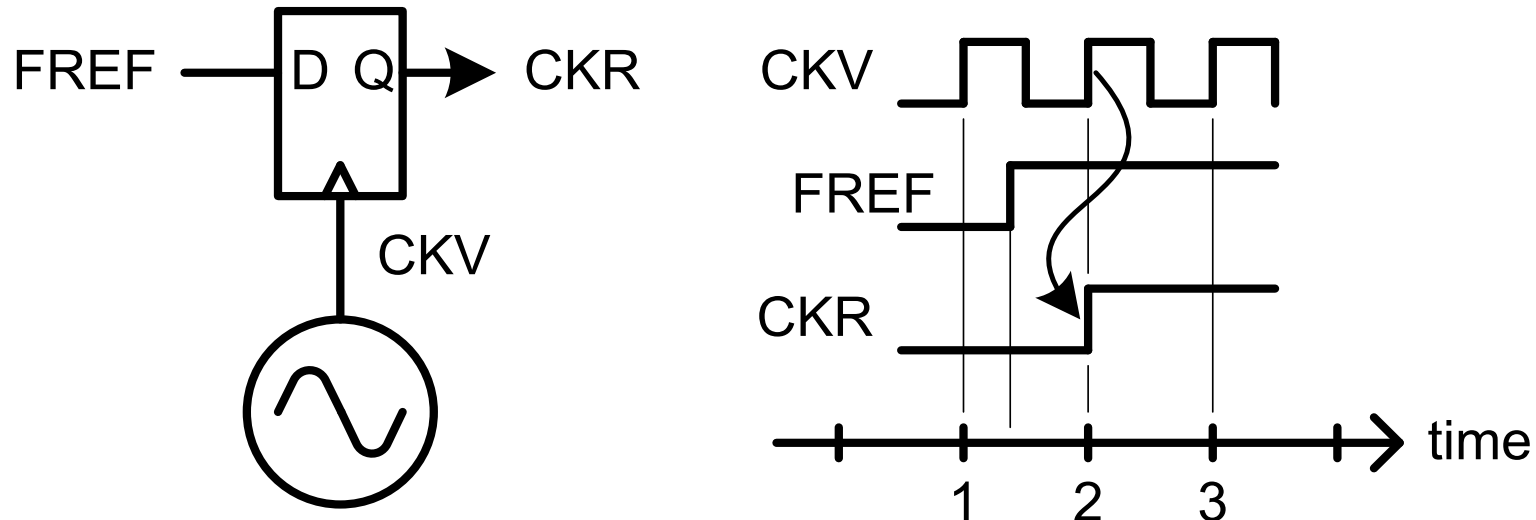
bits							
7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1
0	0	0	0	0	0	1	0
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	1	0	0
0	0	0	0	0	1	0	1
0	0	0	0	0	1	1	0
0	0	0	0	0	1	1	1

Reference Clock Retiming



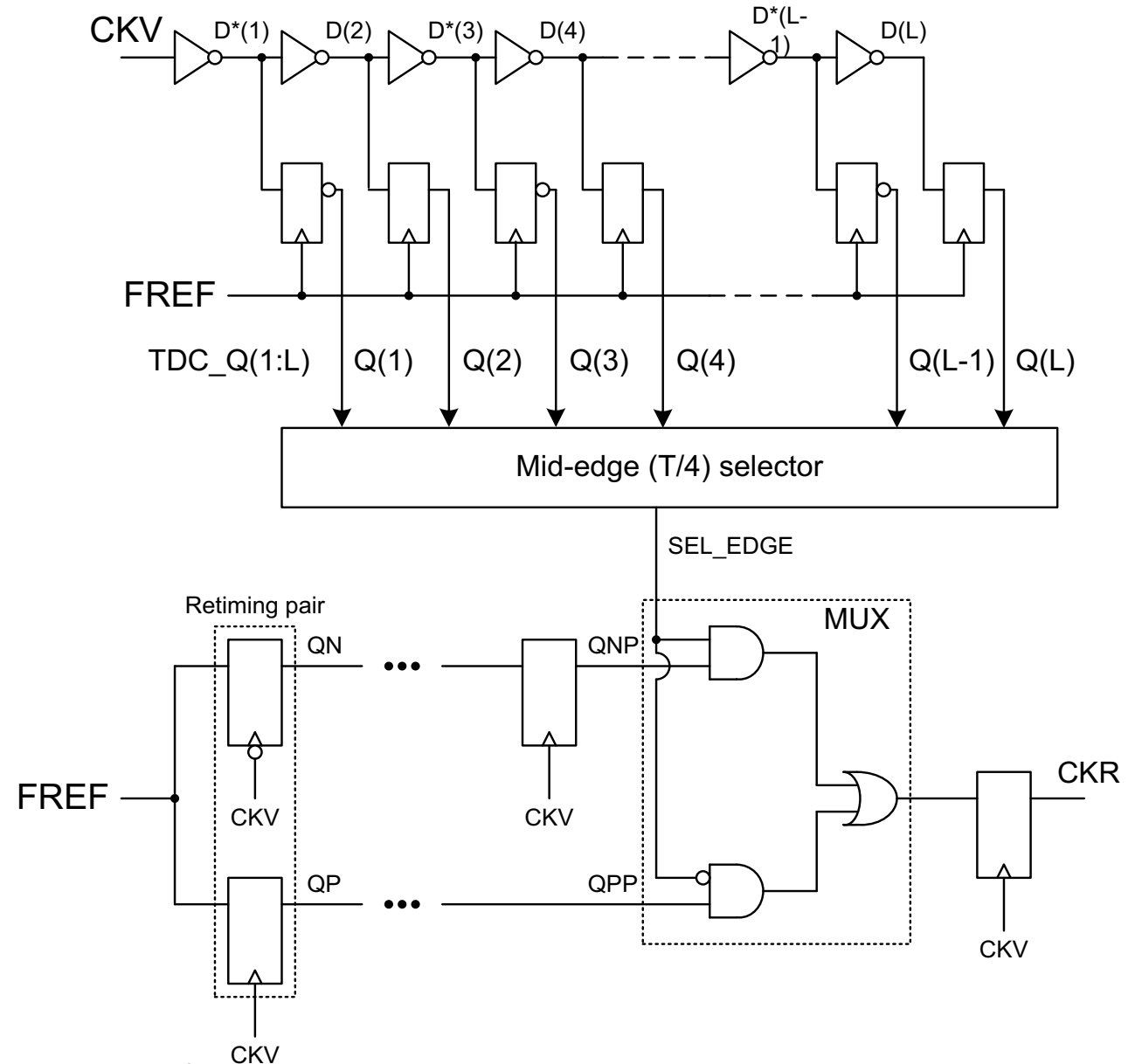
Reference Phase Retiming

- DCO clock and FREF domains are not entirely synchronous despite being in phase lock
- Variable and reference phases cannot be compared in hardware: metastability!
- Solution: Oversampling FREF by CKV and using the resulting CKR

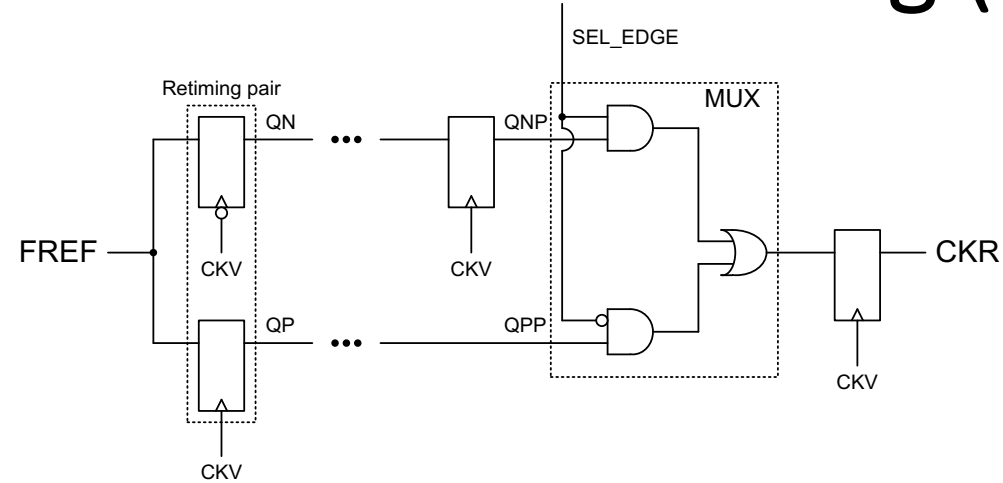


Reference Clock Retiming

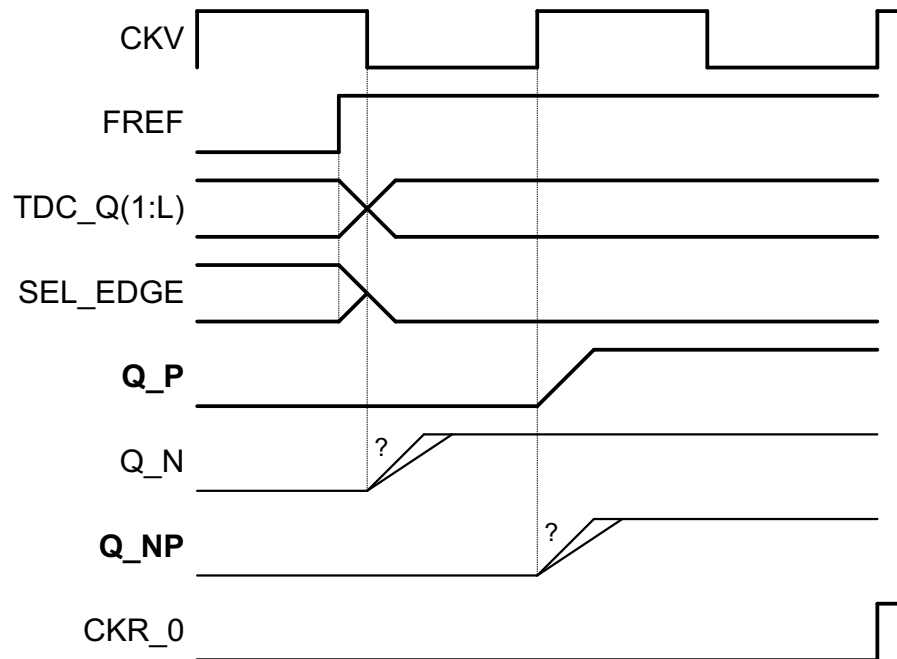
- Key ADPLL idea to synchronize reference and variable clock planes
- Reuses the TDC vector, thus little additional cost
- Avoids metastability by selecting “the better” DCO clock edge
- Can use redundancy in the thermometer-encoded TDC output vector for a high degree of error correction
- Metastability resolution issues similar as in a flash ADC
 - It got solved there! (bubble correction)



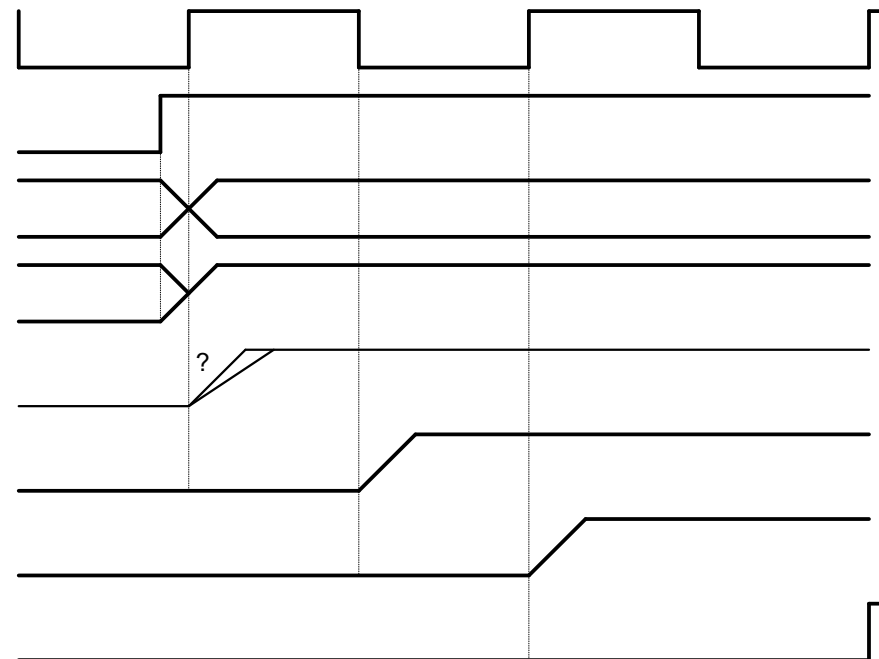
Reference Clock Retiming (cont'd)



Potential metastability of the falling-edge-retimed FREF:

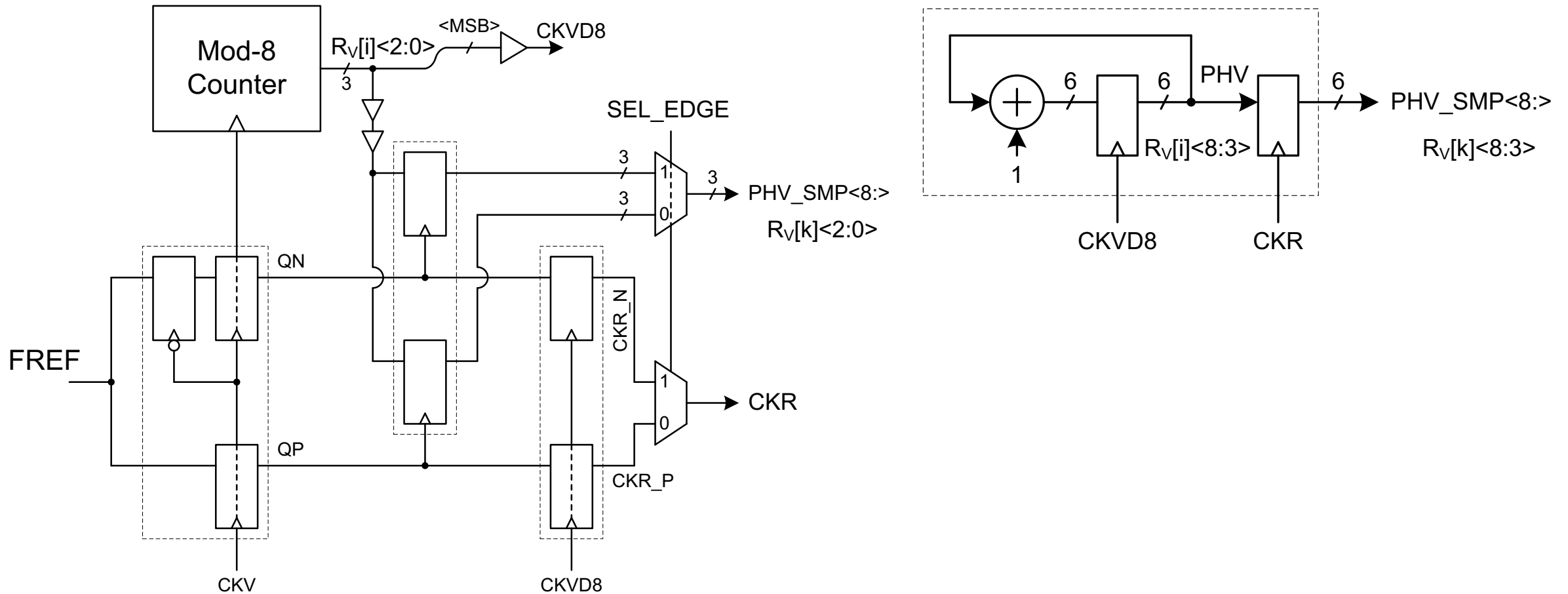


Potential metastability of the rising-edge-retimed FREF:



Improved PV Architecture

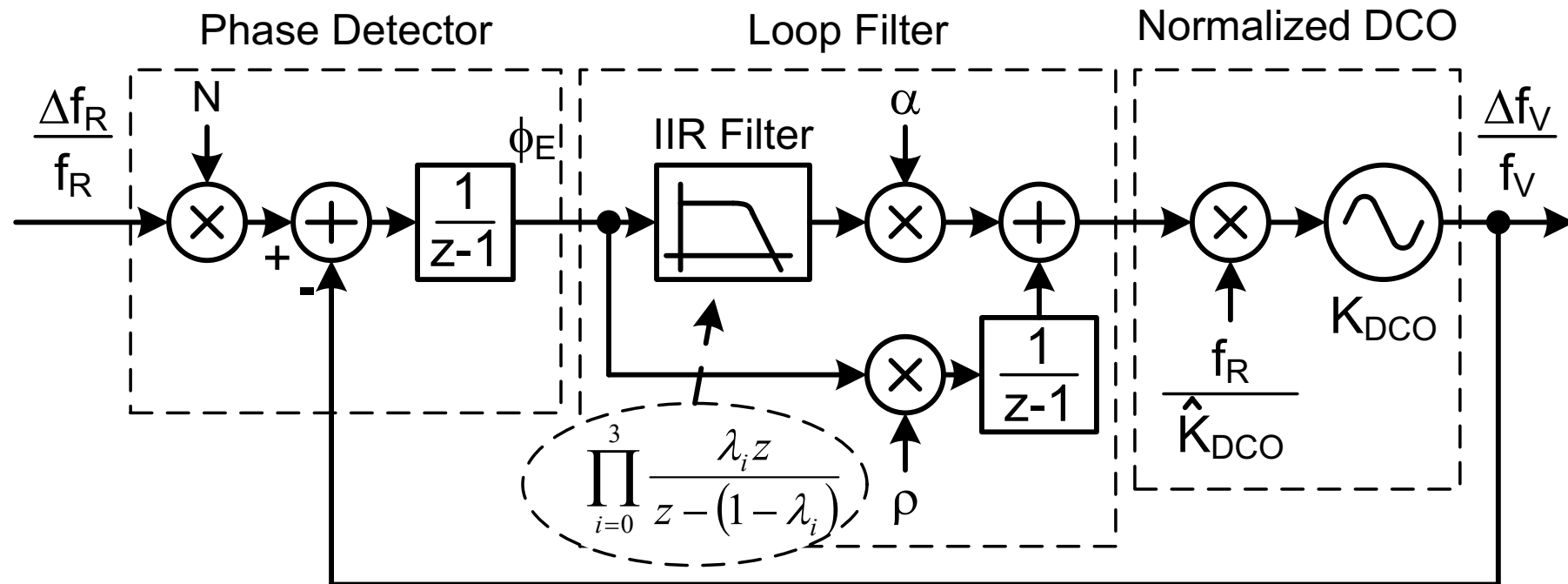
- 8 bits split into lower 3 bits and upper 5 bits



z-Domain Model of the ADPLL

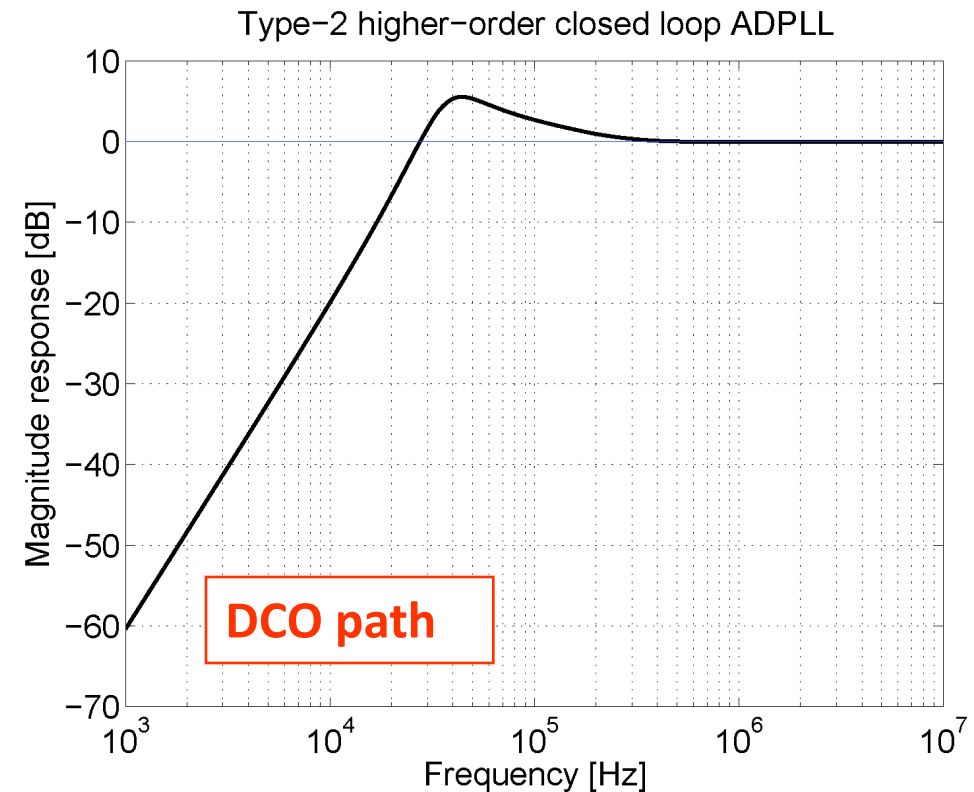
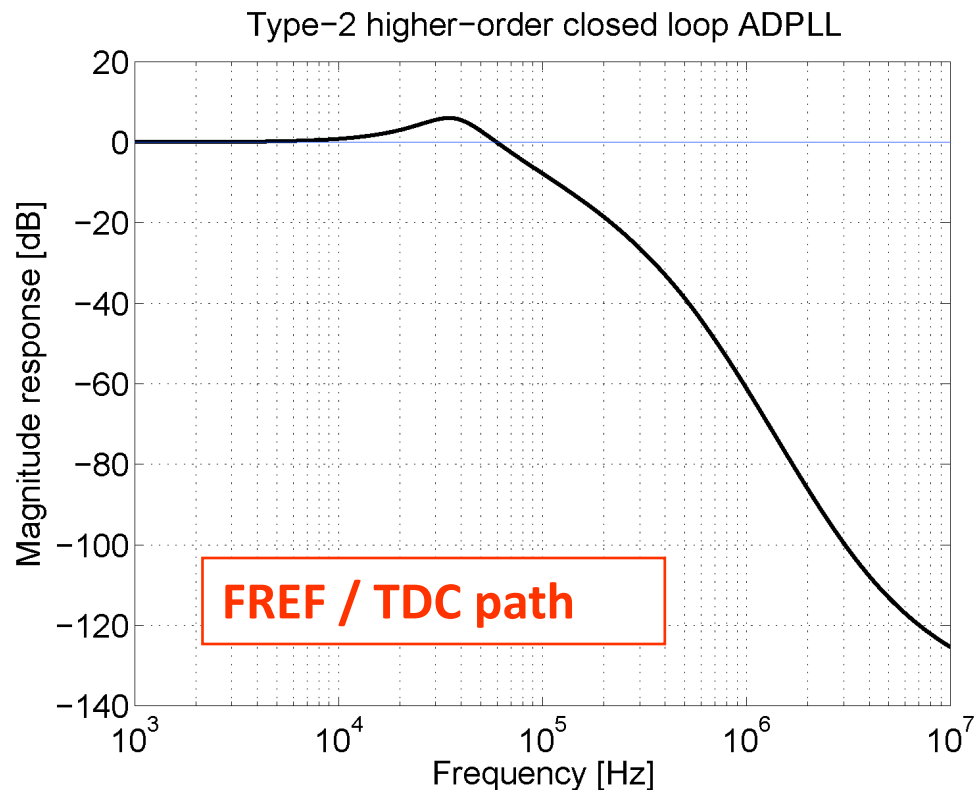
- “knobs” α , ρ , λ_{1-4}
- Type-I or Type-II PLL loop
- 1st through 6th order

$$H_{ol}(s) = \left[\alpha \cdot H_{iir}(s) + \frac{\rho \cdot f_R}{s} \right] \frac{f_R}{s}$$



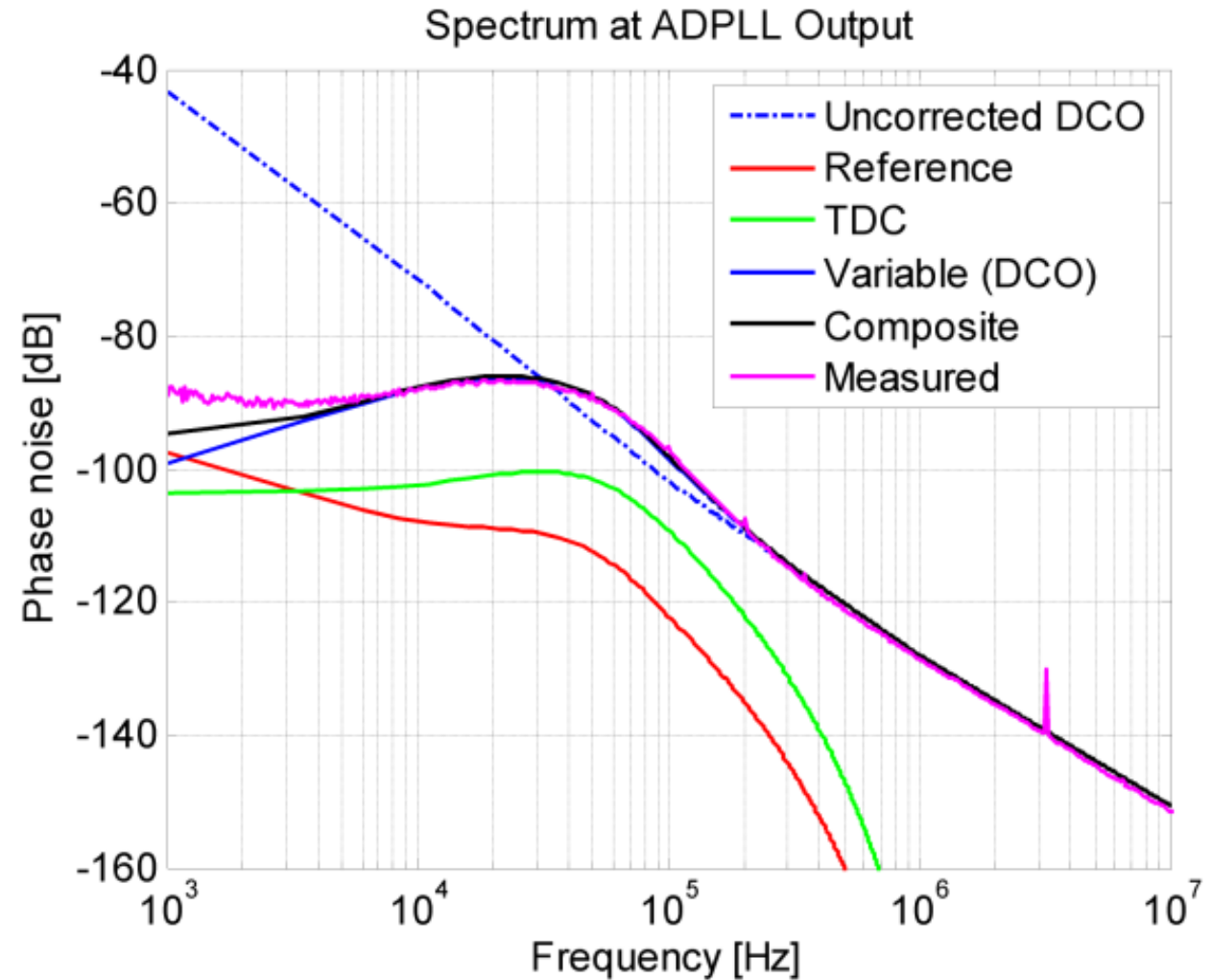
ADPLL Transfer Function

- Type-II 6th-order PLL
- Settings: $\alpha = 2^{-7}$, $\rho = 2^{-15}$, $\lambda = 2^{-[3\ 3\ 3\ 4]}$
- Provides 33 dB of attenuation at 400 kHz
- Provides 40 dB/dec filtering of 1/f DCO noise



Validation of ADPLL Noise Model

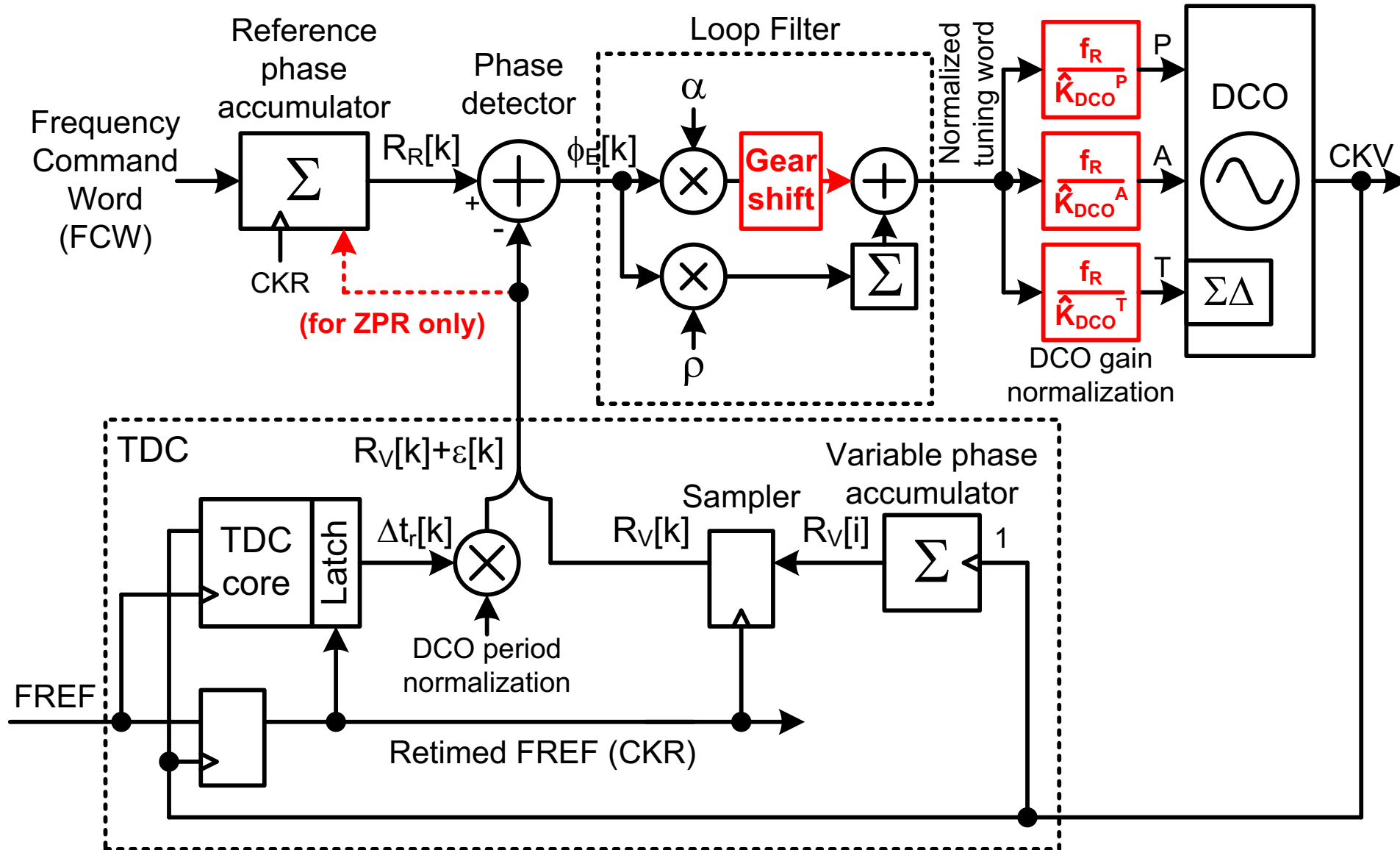
- Measured phase noise superimposed on the noise model
- Theoretical Q-noise level verified!



- [Bashir, RFIC'09]

Ultra-Fast ADPLL Settling

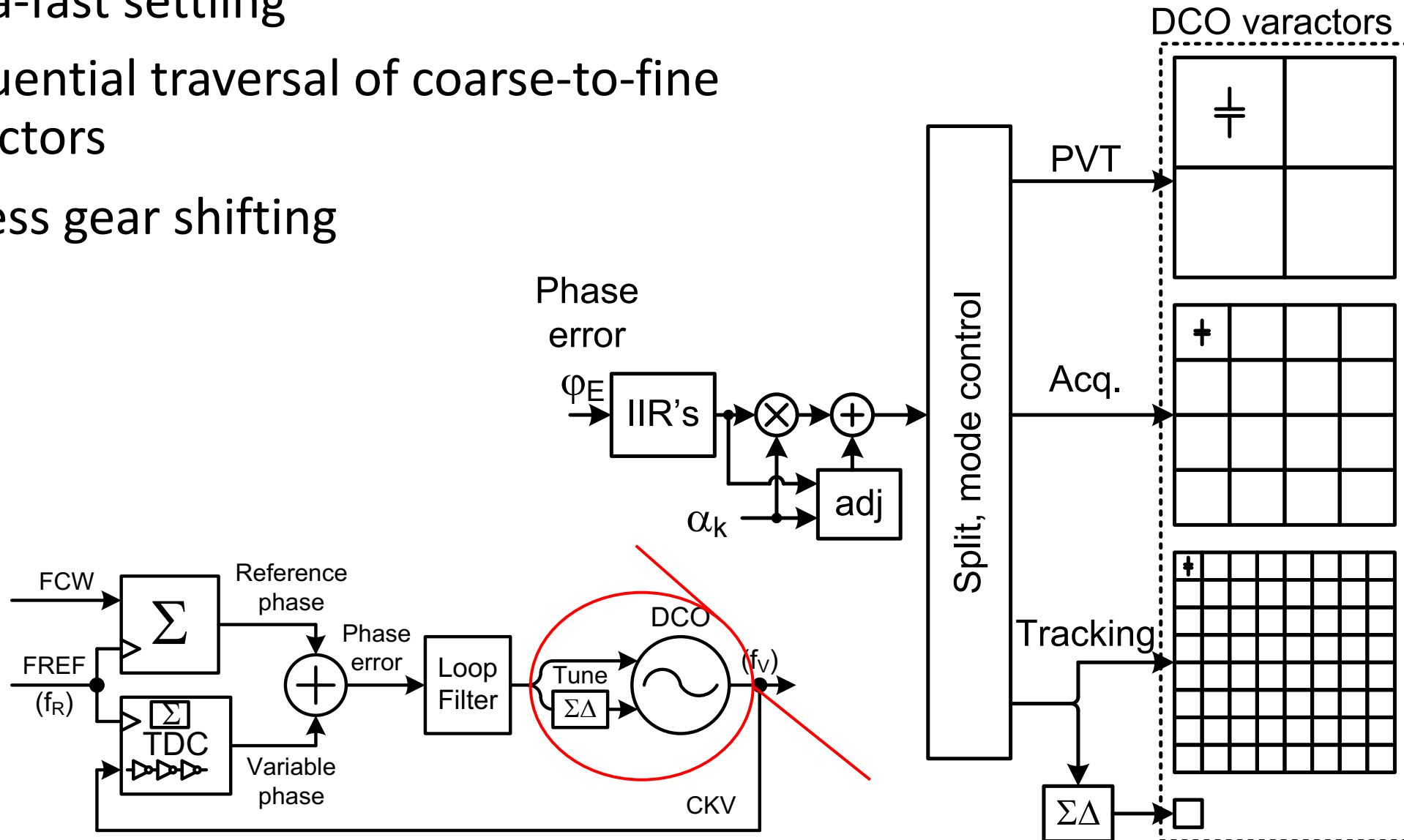
ADPLL with Zero Phase Restart



- 5...20 us to settle

ADPLL Loop Control

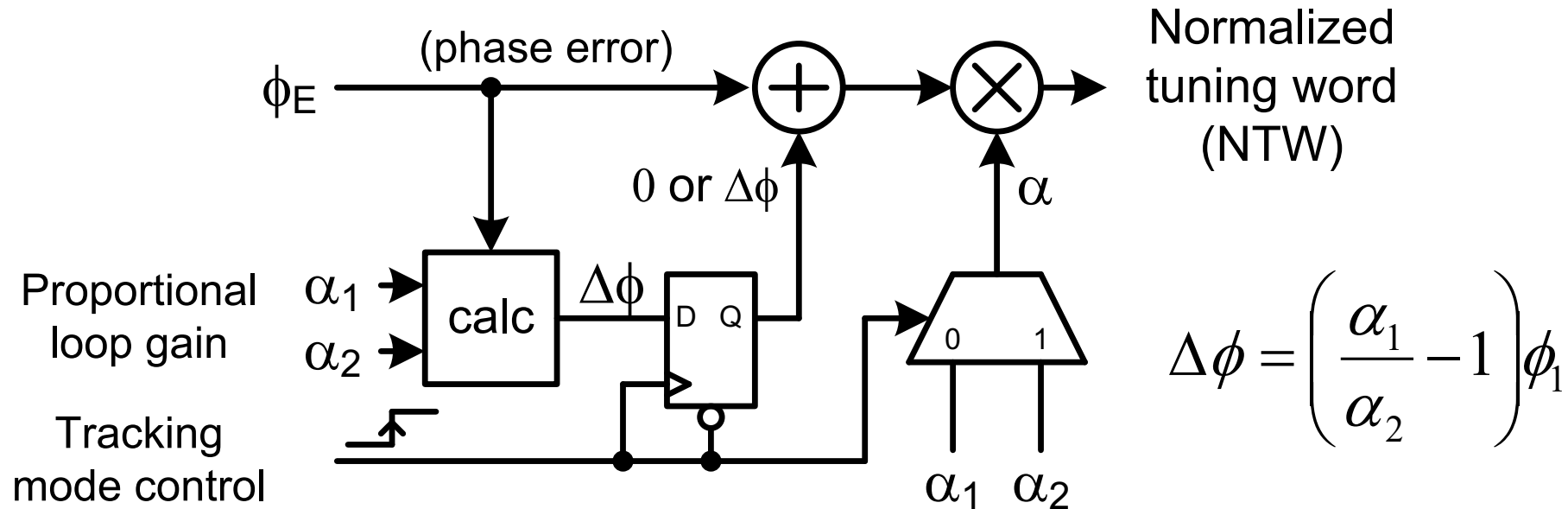
- Ultra-fast settling
- Sequential traversal of coarse-to-fine varactors
- Hitless gear shifting



Gear Shifting of PLL Bandwidth

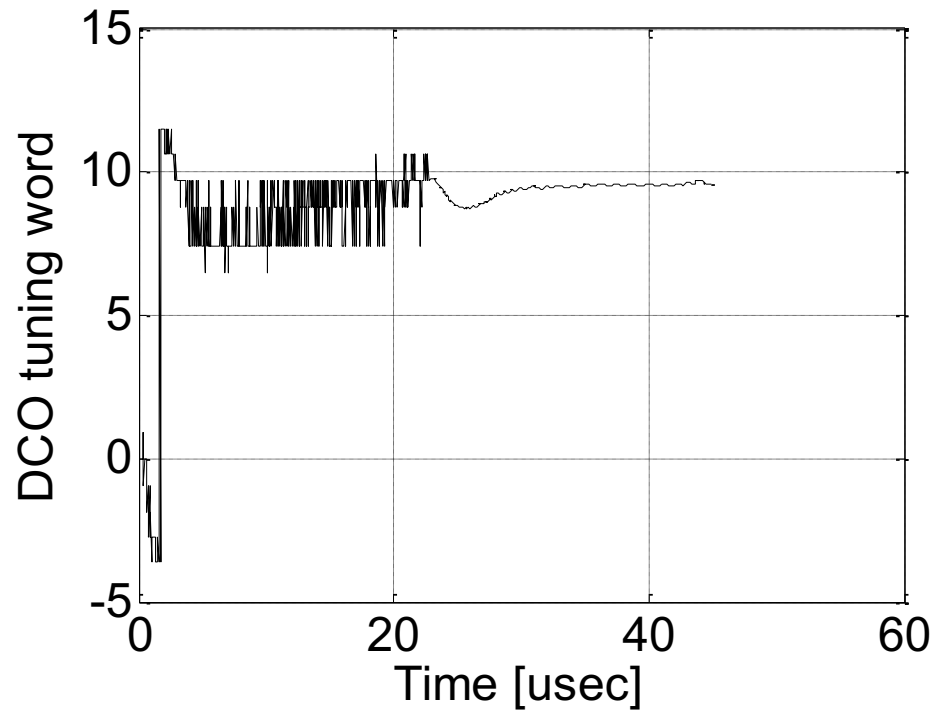
- Executed in tracking mode after the acquisition is completed
- Normalized tuning word continuity before and after the event
- Guarantees no frequency perturbation of the oscillator

$$\alpha_1 \cdot \phi_1 = \alpha_2 \cdot (\phi_1 + \Delta\phi)$$

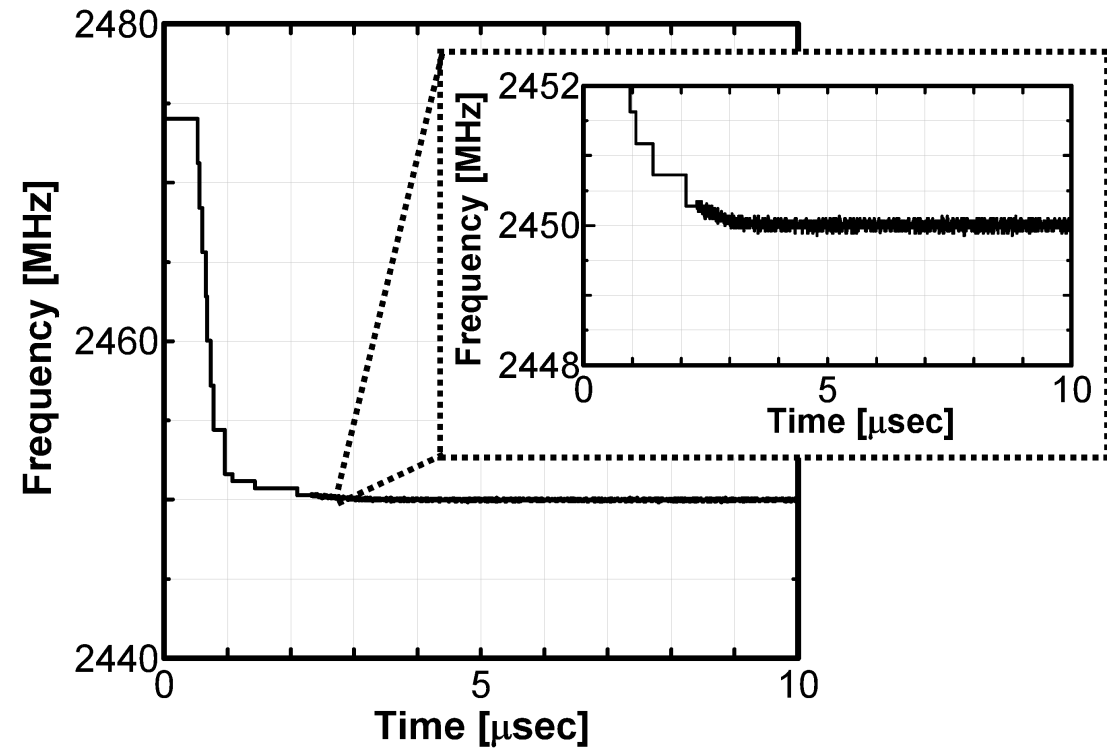


Measured Trajectory during Settling

- Gear shift at 23 μs

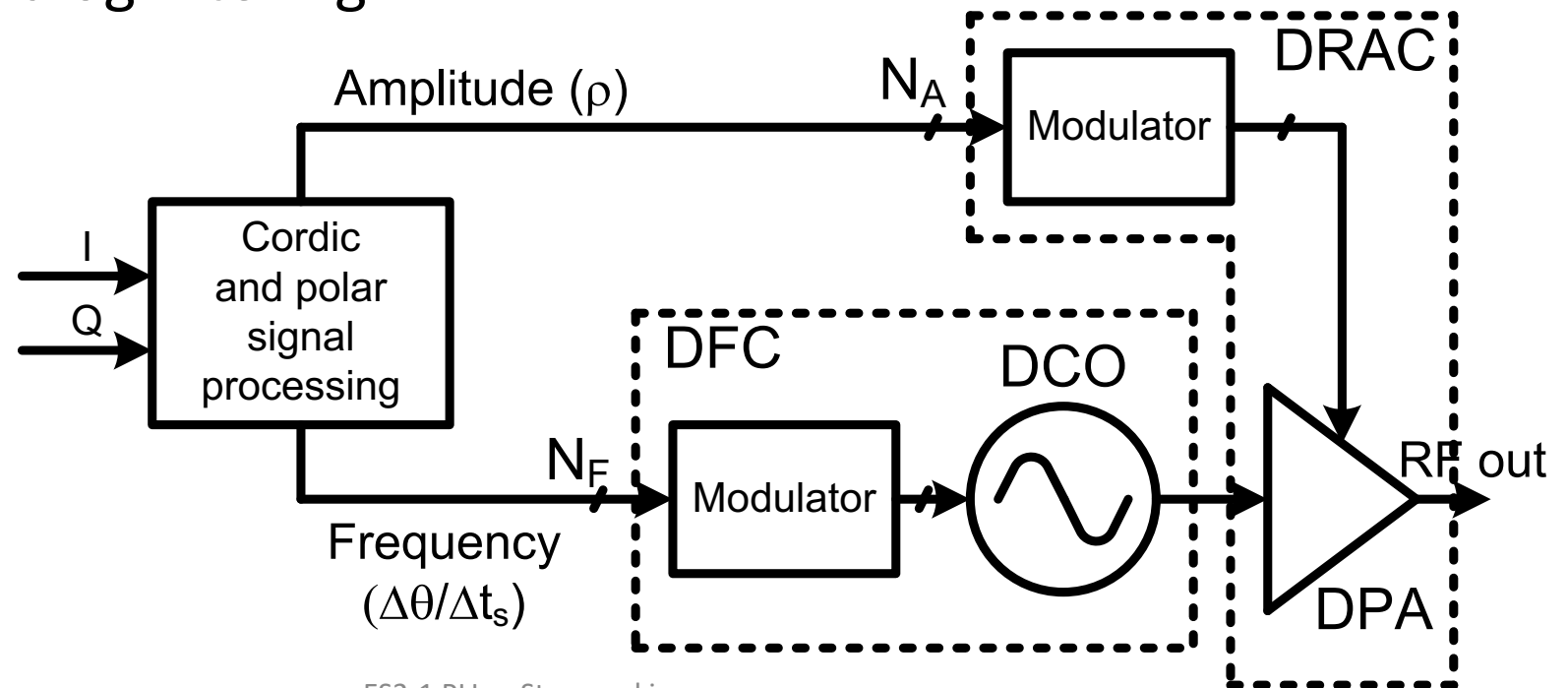
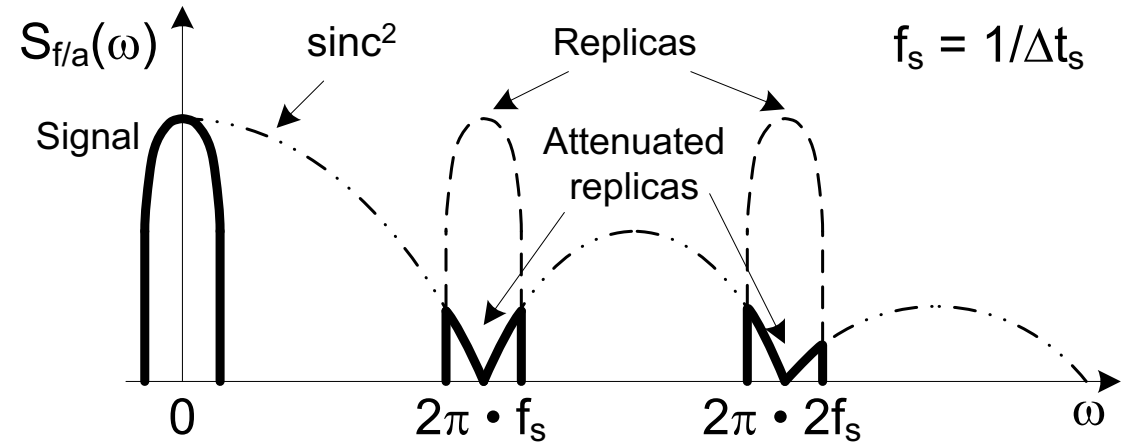


- Type-I ADPLL [Tokairin ISSCC 2010]



All-Digital Polar TX Architecture

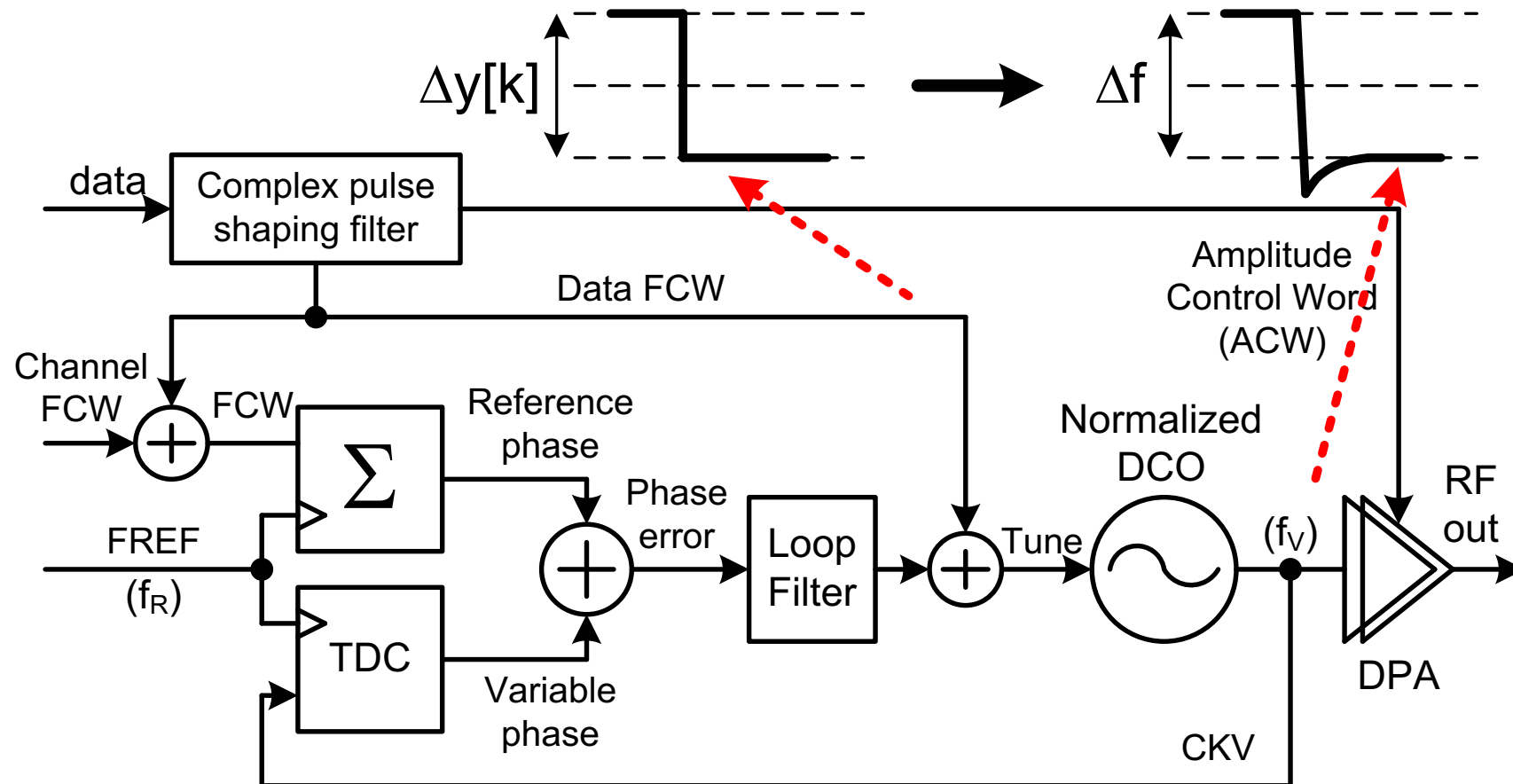
- Dense logic for digital signal processing
- DCO for digital-to-frequency conversion
- DPA for digital-to-RF-amplitude conversion
- No continuous-time analog filtering



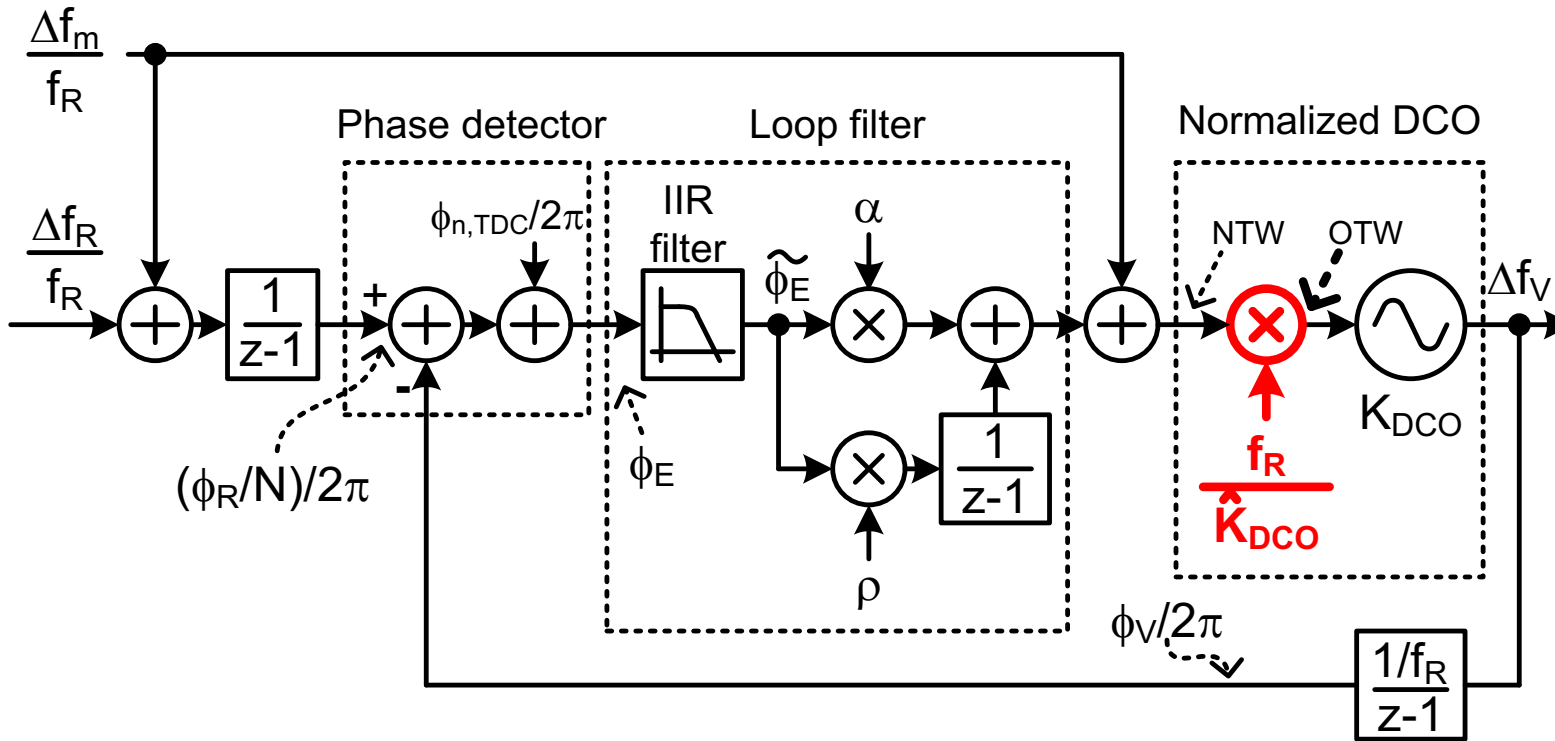
Also see
[Syllaios 2010 TCAS-I]

ADPLL with Wideband Modulation

- Two-point frequency modulation
 - Direct feedforward path – $y[k]$ directly drives the DCO
 - Compensating path – $y[k]$ added to the channel FCW



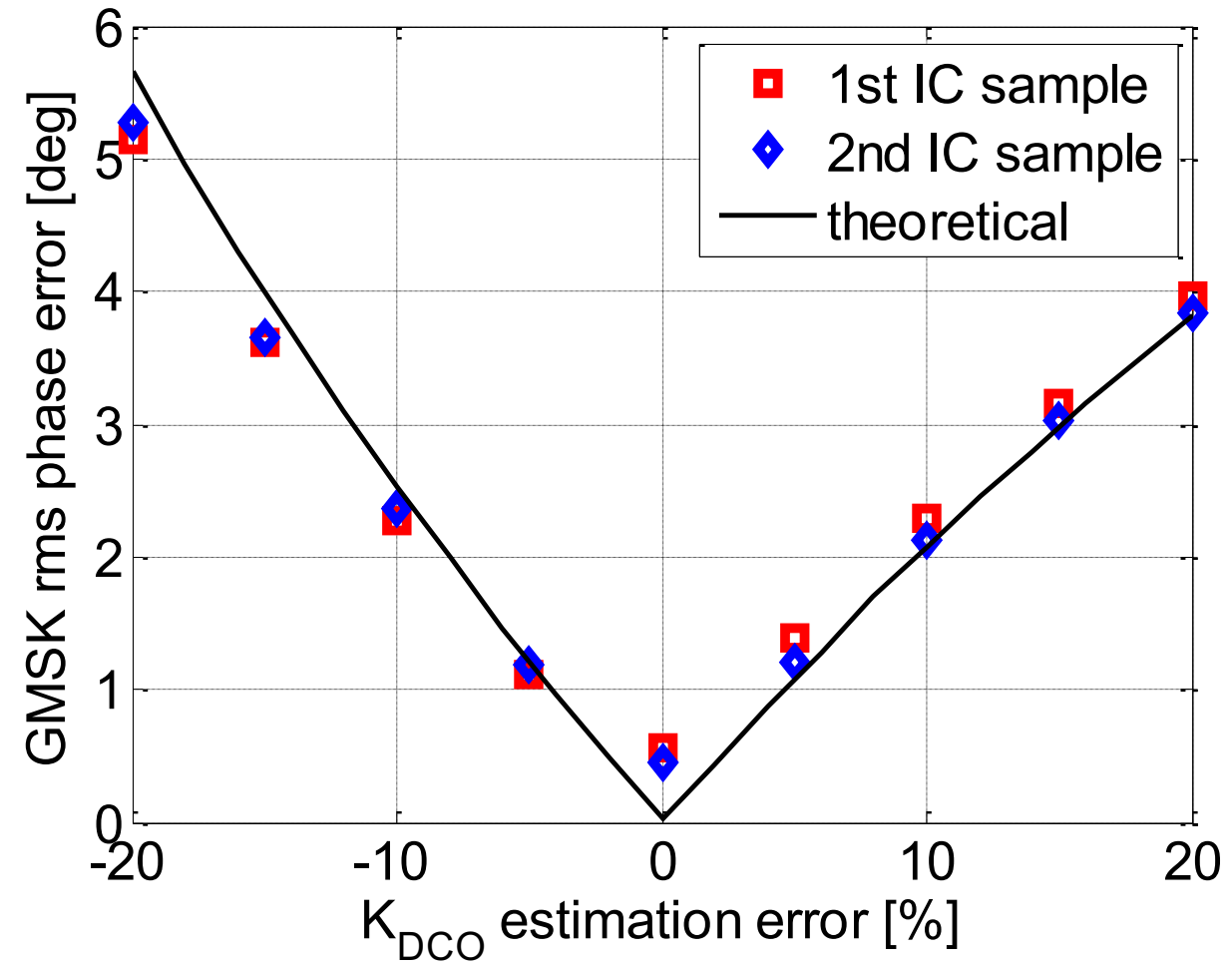
DCO Gain Estimation



- Estimation of RF oscillator gain is critical in low-cost high-volume transceivers
- RX: sets the loop bandwidth
- TX: sets transfer function of the direct frequency modulation path
- Tolerated gain estimation error from less than 1% (CDMA) to several % (GSM, Bluetooth)

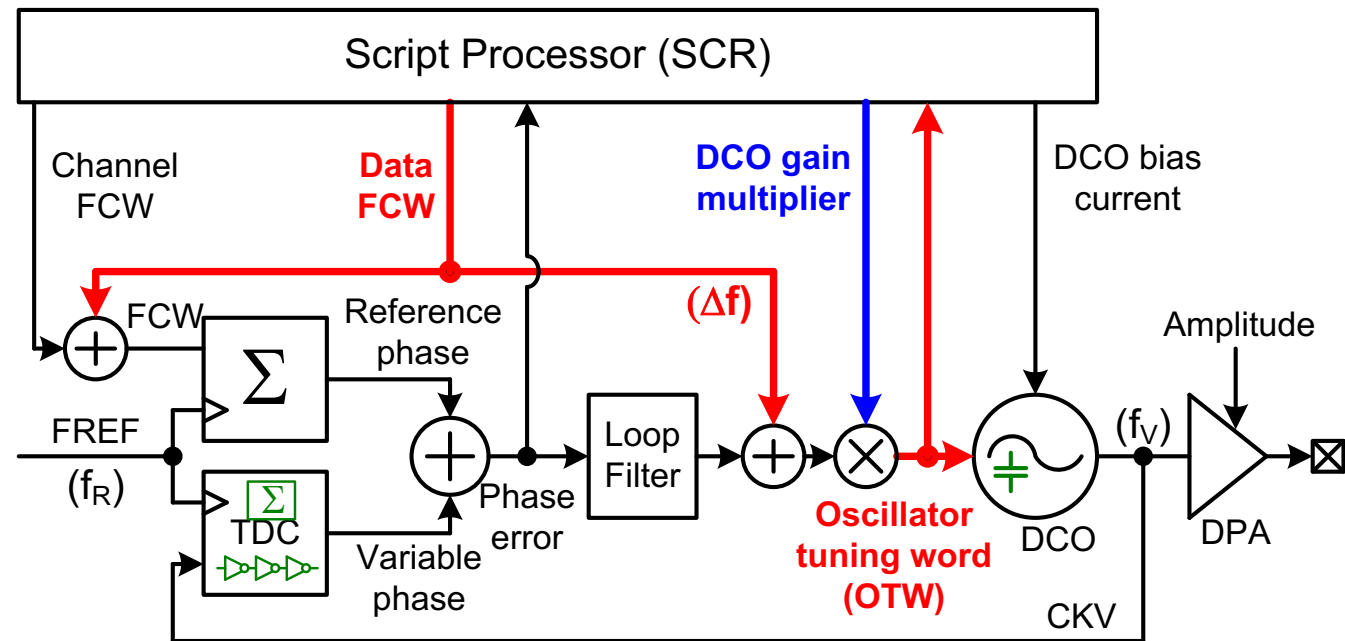
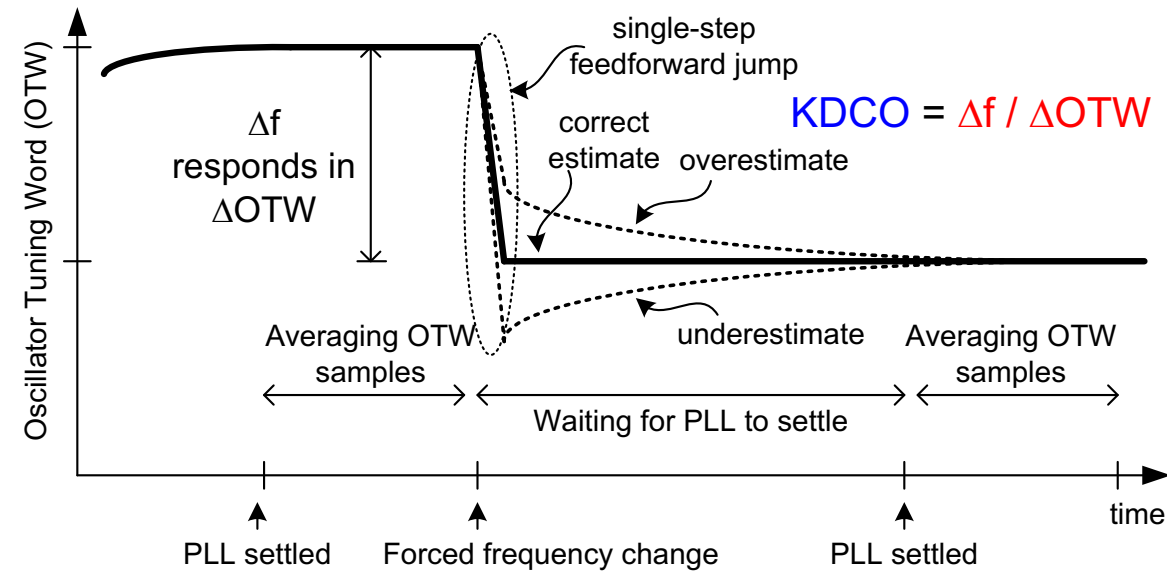
K_{DCO} Error Effect on GSM Modulation

- 2 measured IC samples over z-domain theory
- GSM spec is 5 degrees
- KDCO estimation error of several % allowed



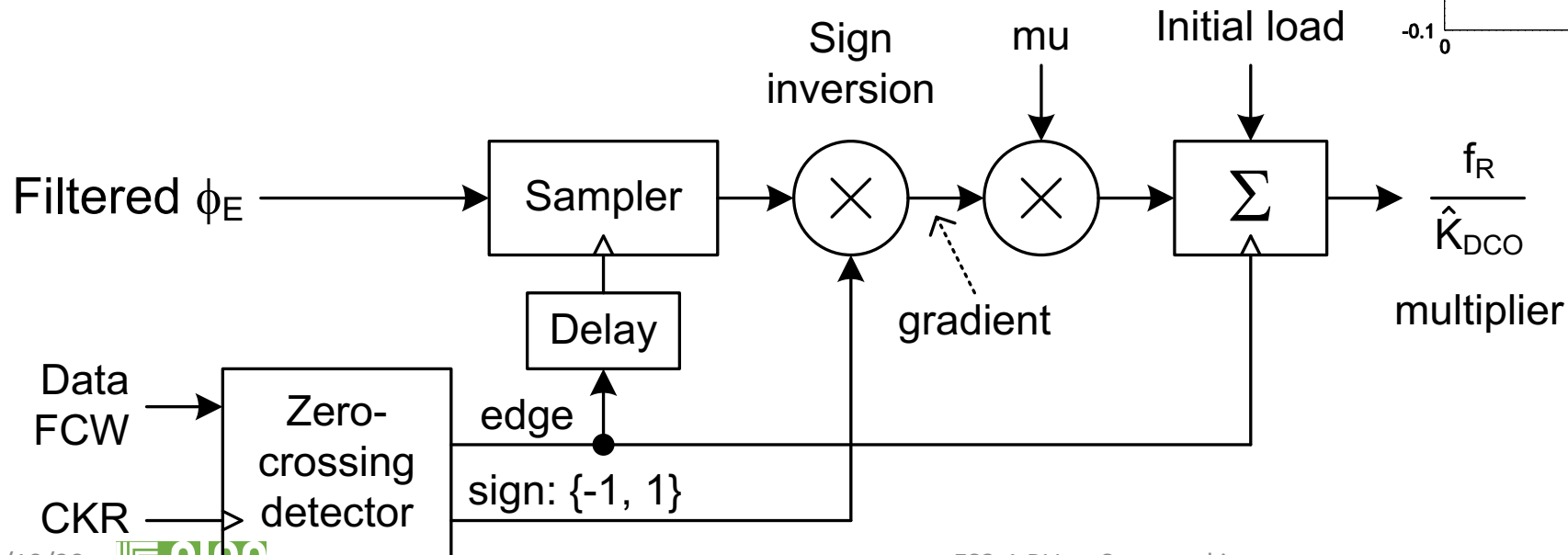
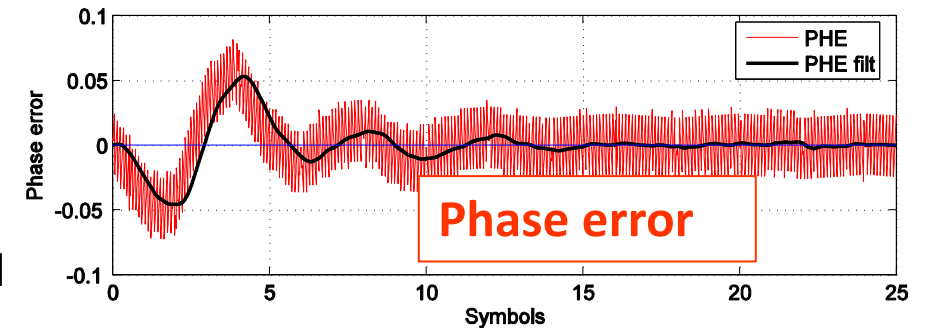
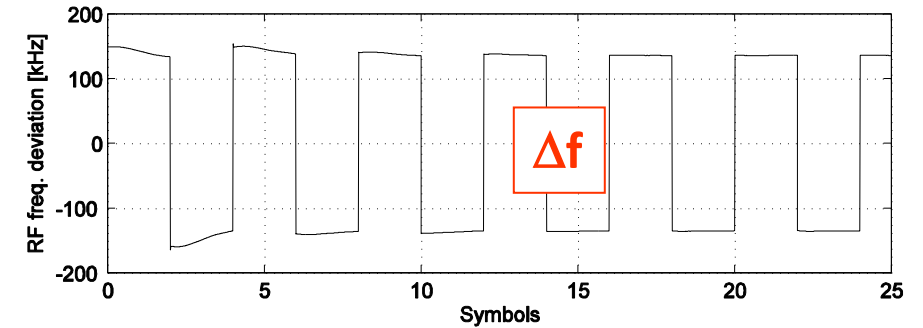
DCO Gain Estimation: Mechanism

- Fully autonomous
- Forces Δf through the PLL
- Measures steady-state ΔOTW
- Could be repeated a few times for better estimation



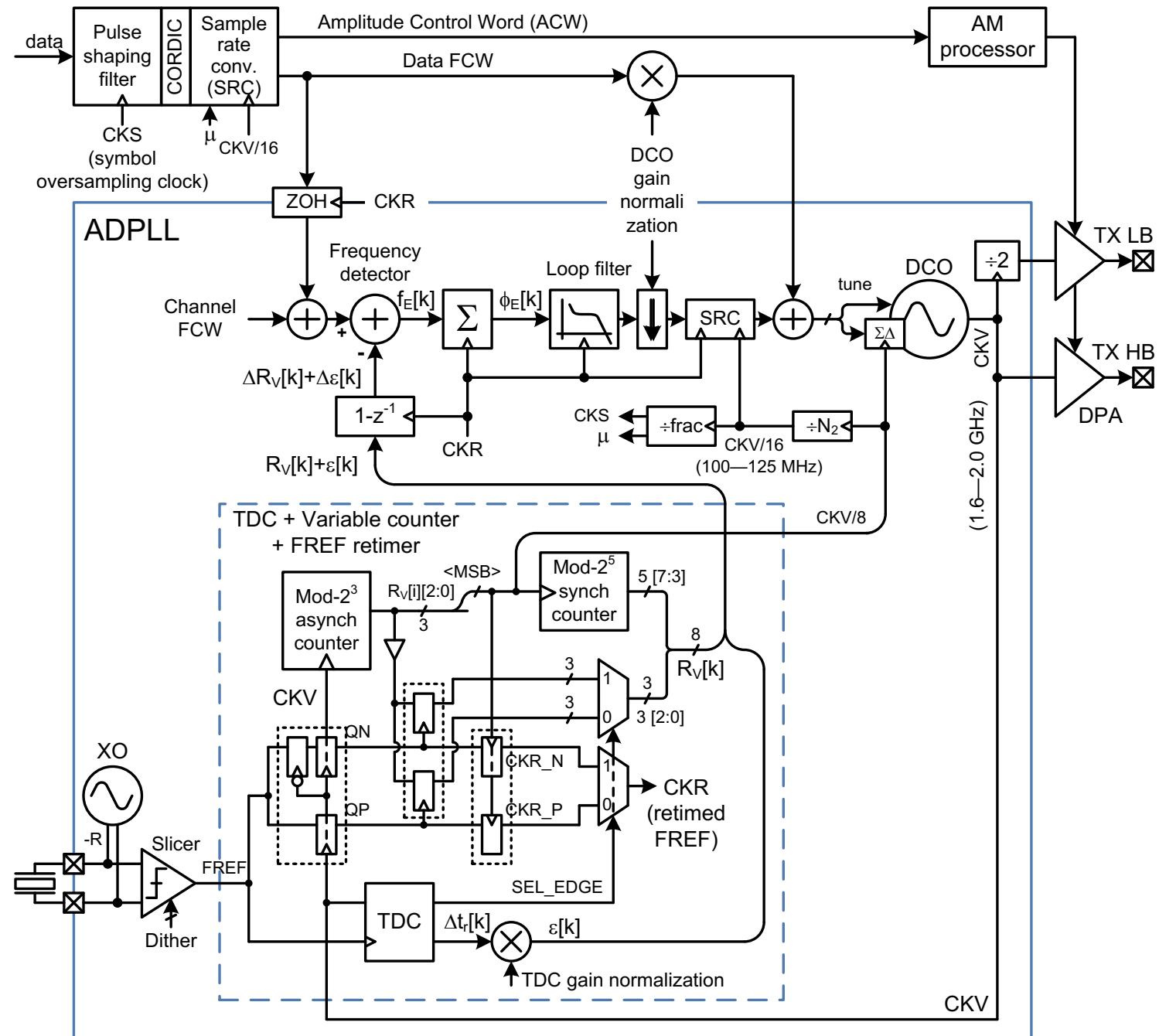
LMS-Based K_{DCO} Adaptation

- Insensitive to the RF oscillator frequency drift
- LMS-like adaptation algorithm
- Examines the filtered phase error when the modulating data is transitioning
- Updates the gain estimate
- Sim: initial K_{DCO} error of +10%



Multi-rate ADPLL

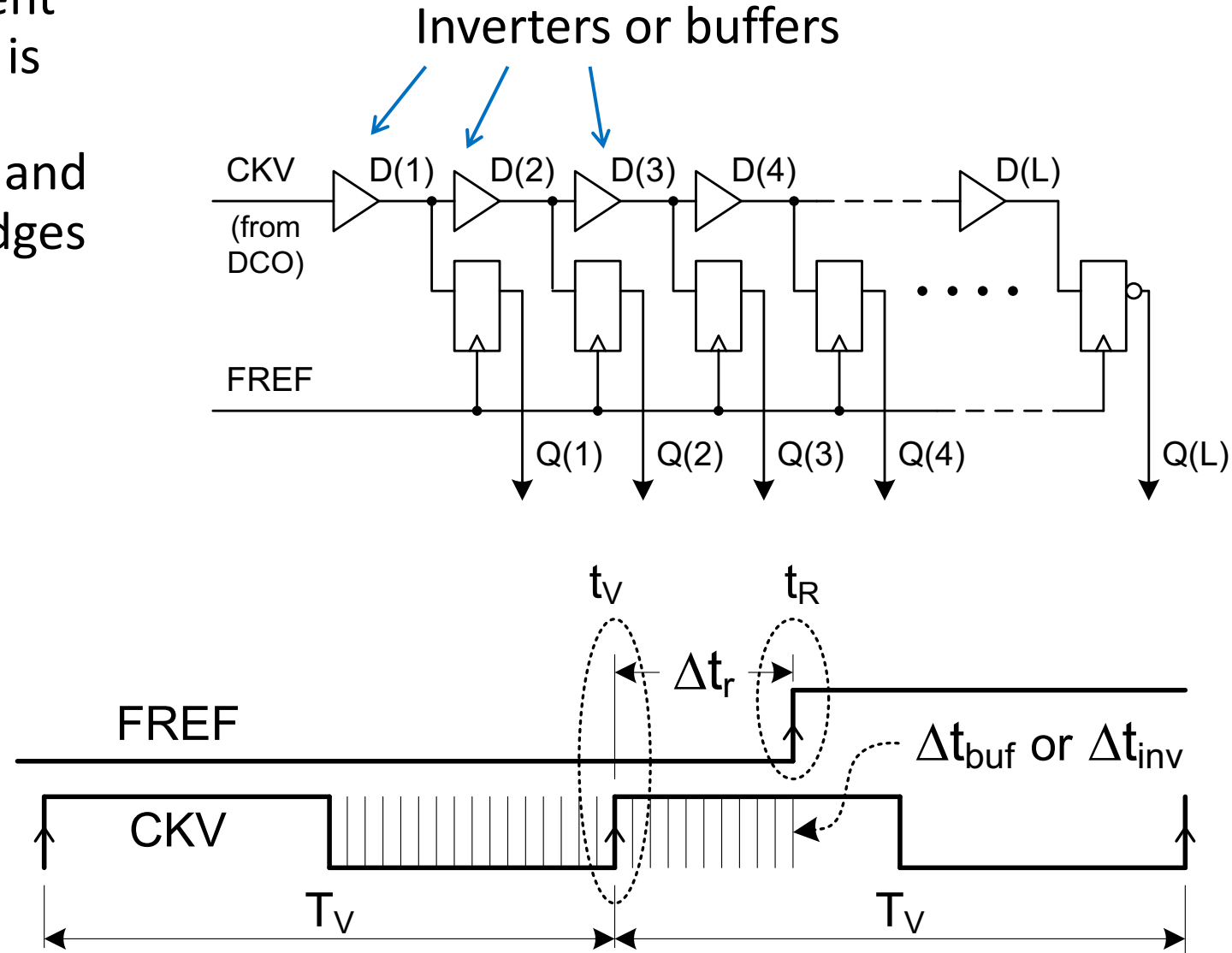
- DCO merges two streams
 - ADPLL error corrections
 - FM data
- FM data is wideband
- ADPLL error corrections ϕ_E are FREF-based
- ADPLL ϕ_E interpolated to be synchronous with FM samples



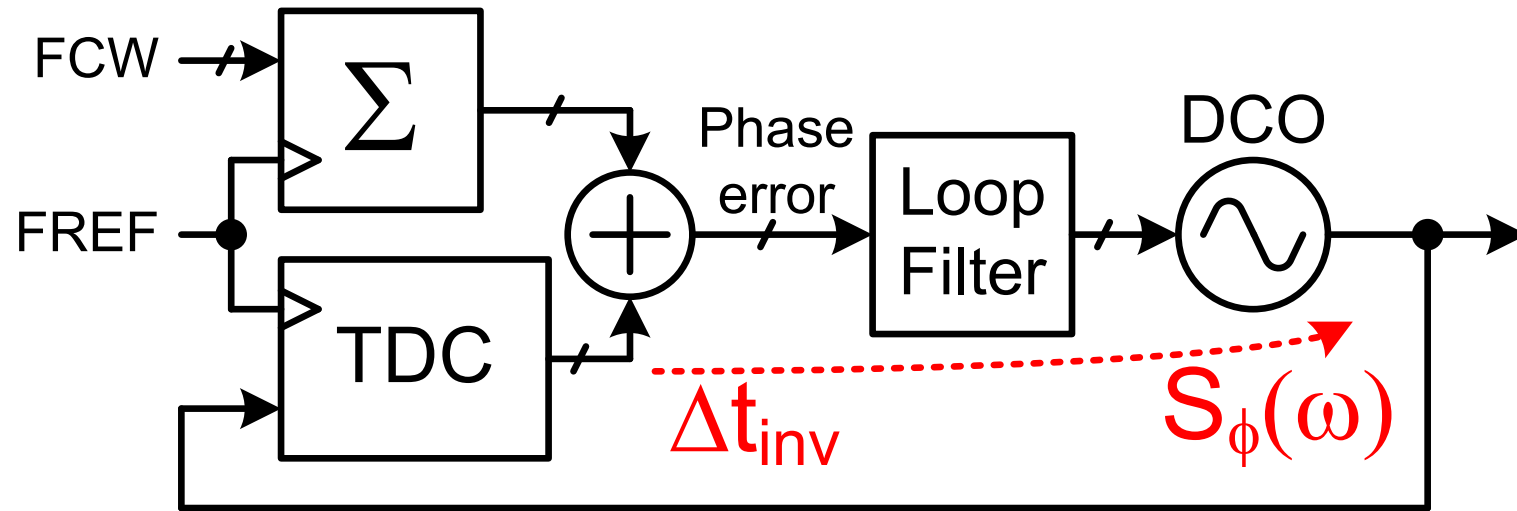
TDC Dithering

TDC Operation

- TDC is an equivalent of ADC where “A” is time difference between variable and reference clock edges



Phase Noise Due to TDC



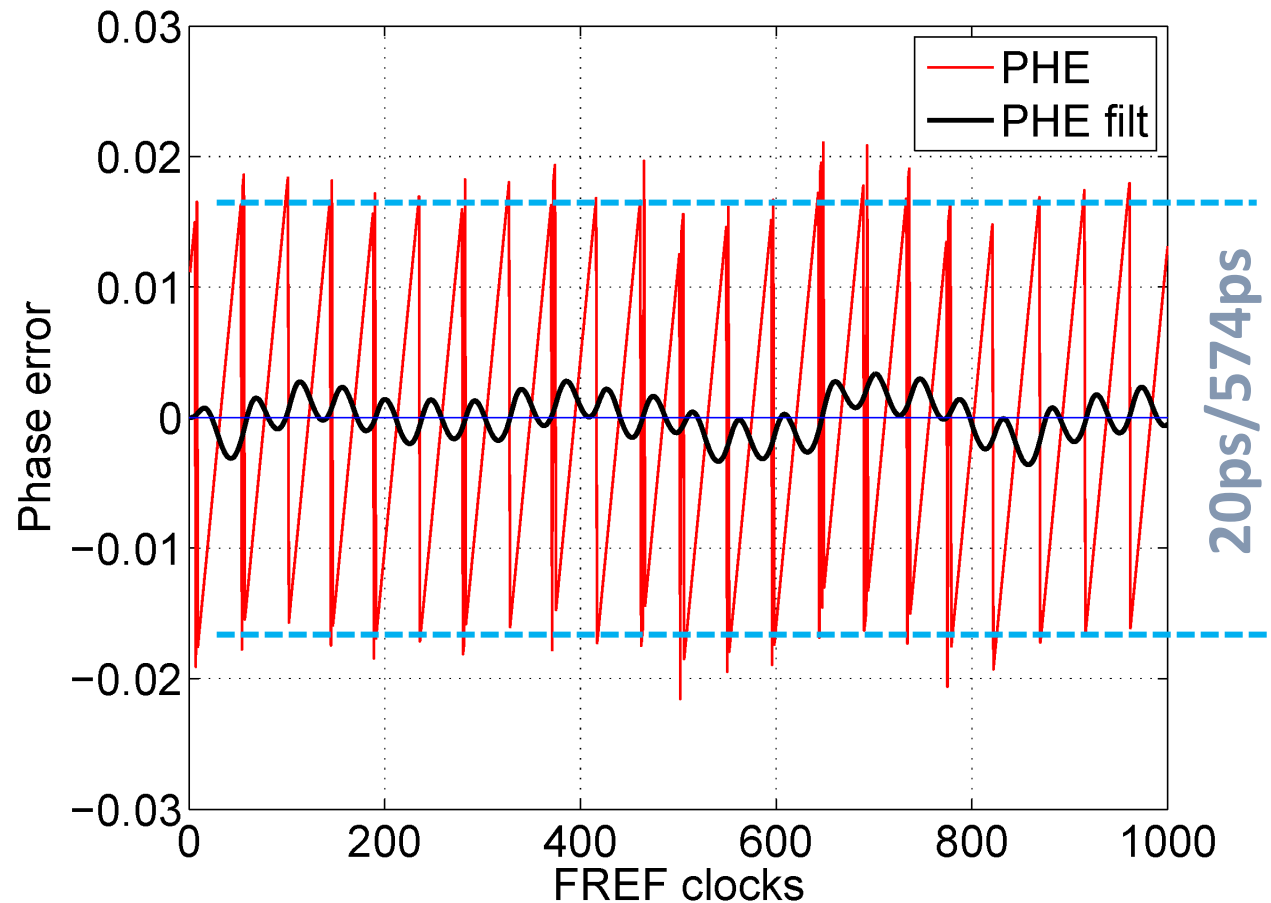
- In-band phase noise at RF output

$$L = \frac{(2\pi)^2}{12} \left(\frac{\Delta t_{inv}}{T_V} \right)^2 \cdot \frac{1}{f_R}$$

- E.g., $\Delta t_{inv}=20\text{ps}$, $f_v=1.8\text{GHz}$, $f_R=26\text{MHz}$, **$L = -97.8\text{dBc/Hz}$**
- Good enough for many wireless systems (at nominal 60-150kHz bandwidth)
- can get only better

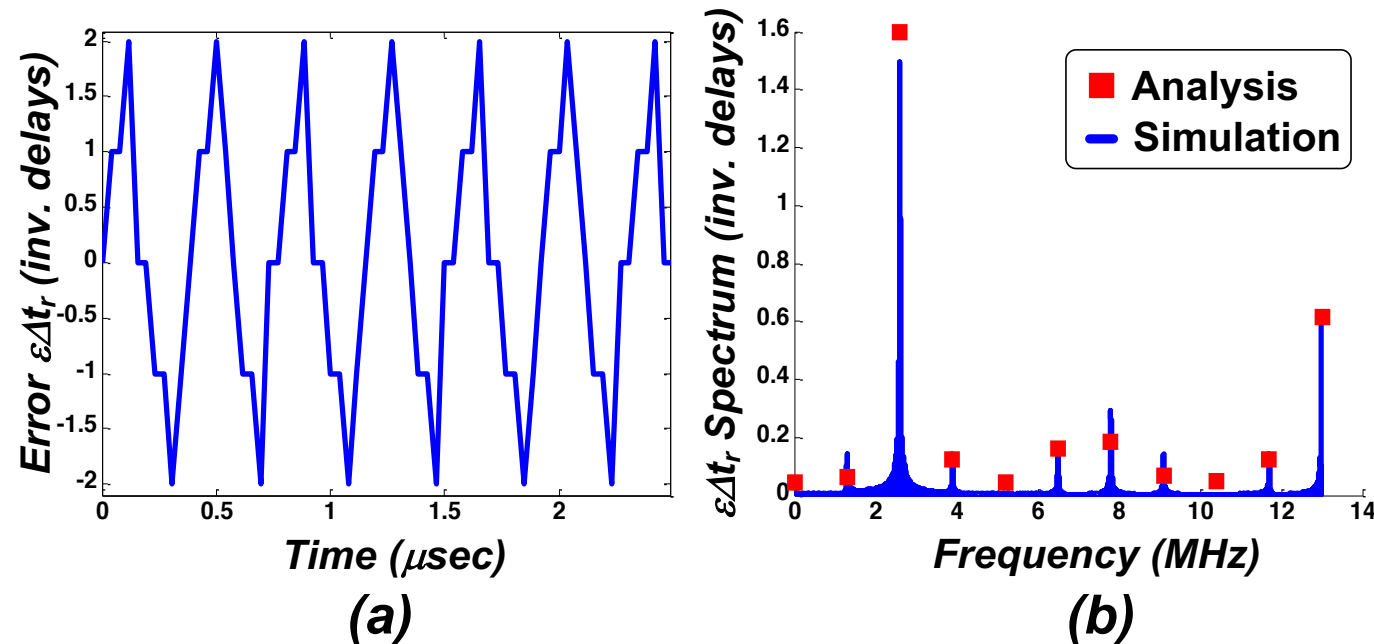
III-Behaved TDC Q-noise

- Red: Phase error
- Black: Filtered phase error
- $t_{\text{inv}} = 20\text{ps}$
- $f_V = 1742.02\text{MHz}$
- $T_V = 574\text{ps}$
- $f_R = 26\text{MHz}$
- $\text{FCW} = 67.000769$
- 20kHz offset from integer-N channel
- Tone at 574kHz
- $20\text{kHz} * 574\text{ps}/20\text{ps} = 574\text{kHz}$
- Main tone is well predicted!



Literature of TDC Q-noise in ADPLL

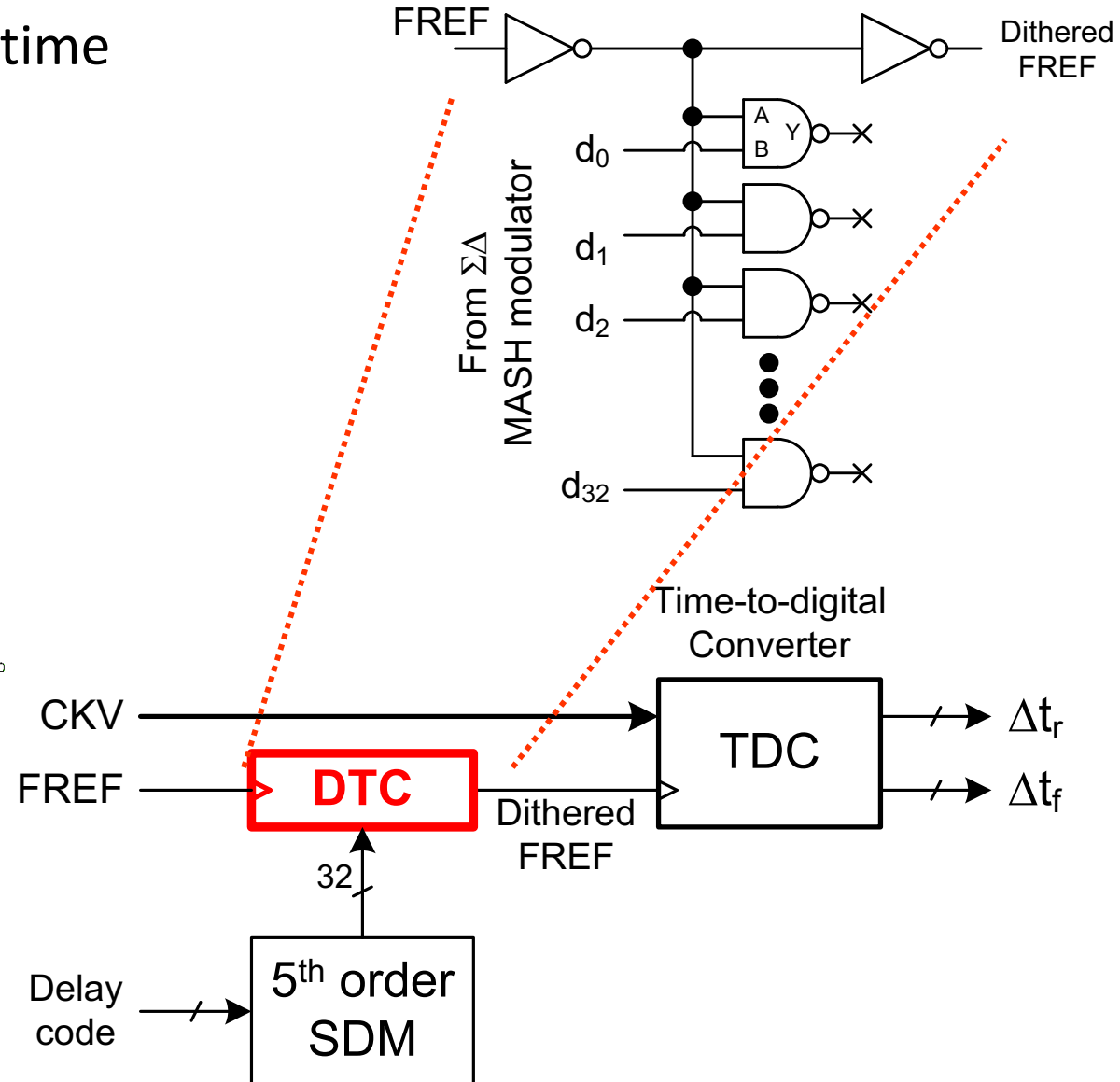
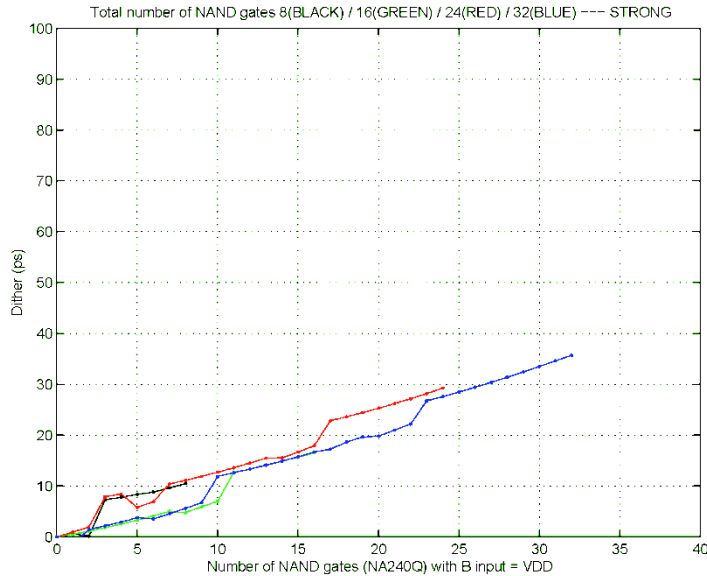
- *Exact* Q-noise analysis of TDC in ADPLL [VAMVA06]



- [VAMVA06] S. D. Vamvakos, R. B. Staszewski, M. Sheba, and K. Waheed, “Noise analysis of time-to-digital converter in all-digital PLLs,” *Proc. of Fifth IEEE Dallas Circuits and Systems Workshop: Design, Application, Integration and Software (DCAS-06)*, pp. 87–90, Oct. 2006, Dallas, TX, USA.

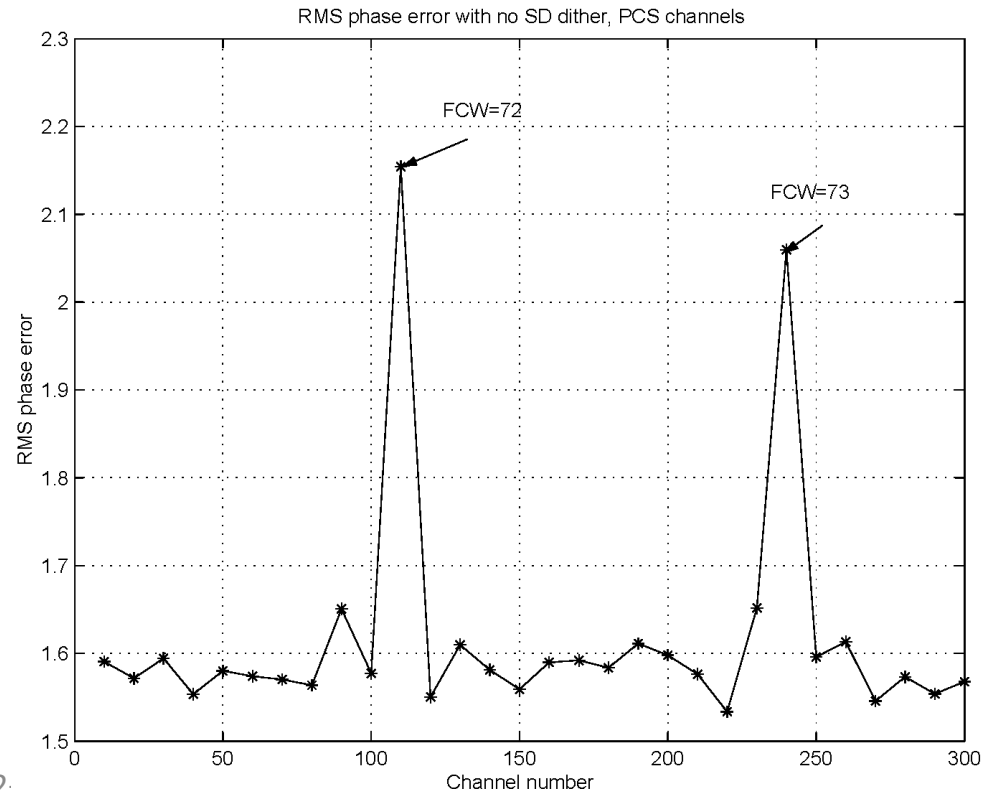
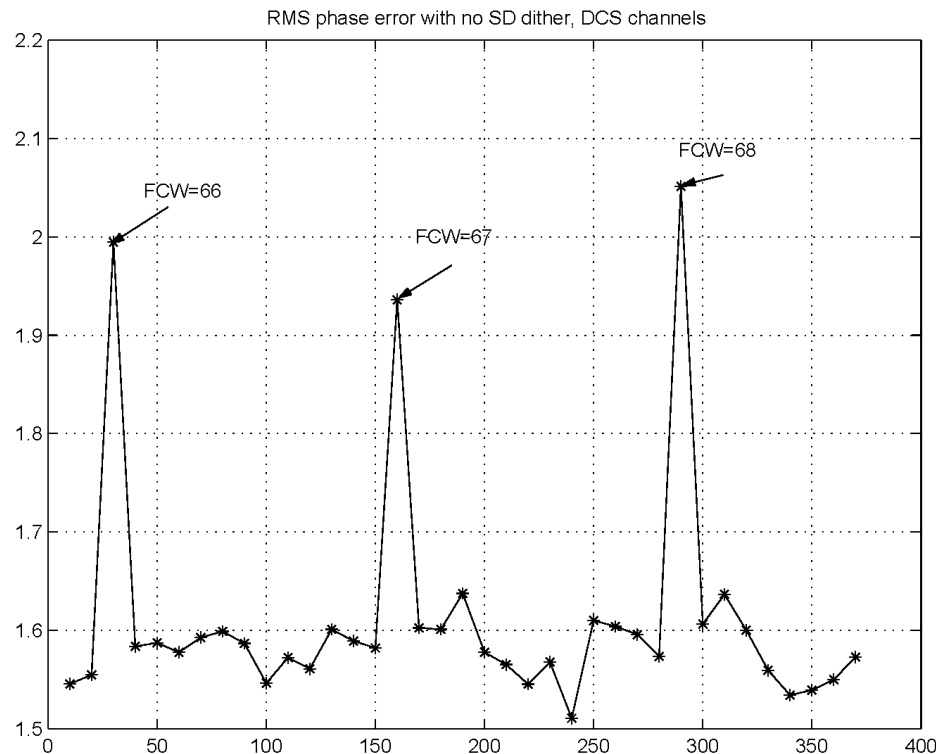
FREF Dithering

- Fine precision digital-to-time converter



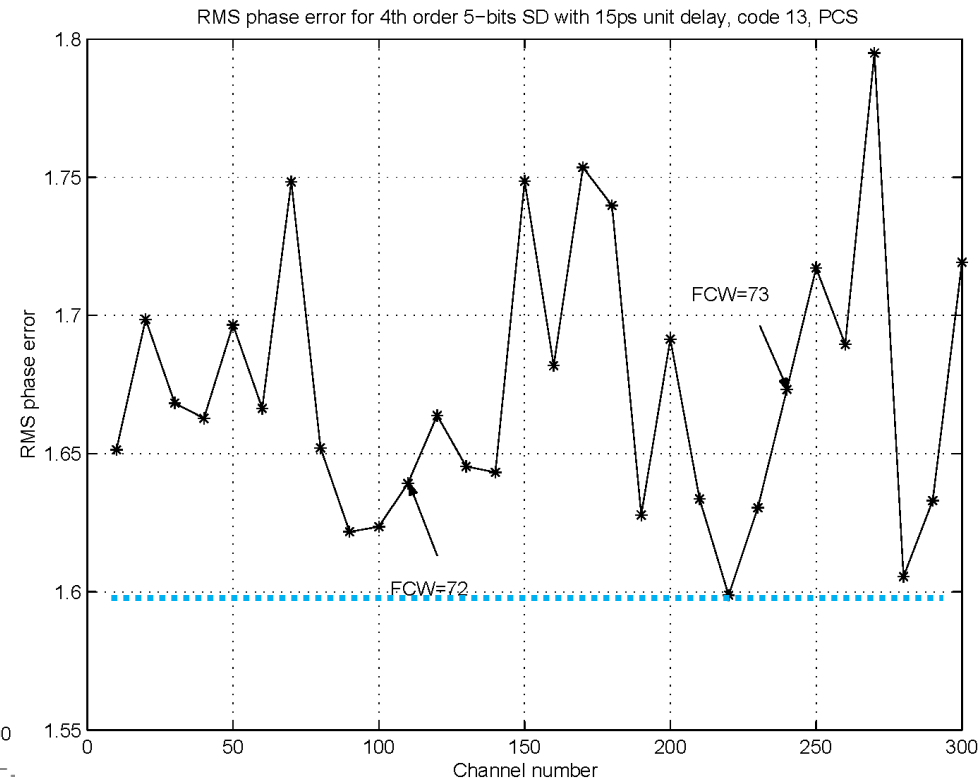
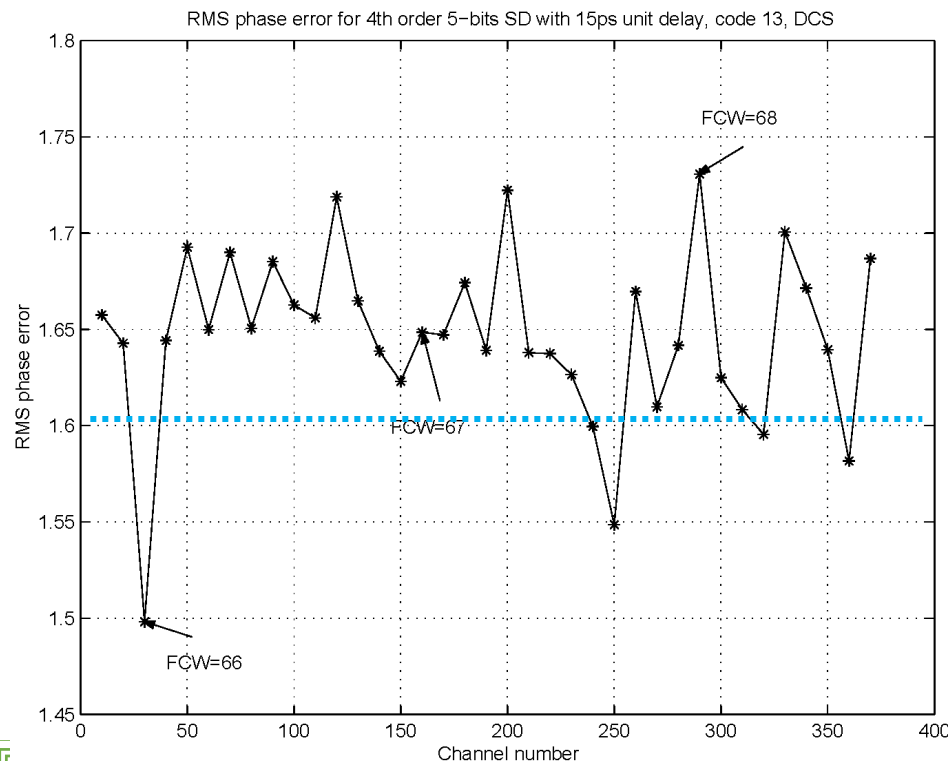
System Simulations: Before

- Shown: Top-level VHDL simulation of the entire transmitter
- Modulation distortion (rms EVM) vs. channels in DCS and PCS bands (~2GHz)
- Good general performance but degraded at integer-N channels



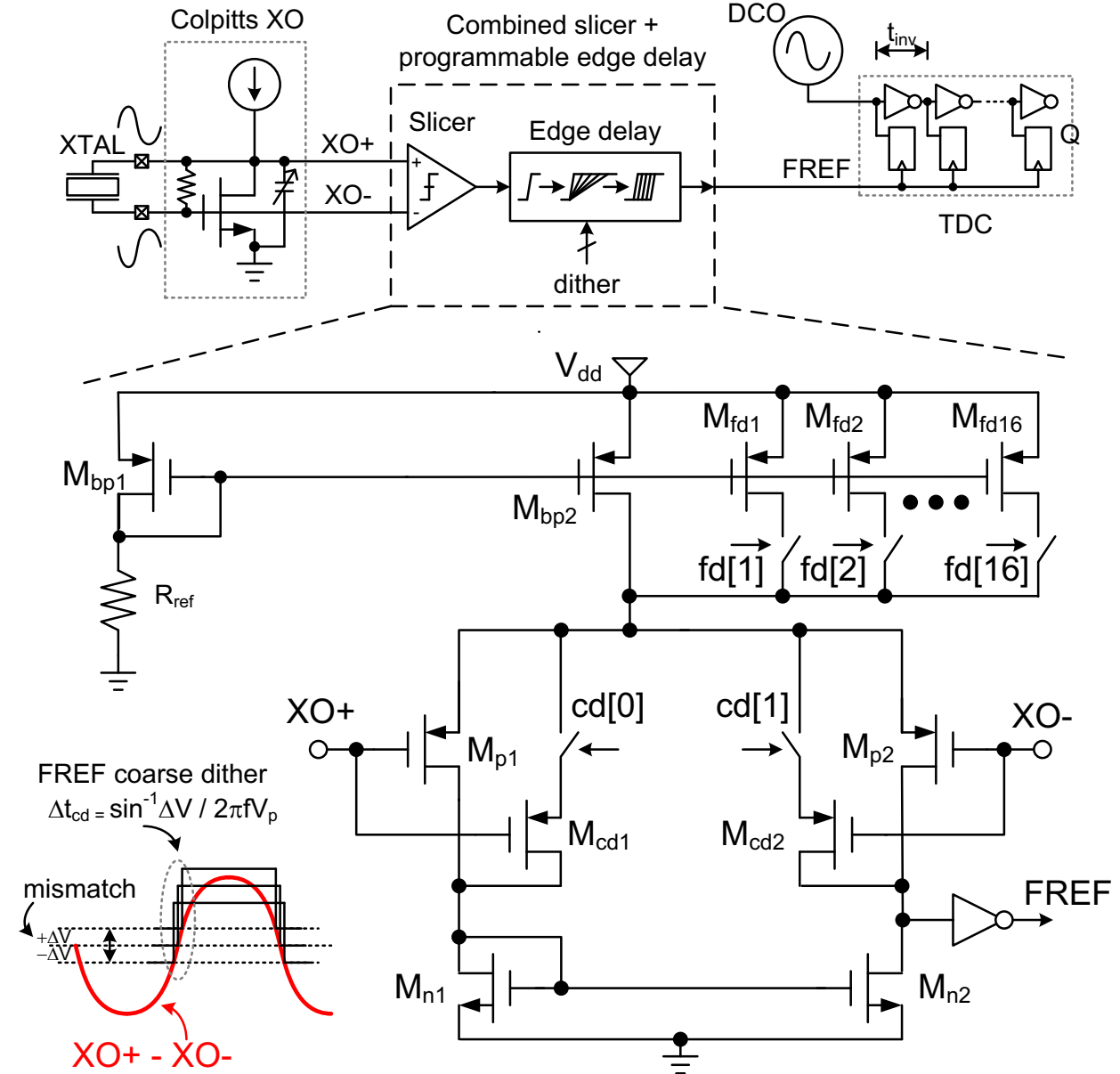
System Simulations: After Fix

- Shown: Top-level VHDL simulation of the entire transmitter after applying the dither to the FREF clock
- Conclusion: The sigma-delta dither is an effective means of breaking the ill-behaved condition of TDC quantization noise

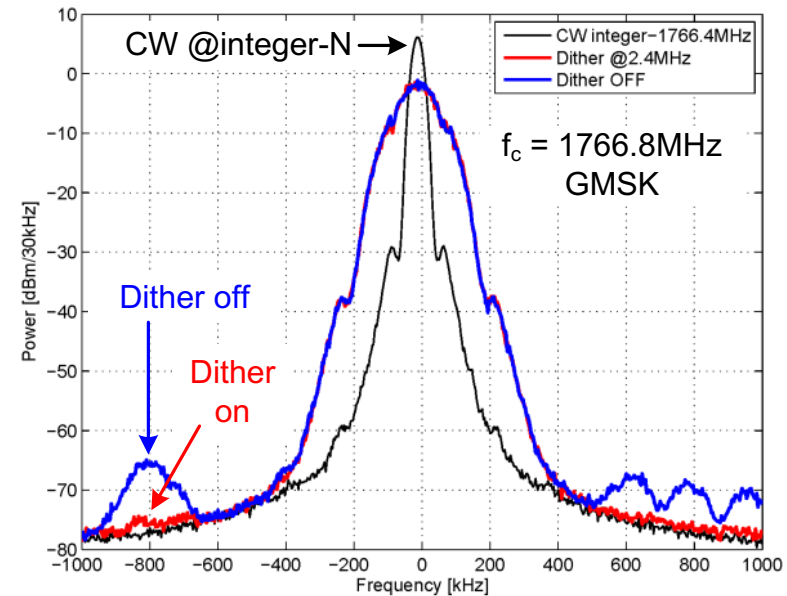
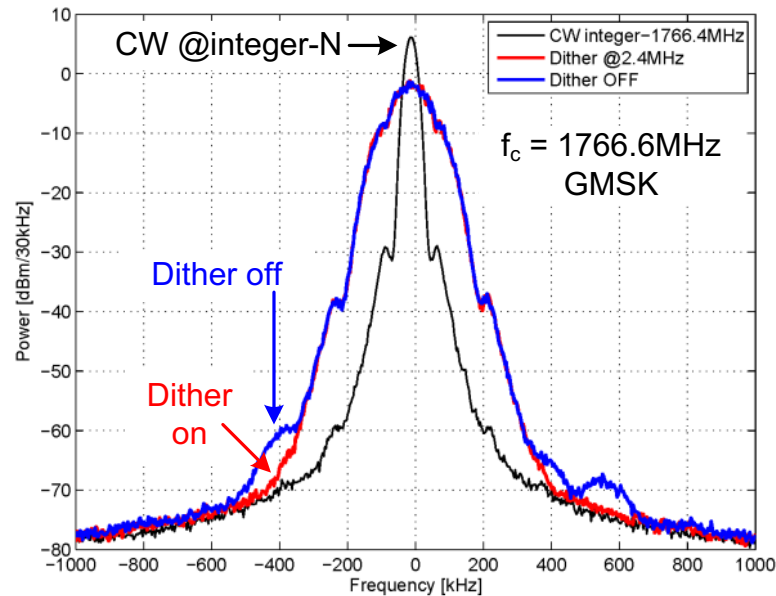
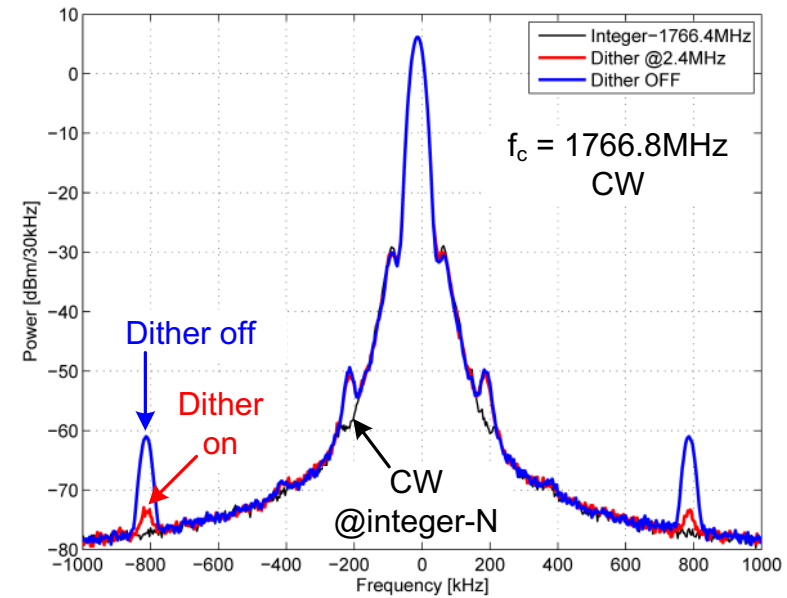
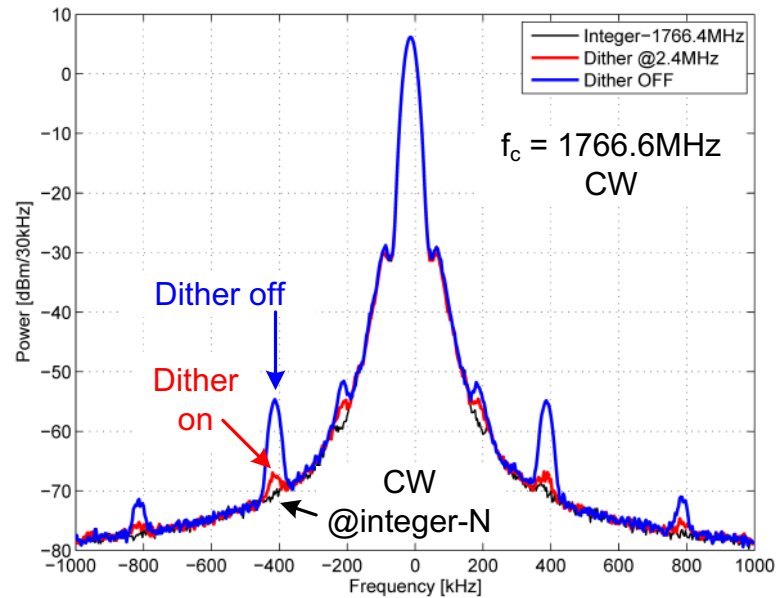


Crystal Oscillator Slicer

- Additional delay w/o noise penalty
- FREF risetime does not get degraded
- Delay controls:
 - coarse (3-level)
 - fine (16-level) delay control

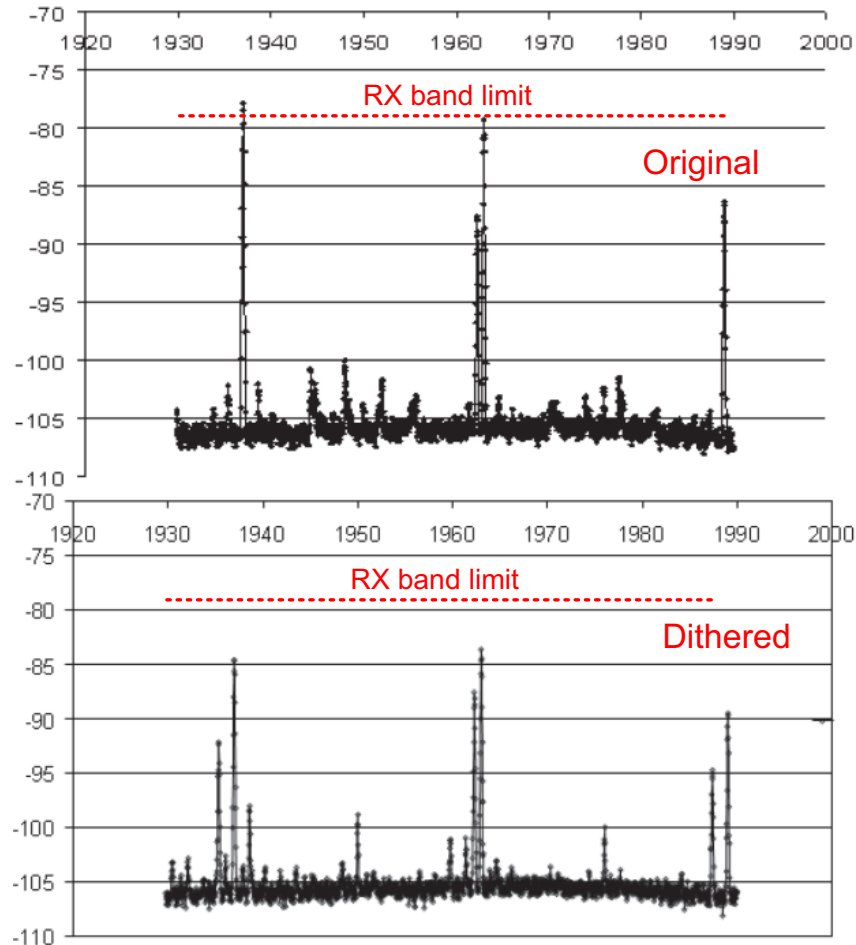
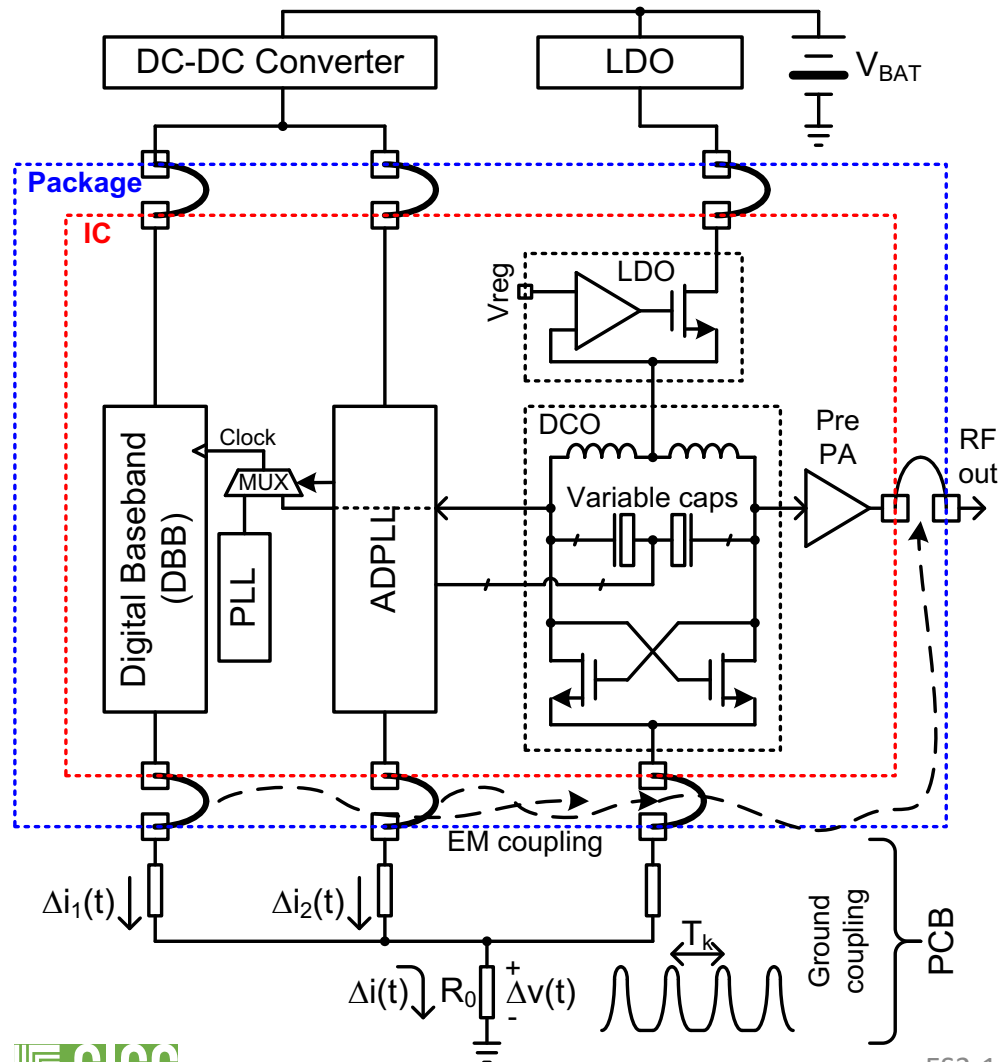


Measured Effect of Coarse Dither

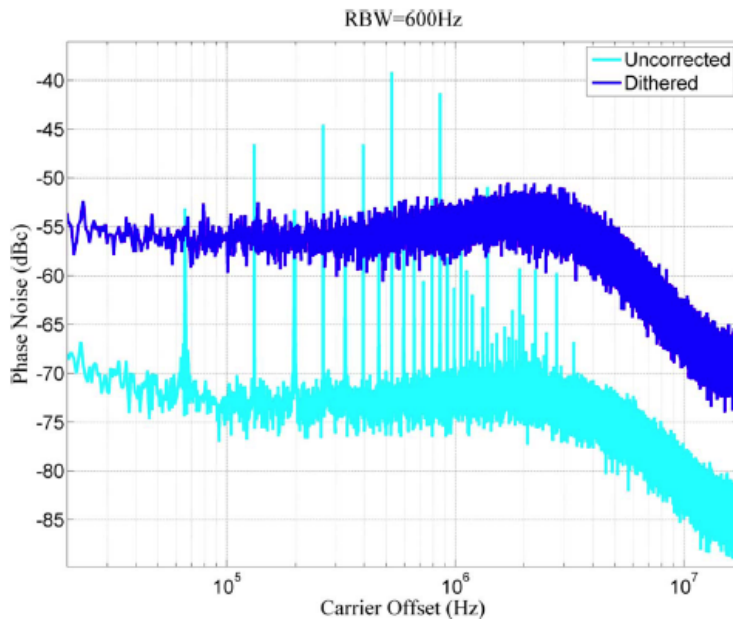
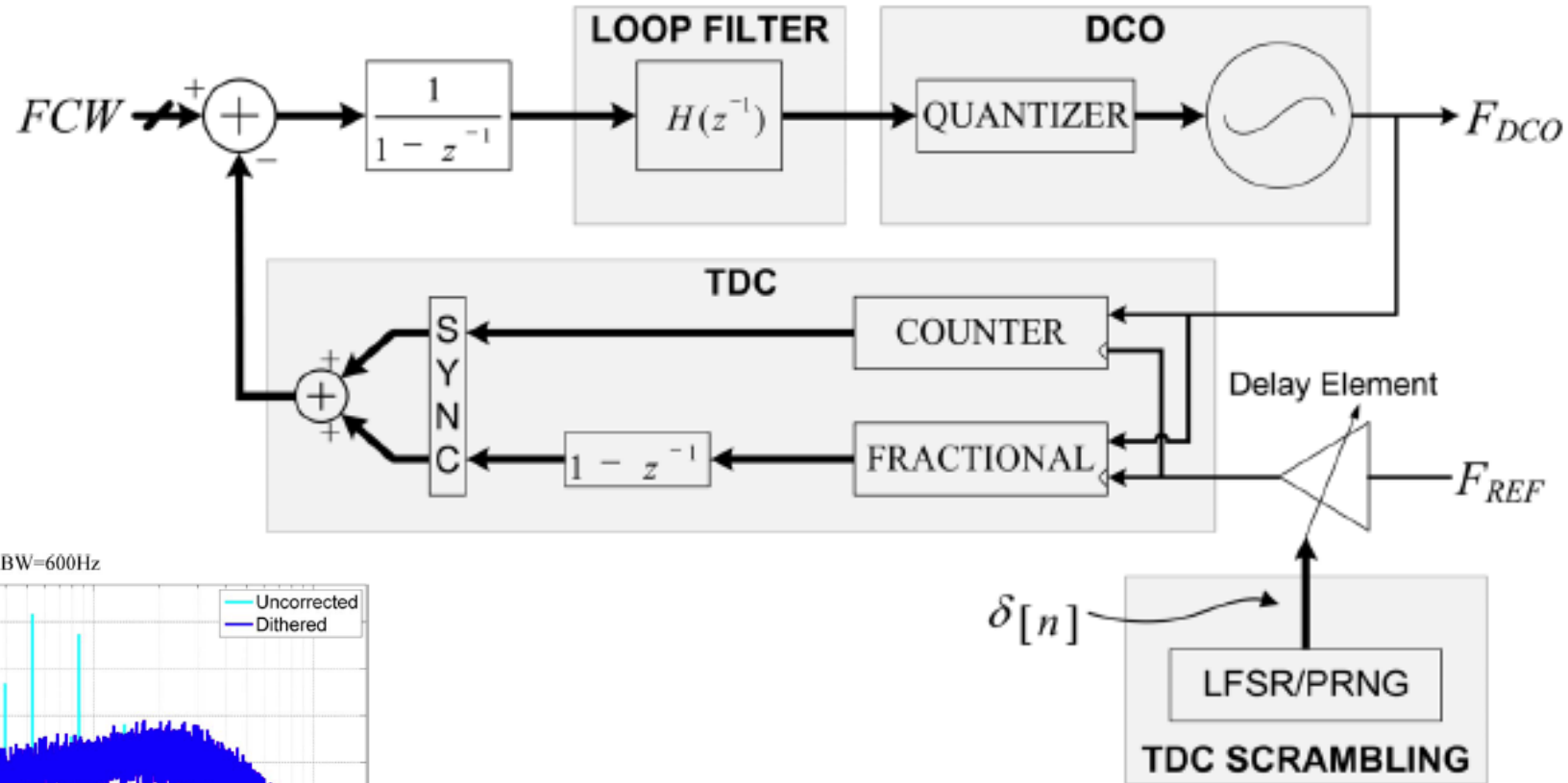


DBB Clock Synchronous to DCO

- Digital clock activity is a strong aggressor
- DCO LC tank is the most sensitive victim



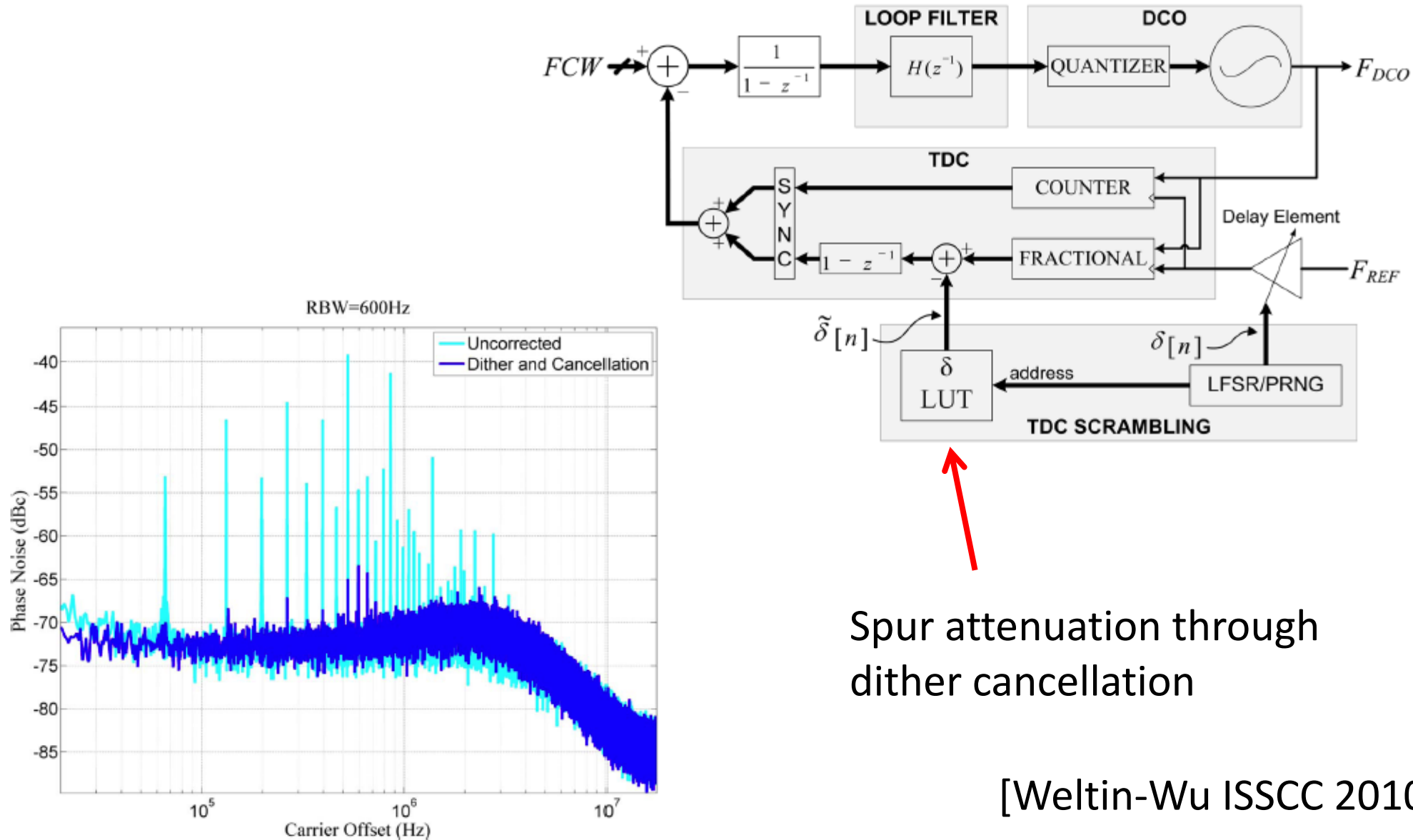
ADPLL with TDC Dither



Spur attenuation through dither
... but noise floor rises

[Weltin-Wu ISSCC 2010]

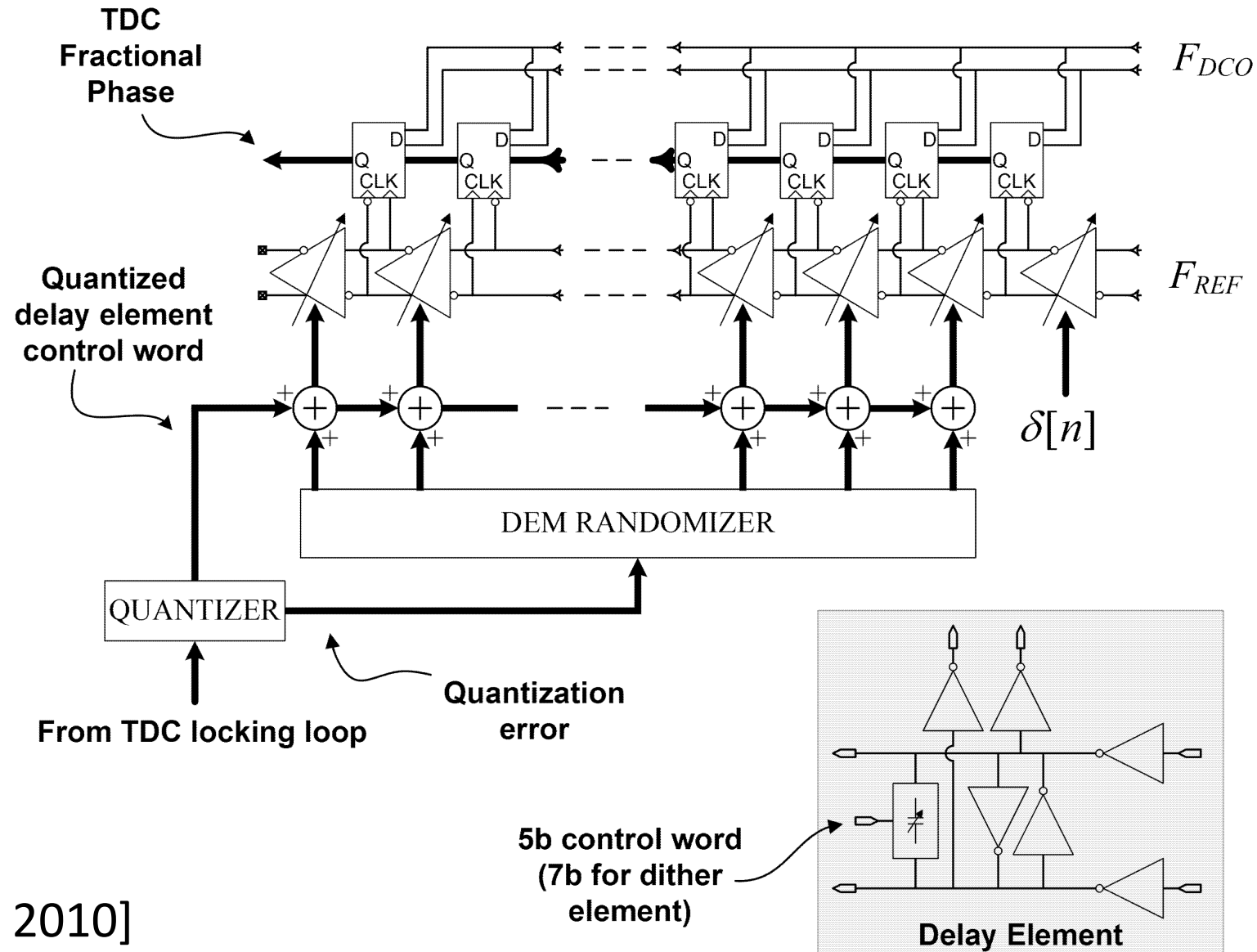
ADPLL with TDC Dither Cancellation



Spur attenuation through dither cancellation

[Weltin-Wu ISSCC 2010]

Fractional TDC with DEM



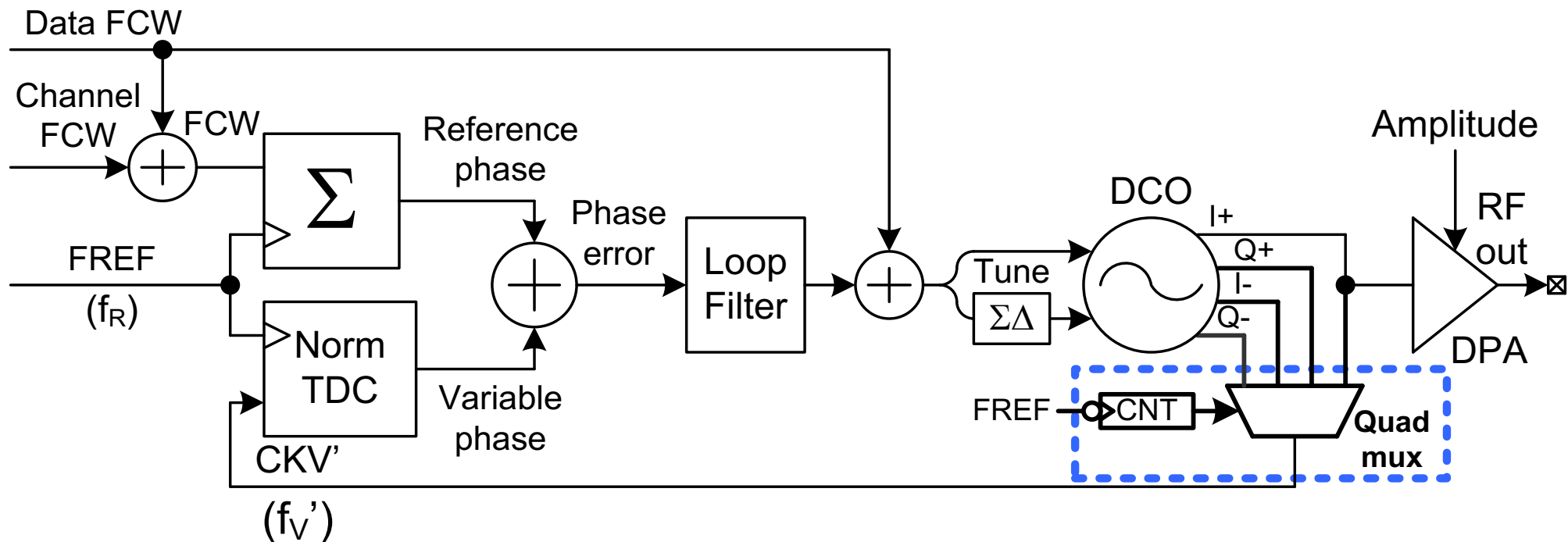
[Weltin-Wu ISSCC 2010]

Is there Another Way to Handle Q-Noise?

- TDC dithering seems to be an effective way to handle near integer-N channel problem
- However, too many cases to consider
 - Stacking up various conditions, such as integer count of inverters per DCO clock cycle
- Some components cannot be completely eliminated
- Can we avoid the problem altogether?
 - Change we the frequency relationship away from integer-N?
 - Yes!

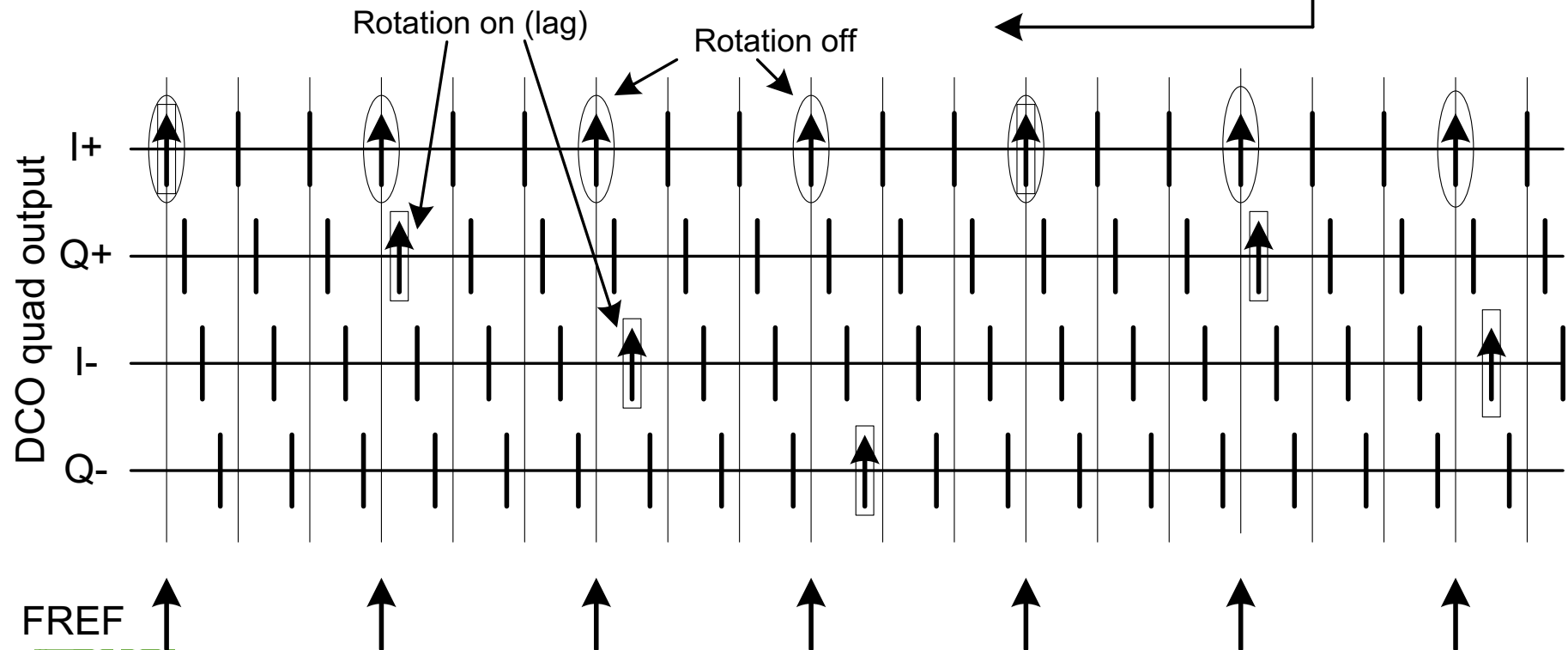
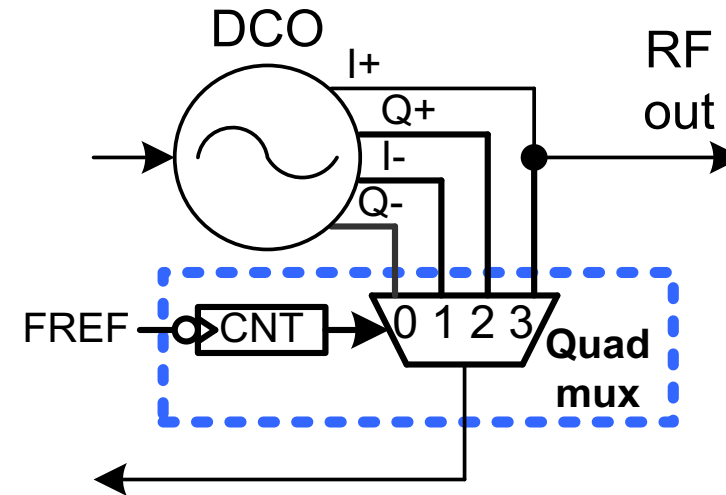
Offset ADPLL Architecture

- Frequency translation in the feedback path
 - Rotation of DCO quadrature phases
- Straightforward digital implementation
 - 4:1 mux and 2-bit counter
 - Performed at falling FREF edges



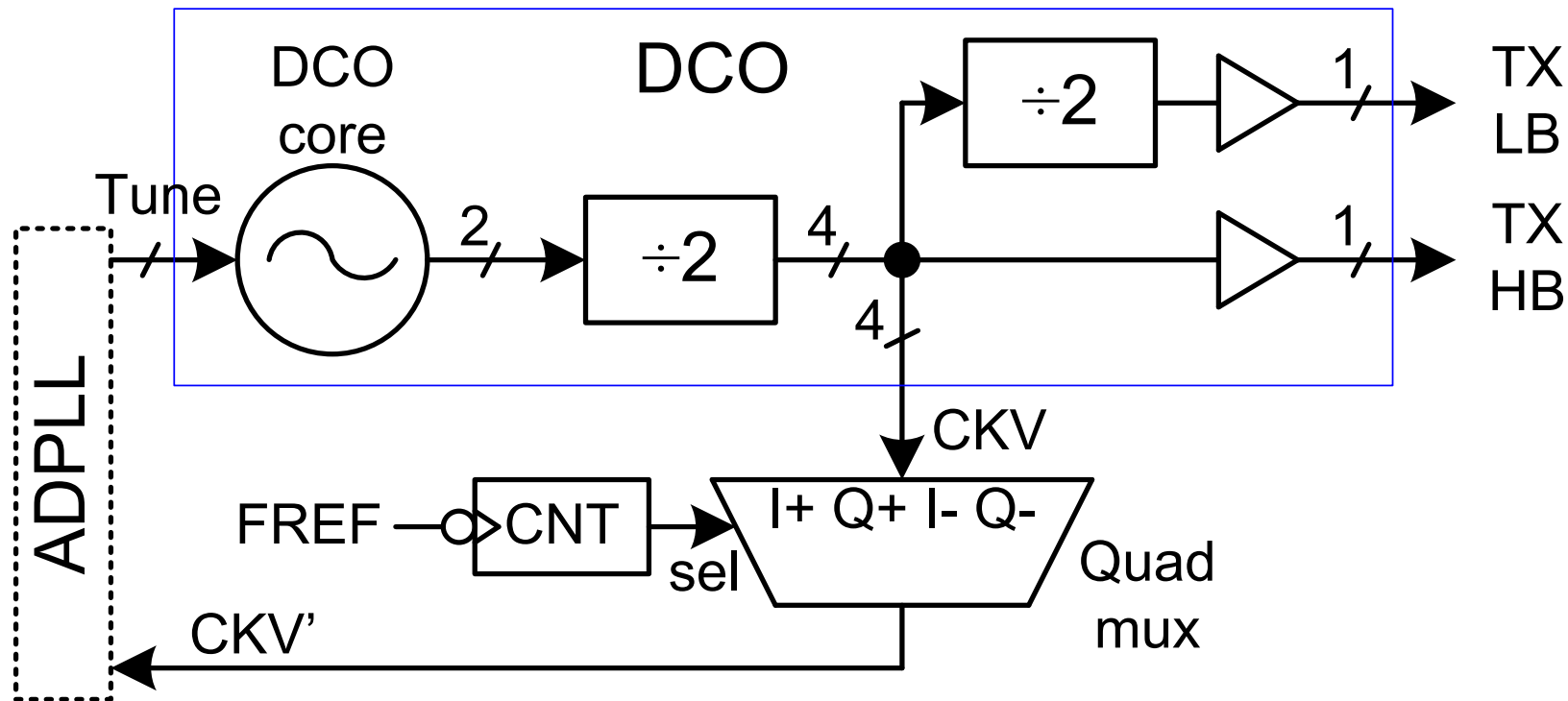
DCO Quadrature Phase Rotation

- Simple example with $FCW=3$
- Changes the frequency by $f_R/4$
 - $FCW' = 3\frac{1}{4}$



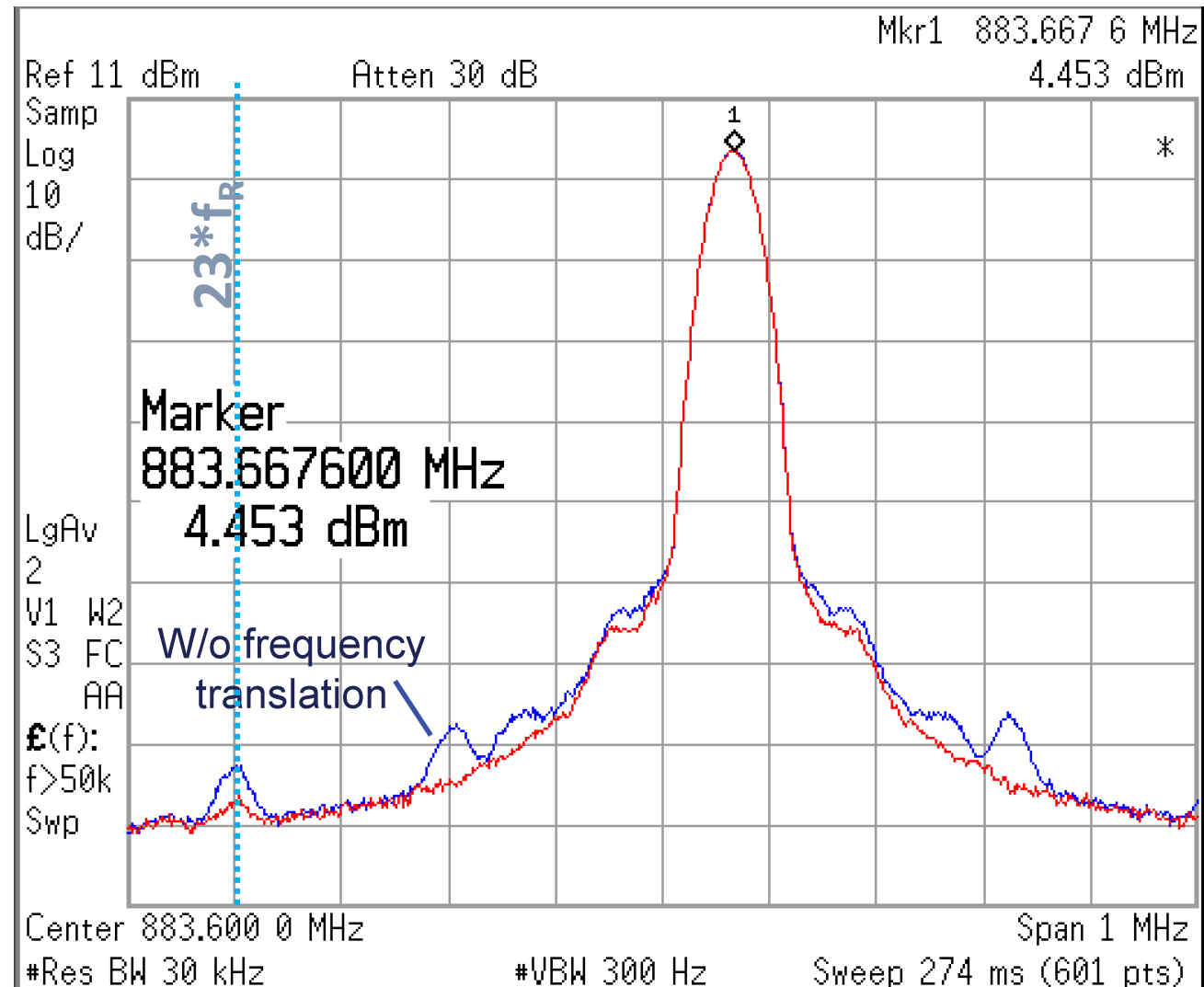
DCO Details

- DCO core operates at 2x high-band and 4x low-band frequency
- First divider produces quadrature clocks
- Quad mux produces frequency translation by $\pm f_R/4$



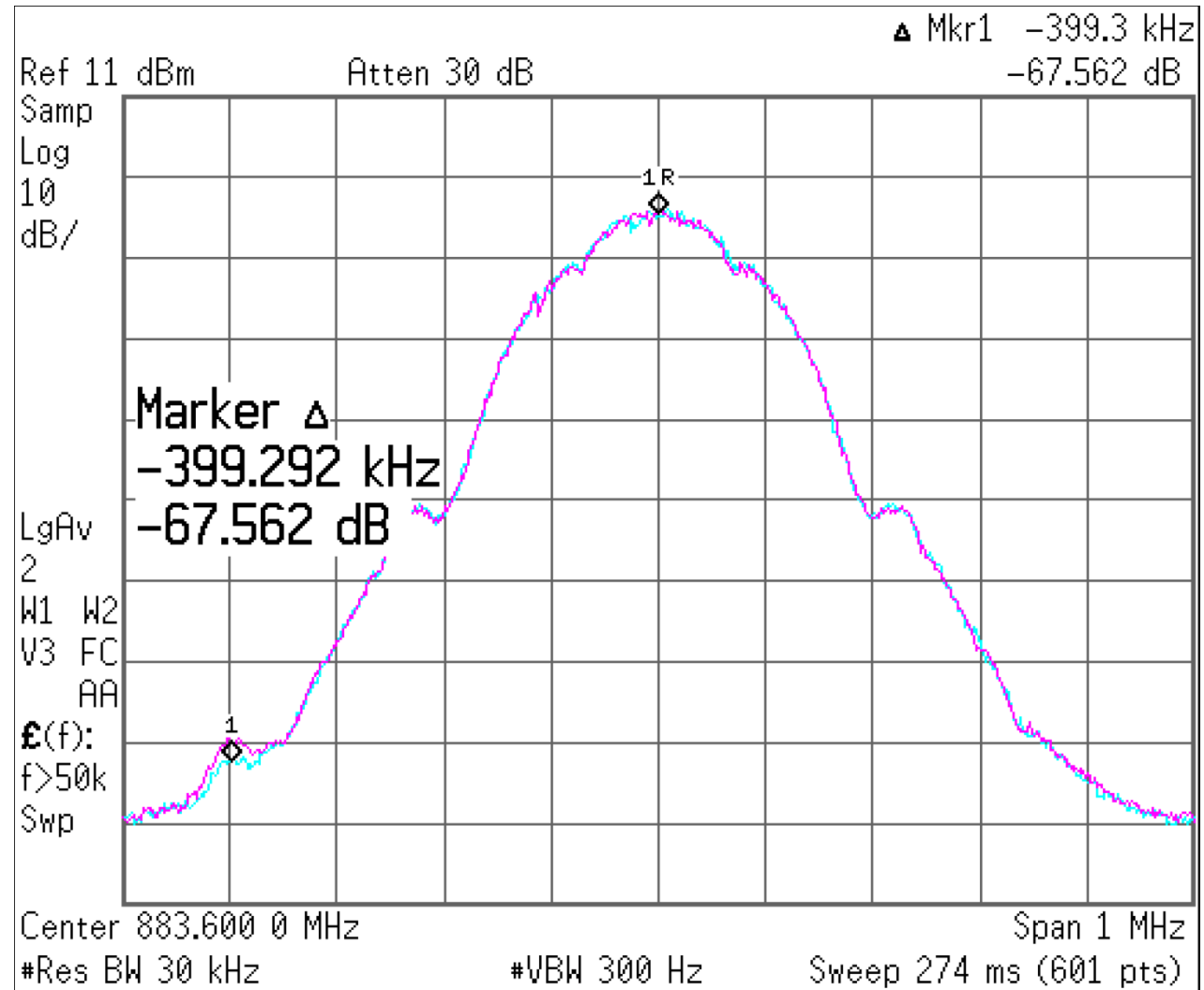
Measured Un-modulated Spectrum

- Blue: reference
- Red: with feedback translation
- $f_R = 38.4$ MHz
- 883.67MHz carrier
- 467.7 kHz away from integer-N
- Remaining small spur is direct FREF harmonic coupling



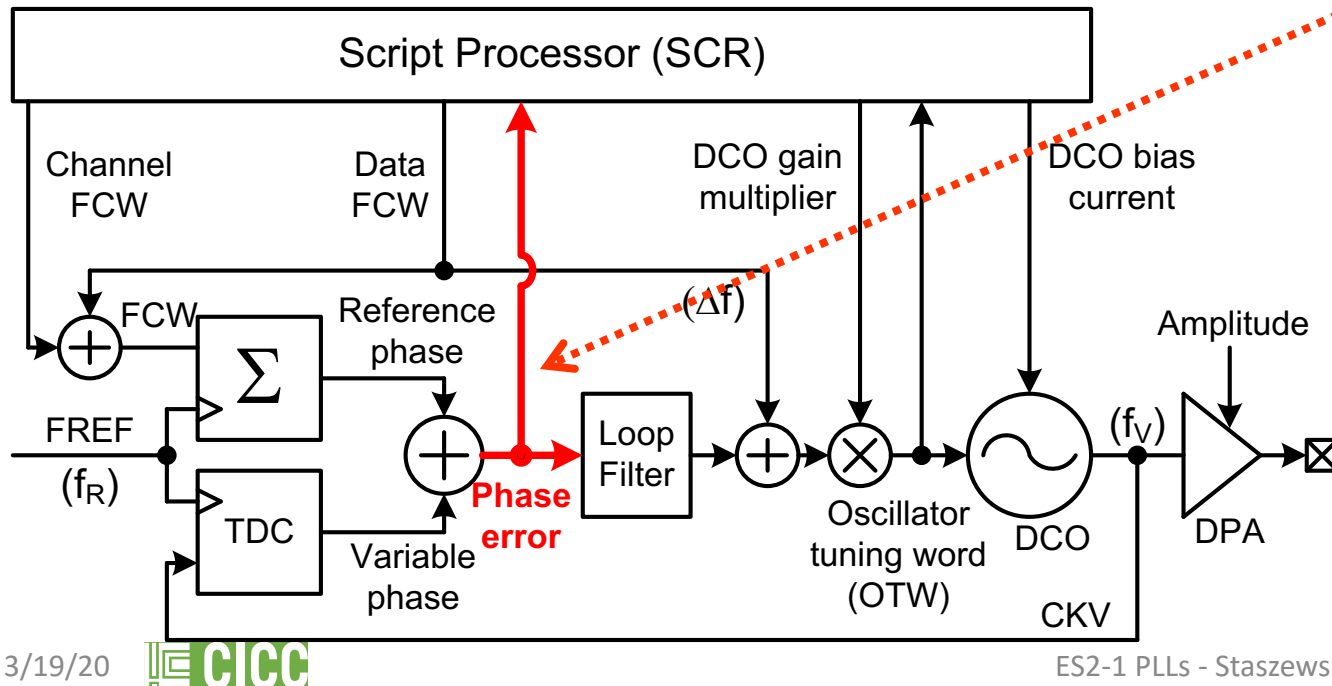
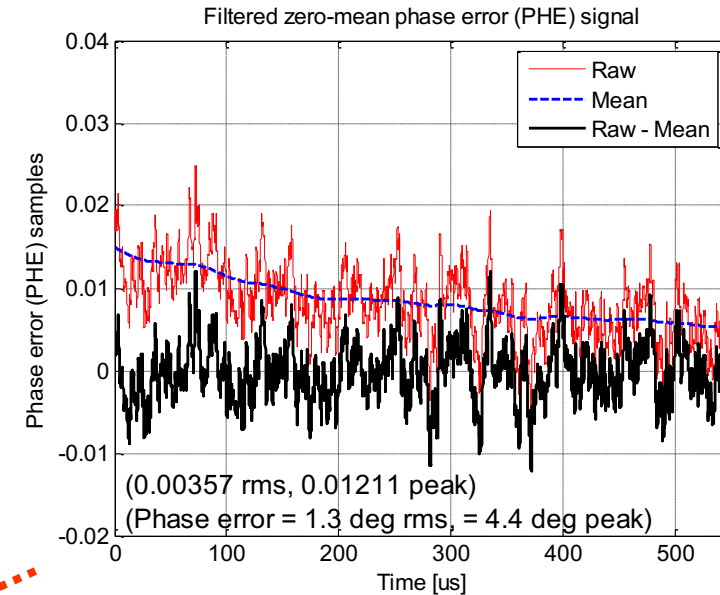
Measured Modulated Spectrum

- Violet: reference
- Blue: with frequency translation
- 2-3 dB improvement at 400 kHz offset to -67.5 dBc
- Modulation distortion: $2.2^\circ \rightarrow 1.8^\circ$



Digital Assistance of RF: Digital Phase Error

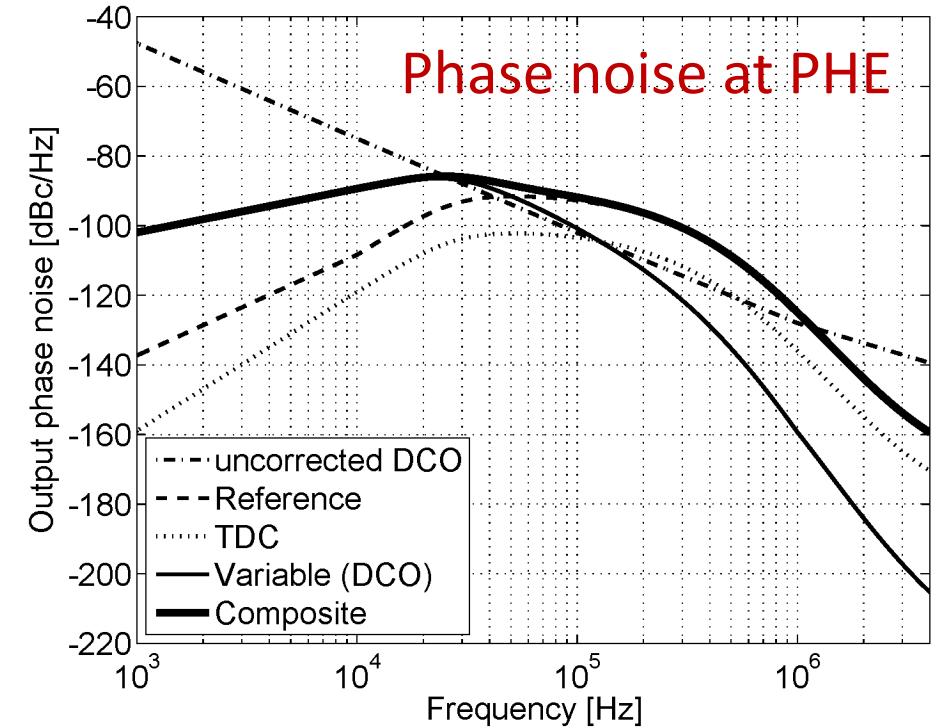
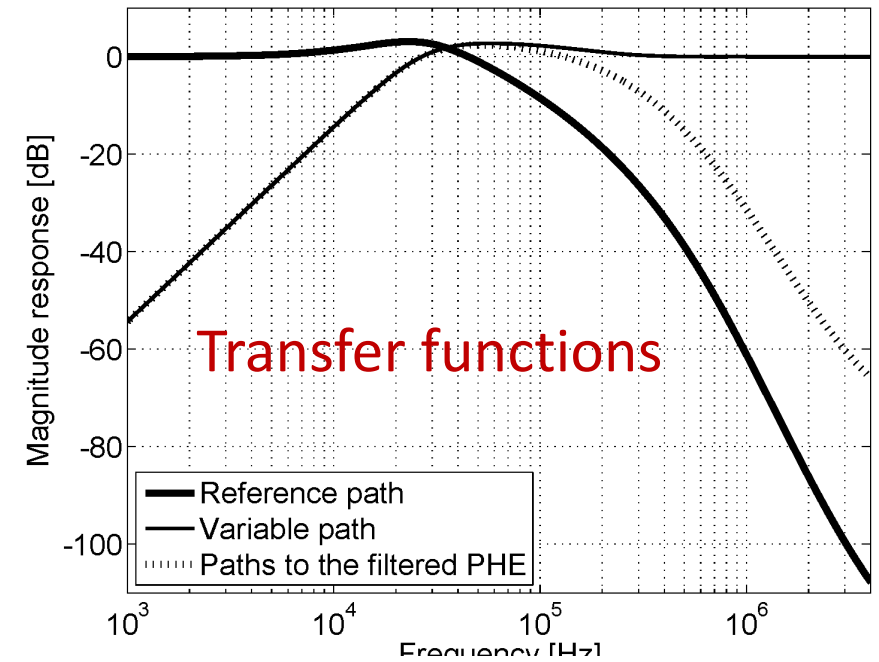
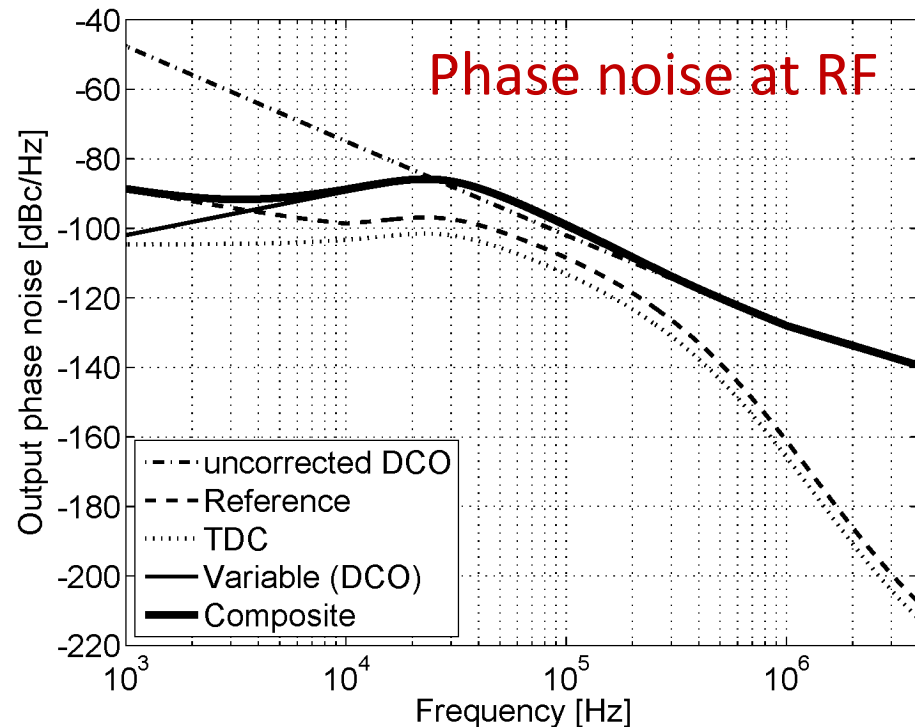
- Digital processing of phase error is useful in determining DCO noise performance
 - natural noise and interference
- Phase error statistics to ascertain RF output performance



- RMS, peak of phase error (PHE) directly corresponds with RMS, peak of GMSK phase trajectory error

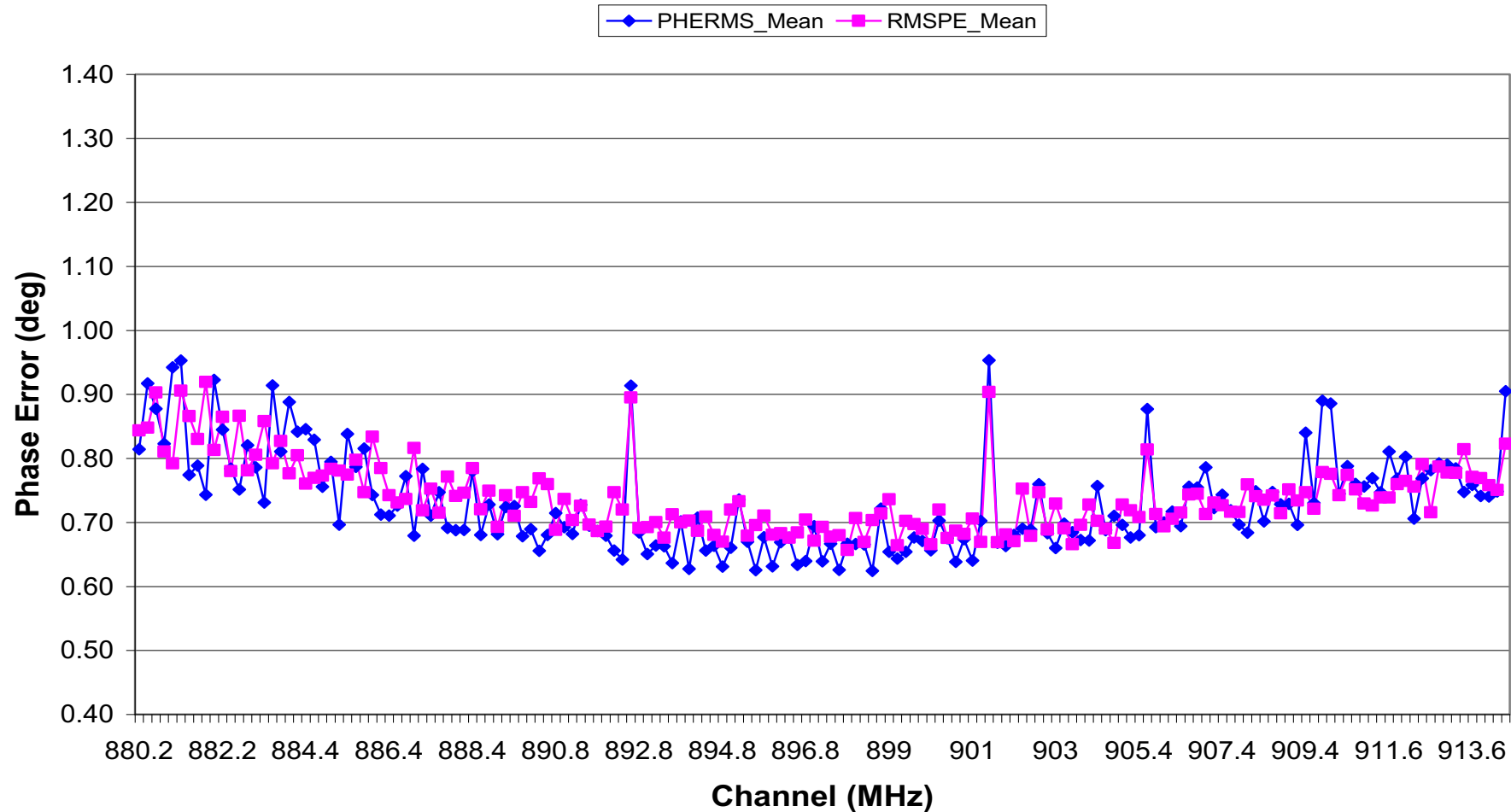
PHE as a Proxy

- PHE signal can estimate the RF output phase noise (at frequencies where it really matters!)
- Its transfer function matches well that of the DCO to beyond the loop bandwidth
- Measurements show very good correlation



Phase Error Calculation vs. Measurement

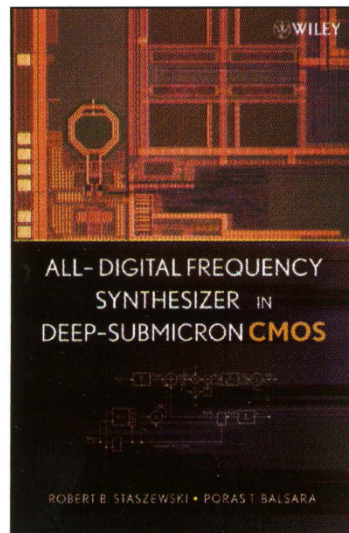
Measured and Estimated Mean RMS Phase Error in GSM900 Band



- Very high correlation between BIST and external measurements

Book on ADPLL

- R. B. Staszewski and P. T. Balsara, “All-Digital Frequency Synthesizer in Deep-Submicron CMOS,” New Jersey: John Wiley & Sons, 2006



All-Digital Frequency Synthesizer in Deep-Submicron CMOS

best seller

Robert B. Staszewski, *Texas Instruments*; Poras T. Balsara, *Univ. of Texas at Dallas*

Learn the techniques for designing and implementing an all-digital RF frequency synthesizer. This innovative book sets forth digitally intensive design techniques that lead the way to the development of low-cost, low-power, and highly integrated circuits for RF functions in deep submicron CMOS

processes. Furthermore, the authors demonstrate how the architecture enables readers to integrate an RF front-end with the digital back-end onto a single silicon die using standard ASIC design flow. Taking a bottom-up approach that progressively builds skills and knowledge, the book begins with basic concepts of frequency synthesis and then guides the reader through an all-digital RF frequency synthesizer design.

Hardcover 261 pages 2006 ISBN 978-0-471-77255-2 USD \$125.00/CAD\$137.99/£65.95/€94.90

Online Book See page 6 for information. ISBN 978-0-470-04195-6

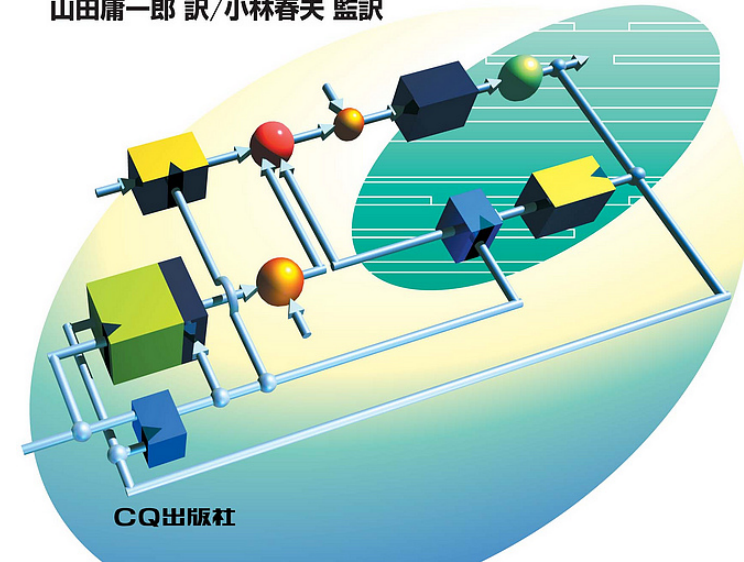
Japanese Translation

ディープ・サブミクロン CMOSプロセスで実現する
Semiconductor Series 半導体シリーズ

ALL-DIGITAL FREQUENCY SYNTHESIZER

完全デジタル PLL回路の設計

Robert Bogdan Staszewski/Poras T. Balsara 著
山田庸一郎 訳/小林春夫 監訳



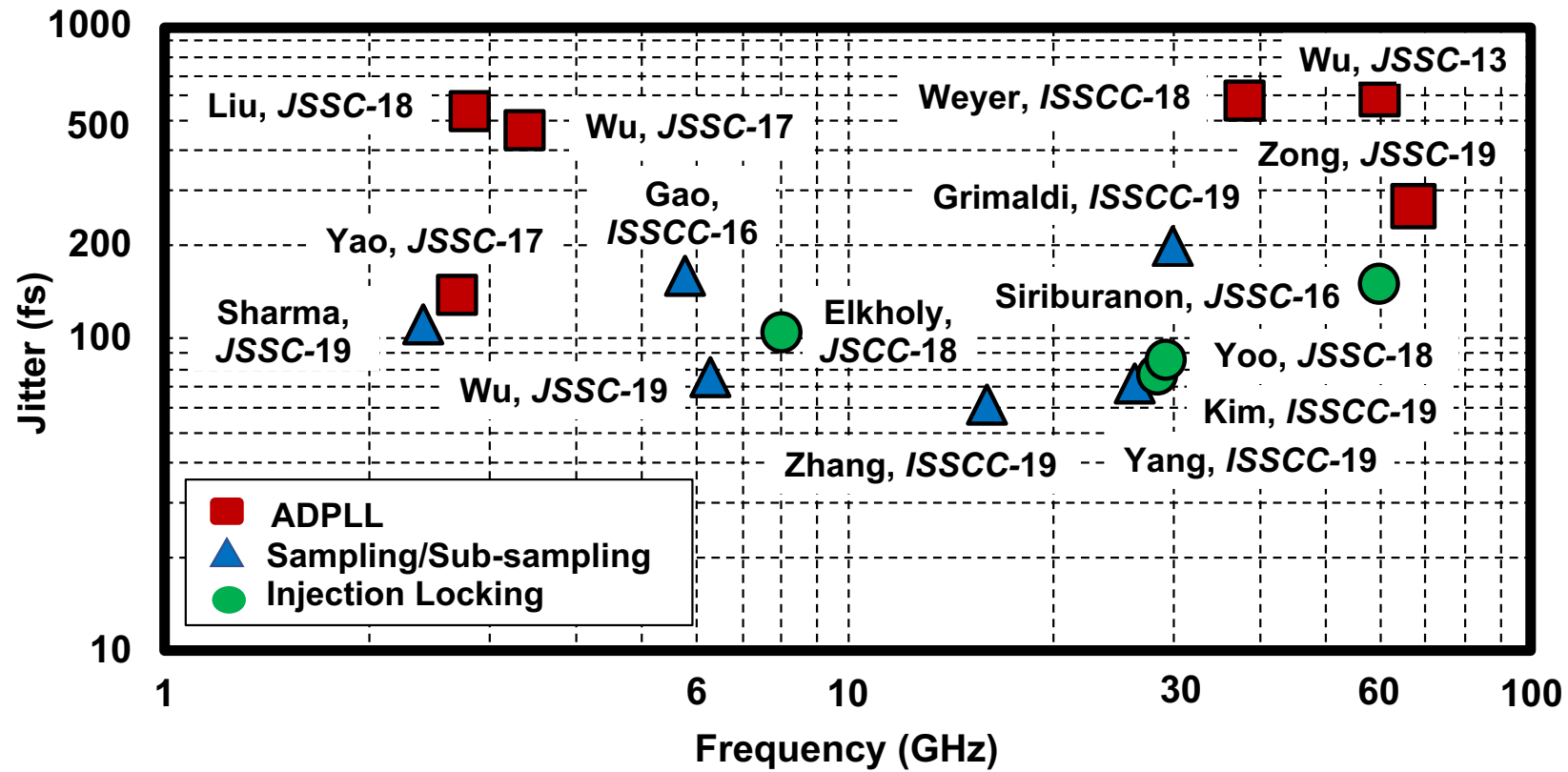
Fundamental TI Patents on ADPLL

US patent	Date	TI inventors	Title
6,326,851	2000-06-26	R.B. Staszewski, D. Leipold	Digital Phase-Domain PLL Frequency Synthesizer
6,429,693	2000-06-30	R.B. Staszewski, D. Leipold	Digital Fractional Phase Detector
6,658,748	2000-10-05	D. Leipold, R.B. Staszewski	Digitally-controlled I-c oscillator
6,809,598	2000-10-24	R.B. Staszewski, D. Leipold, K. Maggio	Hybrid of Predictive and Closed Loop Phase-Domain Digital PLL Architecture
6,414,555	2001-02-22	R.B. Staszewski, D. Leipold, K. Maggio	Frequency Synthesizer

Conclusion - ADPLL

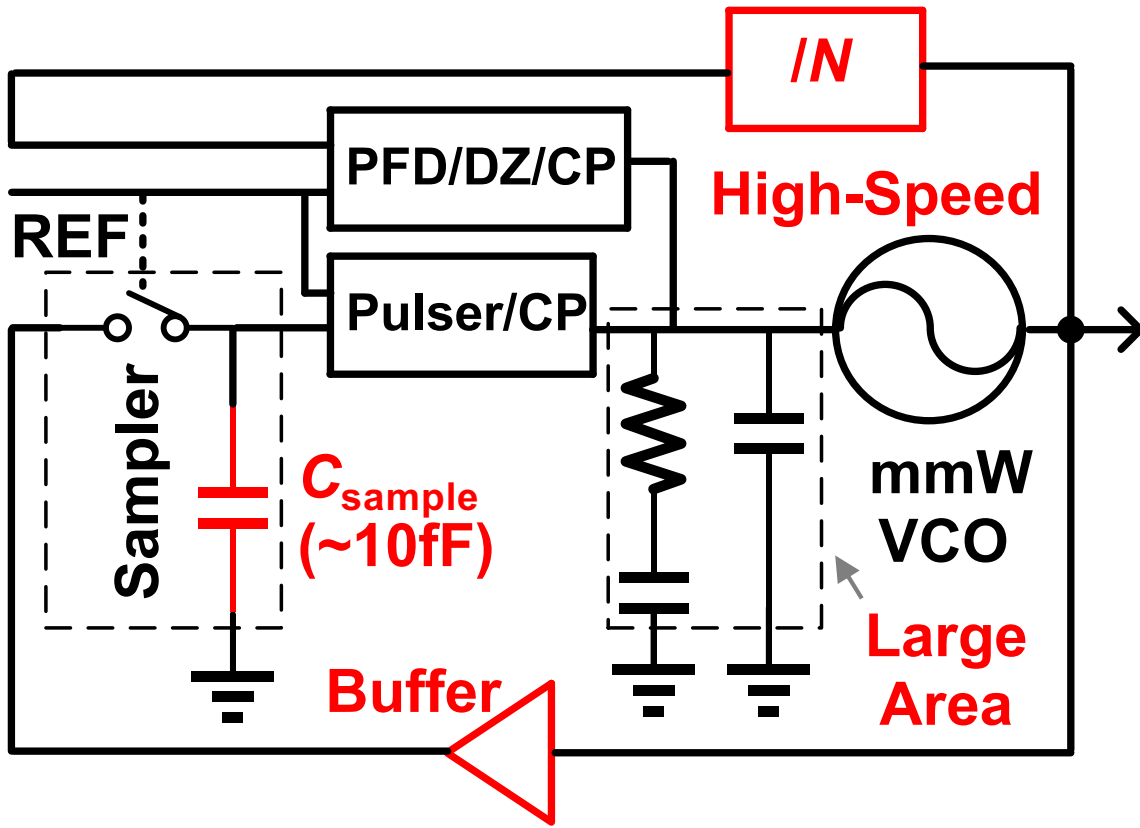
- Presented technical evolution of an all-digital phase-locked loop (ADPLL)
- ADPLL has evolved sophisticated features
 - Direct frequency modulation
 - Dithering of DCO and TDC
 - Improvement of TDC resolution and its quality
 - Wide bandwidth ADPLL
- ADPLL proven through high-volume production
- ADPLL is subject of intense research in academia and industry
- The future of ADPLL is still exciting

PLL Architecture Review for Sub-100fs Jitter



- In ADPLL, jitter limited significantly by TDC resolution (e.g. $\sim 1\text{ps}$)
- Sub-sampling (SS) & injection locking (IL)
 - In-band PN limited mainly by ref. noise
 - Increasingly popular for 5G mmW frequency generation

SS-PLL in mmW Frequency Generation



Sub-sampling PLL

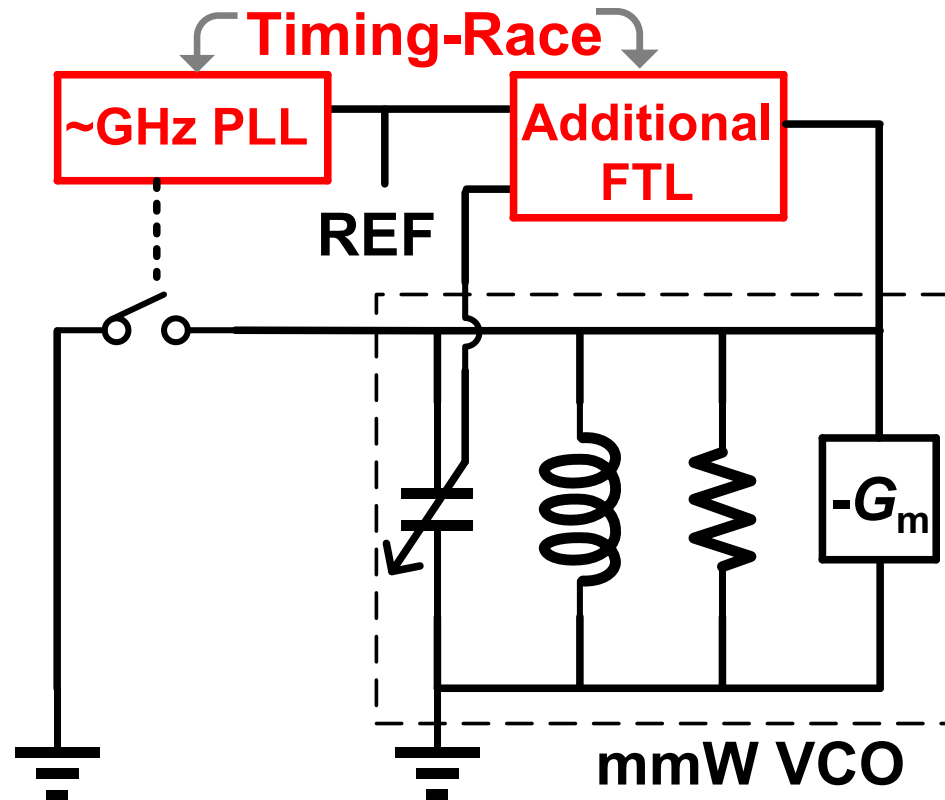
[Yang, ISSCC-19]

[Zhang, ISSCC-19]

[Grimaldi, ISSCC-19]

- Freq. locking loop (FLL) required
 - Enhance PVT robustness 😊
 - High-speed divider 😞
 - High power 😞
- High-speed isolating buffer
 - Isolate mmW VCO with sampler 😊
 - Reducing ref. spur
 - High power 😞
- Large area for analog loop filter 😞
- Poor PN & flicker PN in mmW VCO 😞
 - Low-Q tank, unknown CM path

IL-PLL in mmW Frequency Generation



Injection-Locking PLL w/ FTL

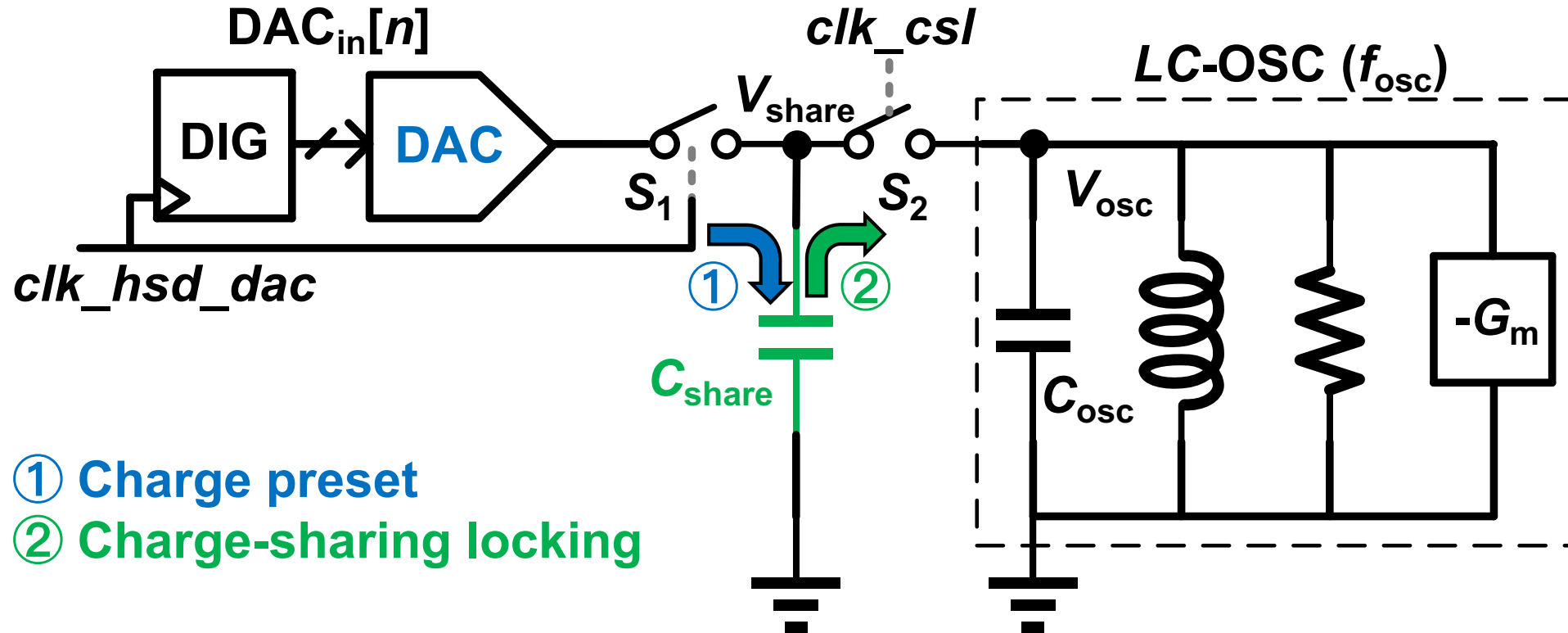
[Elkholy, JSSC-18]

[Yoo, JSCC-18]

[Kim, ISSCC-19]

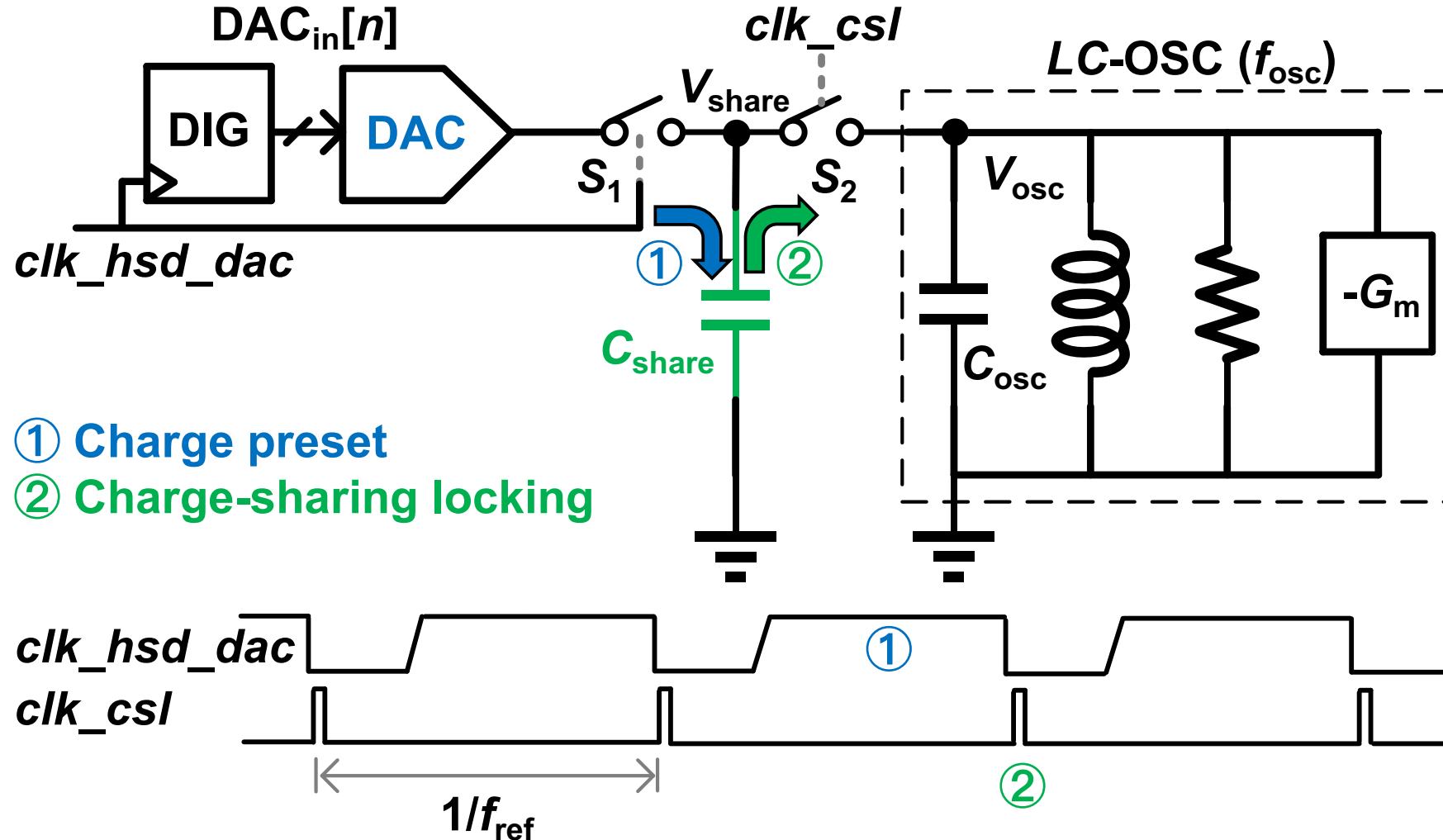
- ~GHz PLL w/ low jitter as injection ref.
 - Fully suppress poor PN in mmW VCO
 - High power 😞
- Feedforward phase correction
 - No isolating buffer 😊
 - No loop delay, large BW 😊
- Additional freq. tracking loop (FTL)
 - Enhance PVT robustness 😊
 - Timing-race between IL & FTL 😞
 - Phase error may be corrected by IL before FTL senses it 😞

Concept of Charge-Sharing Locking (CSL)



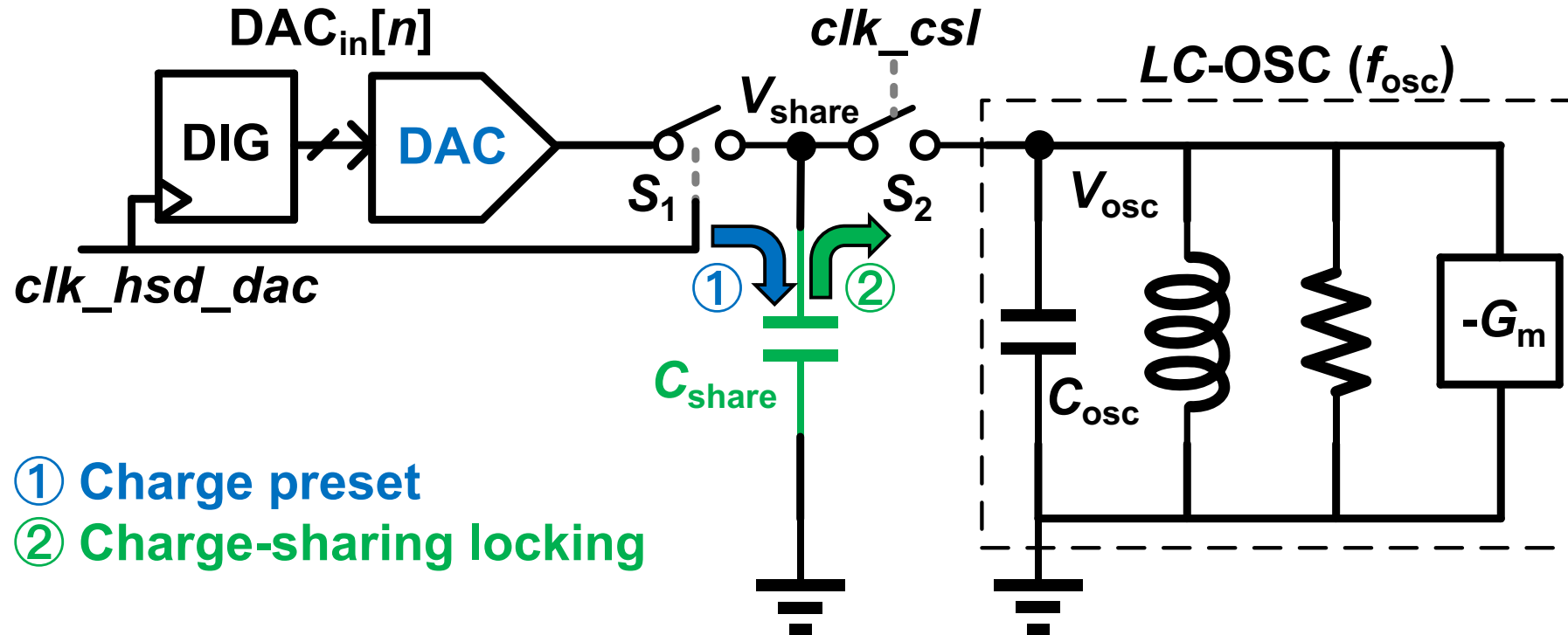
- For integer- N , $DAC_{in}[n]$ is constant
- Can naturally support fractional- N if $DAC_{in}[n]$ time-variant

Concept of Charge-Sharing Locking (CSL)



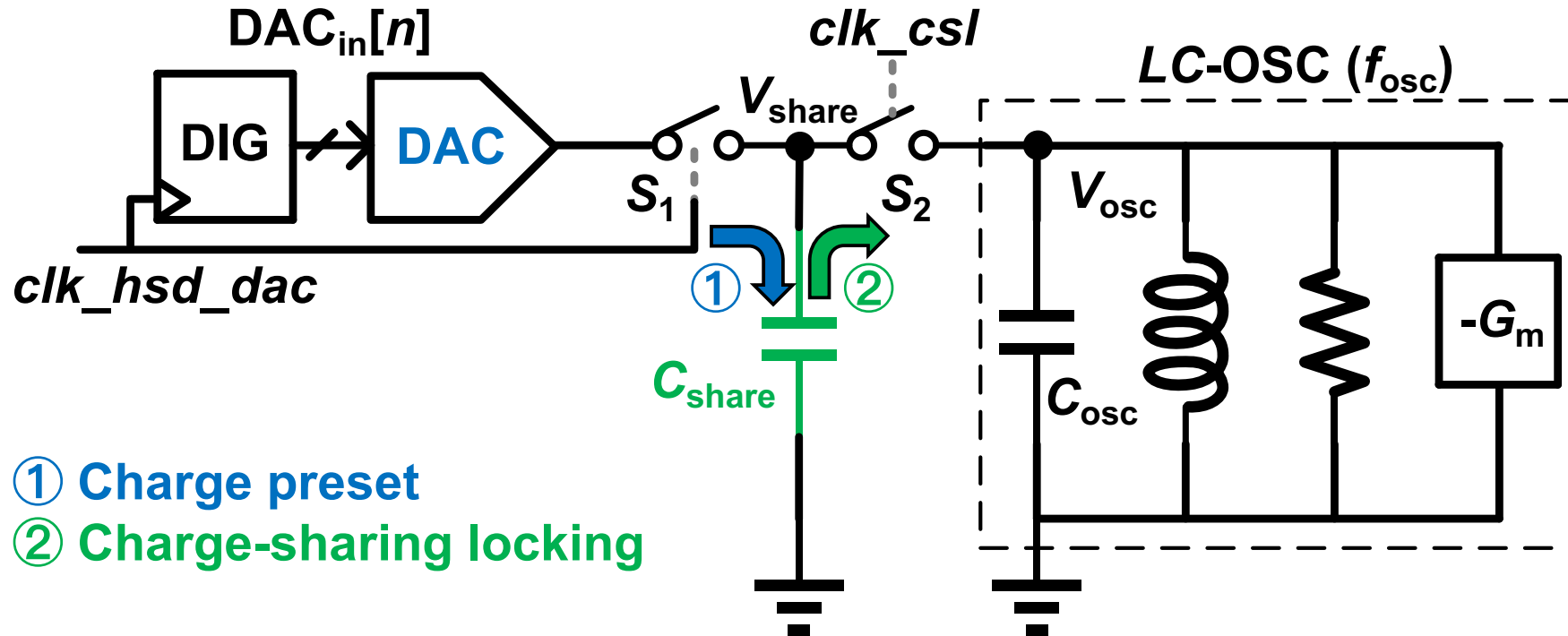
- **Charge preset:** S_1 ON, S_2 OFF, DAC preset C_{share} to expected V_{share}
- **CSL:** S_1 OFF, S_2 ON, C_{share} charge shared w/ C_{osc}

How Large C_{share} for Charge-Sharing Locking?



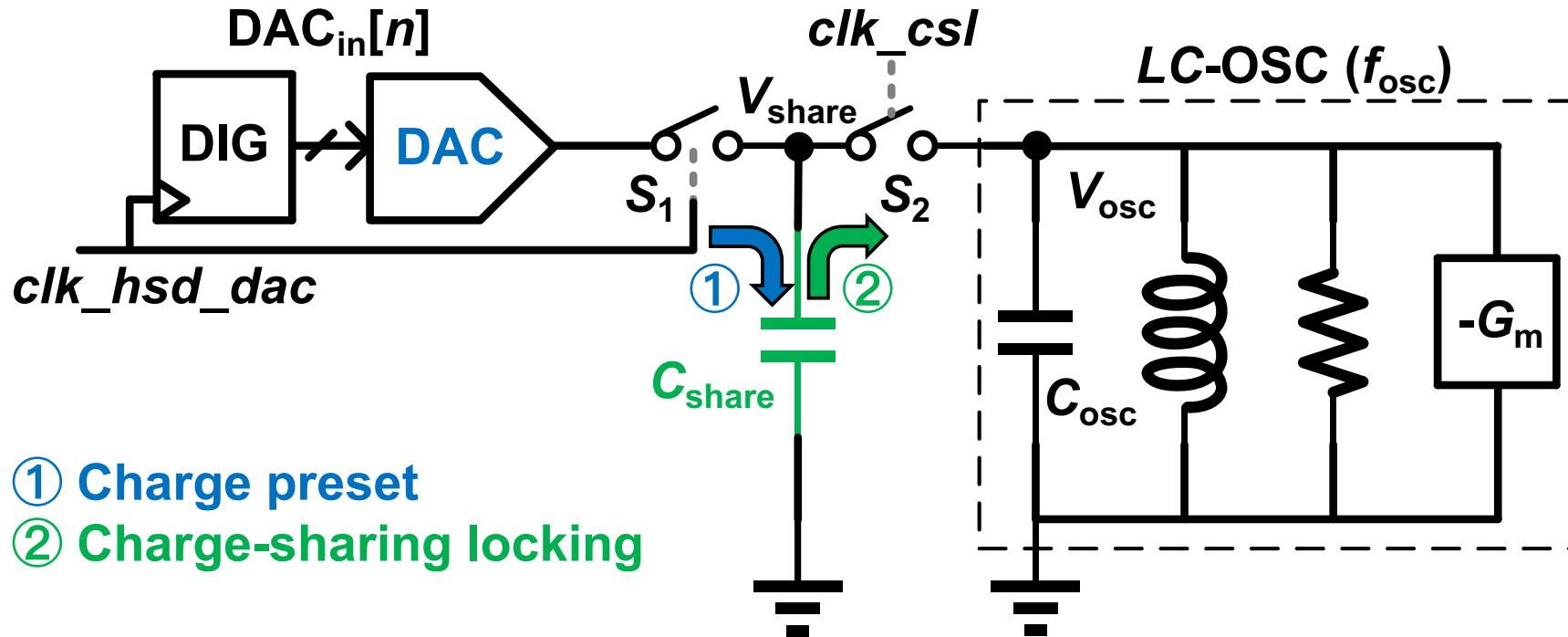
- $\Delta V_{\text{share}} \approx \frac{V_{\text{share}} C_{\text{share}} + V_{\text{osc}} C_{\text{osc}}}{C_{\text{share}} + C_{\text{osc}}} - V_{\text{share}} = (V_{\text{osc}} - V_{\text{share}}) \frac{C_{\text{osc}}}{C_{\text{share}} + C_{\text{osc}}}$
- If $C_{\text{share}} \ll C_{\text{osc}}$, $V_{\text{share}} + \Delta V_{\text{share}} \approx V_{\text{osc}}$
 - Charge on C_{share} CANNOT correct the phase of V_{osc}
 - But, only sample the V_{osc} to C_{share} , like “*sub-sampling*”

How Large C_{share} for Charge-Sharing Locking?



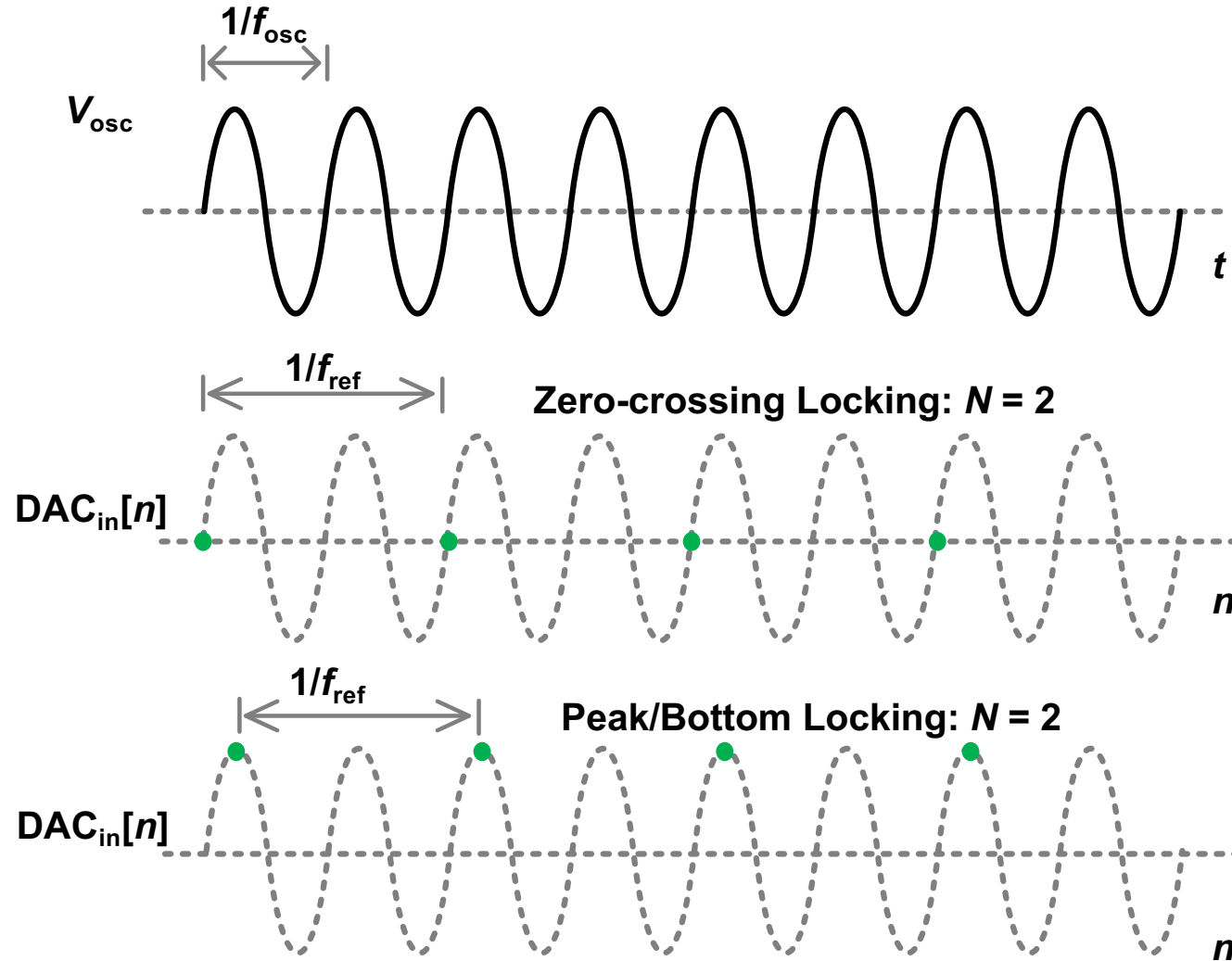
- $\Delta V_{\text{osc}} \approx \frac{V_{\text{share}} C_{\text{share}} + V_{\text{osc}} C_{\text{osc}}}{C_{\text{share}} + C_{\text{osc}}} - V_{\text{osc}} = (V_{\text{share}} - V_{\text{osc}}) \frac{C_{\text{share}}}{C_{\text{share}} + C_{\text{osc}}}$
- If $C_{\text{share}} \gg C_{\text{osc}}$, $V_{\text{osc}} + \Delta V_{\text{osc}} \approx V_{\text{share}}$
 - Charge on C_{share} (like voltage source) can force $V_{\text{osc}} = V_{\text{share}}$
 - If $V_{\text{share}} = V_{\text{DD}}$, V_{SS} , or V_{cm} , like conventional “*Injection-Locking*”

How Large C_{share} for Charge-Sharing Locking?



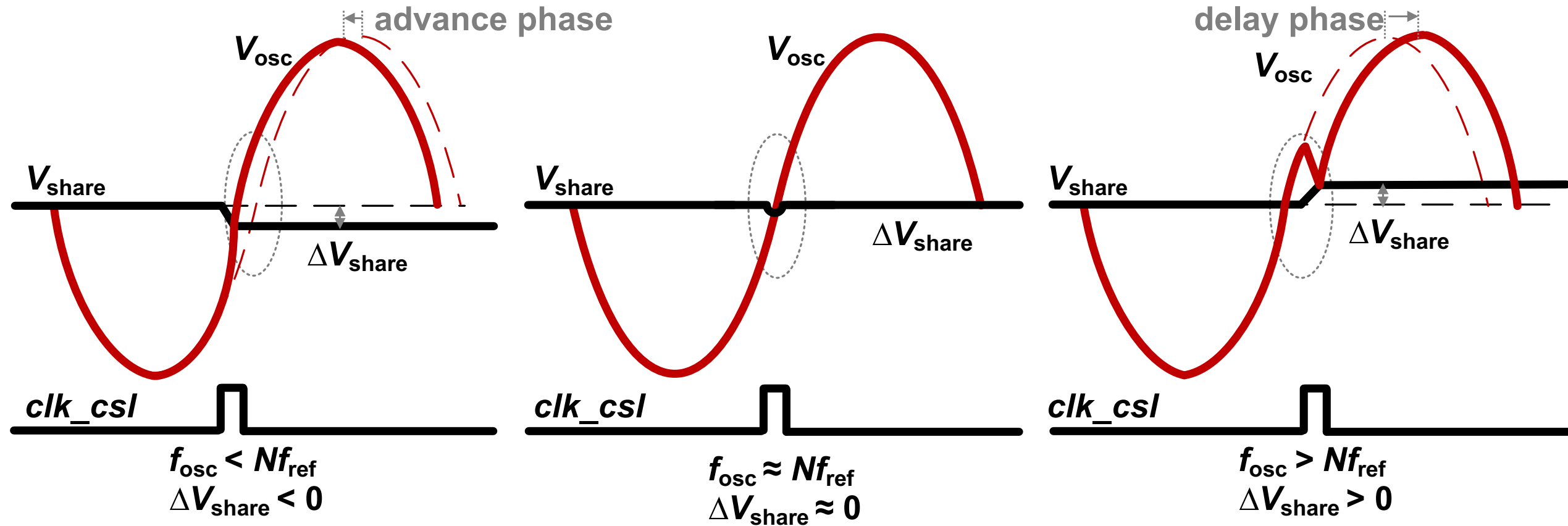
- If $C_{\text{share}} \approx C_{\text{osc}}$, $\Delta V_{\text{share}} \approx \frac{1}{2} (V_{\text{osc}} - V_{\text{share}})$, $\Delta V_{\text{osc}} \approx \frac{1}{2} (V_{\text{share}} - V_{\text{osc}})$
- ΔV_{osc} will correct the phase error of V_{osc}
- ΔV_{share} records the charge to correct phase error, mainly caused by frequency error $f_{\text{osc}} - Nf_{\text{ref}}$ and PN

Concept of Charge-Sharing Locking (CSL)



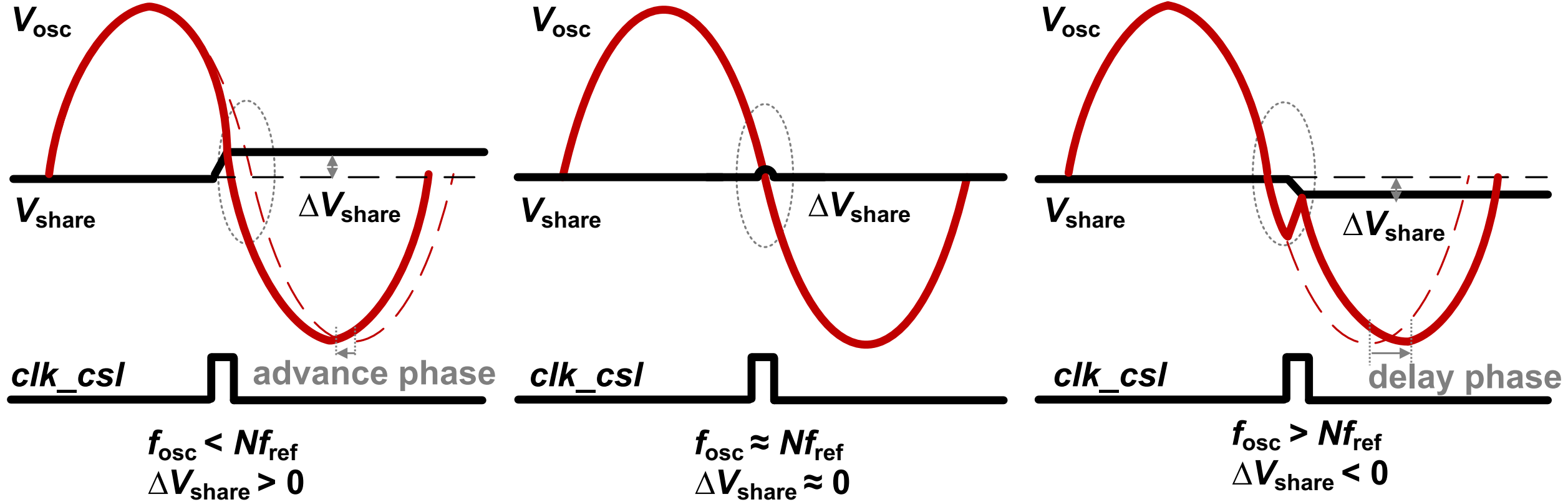
- The combination of DAC and C_{share} can lock on any point of V_{osc} (e.g. zero-crossing point or peak/bottom point)

Lock on Rising Edge of Zero-Crossing



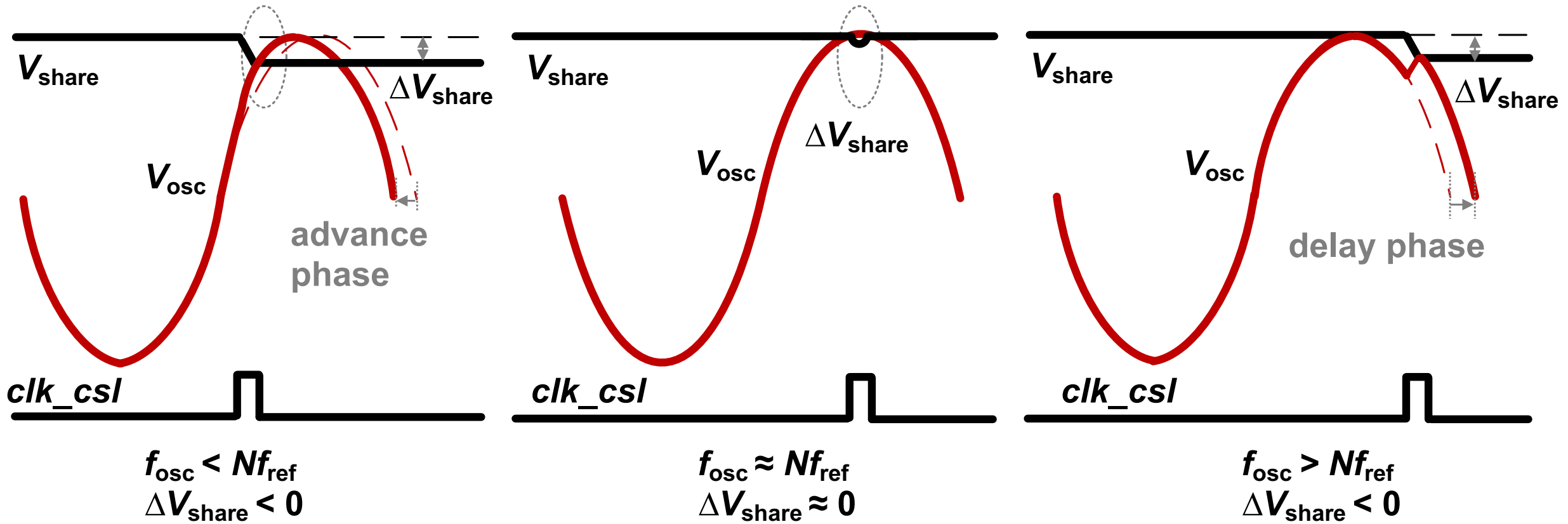
- Magnitude & polarity of ΔV_{share} represent mag. & polarity of $f_{osc} - Nf_{ref}$, which can be used in "frequency tracking loop (FTL)"
- But...

Lock on Falling Edge of Zero-Crossing



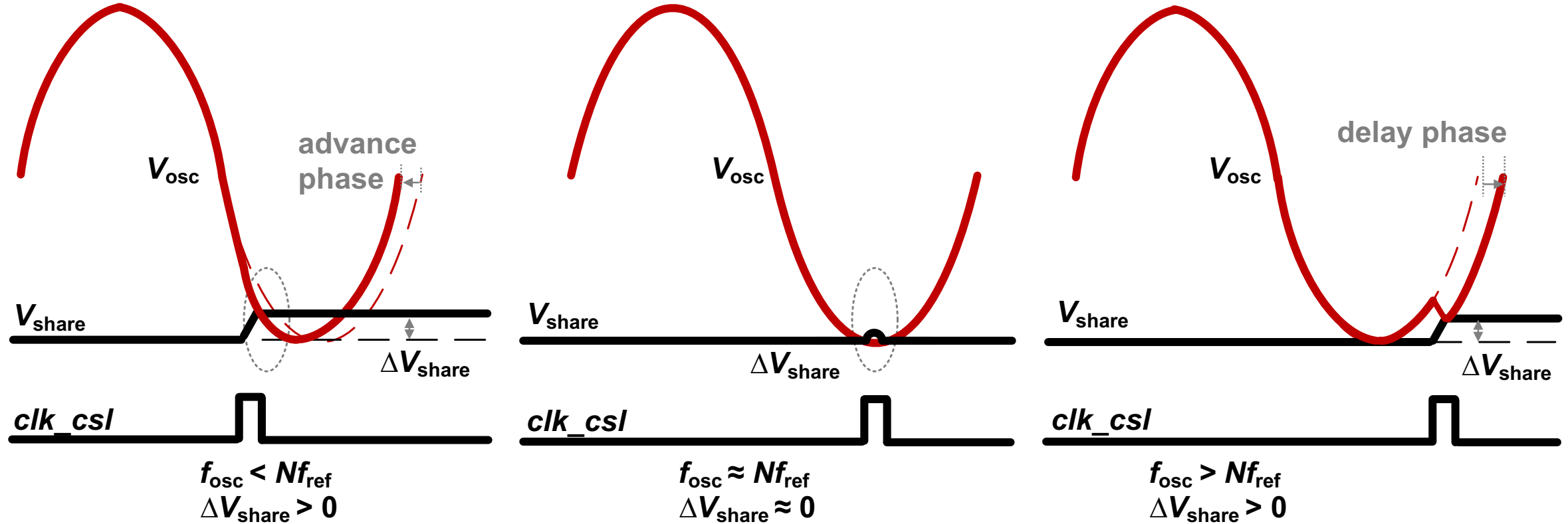
- Polarity relationship between ΔV_{share} & $f_{osc} - Nf_{ref}$ is different when locked on rising or falling edge
- If the initial locking edge is different from the assumed one, it causes positive feedback in FTL (common problem in SS and IL)

Lock on the Peak



- Only one locking point (peak), no ambiguity
- Wider locking range
 - $f_{osc} - Nf_{ref} \neq 0$, locking point close to sensitive ISF region of V_{osc}
- But, ΔV_{share} **CANNOT** distinguish the polarity of $f_{osc} - Nf_{ref}$

Lock on the Bottom



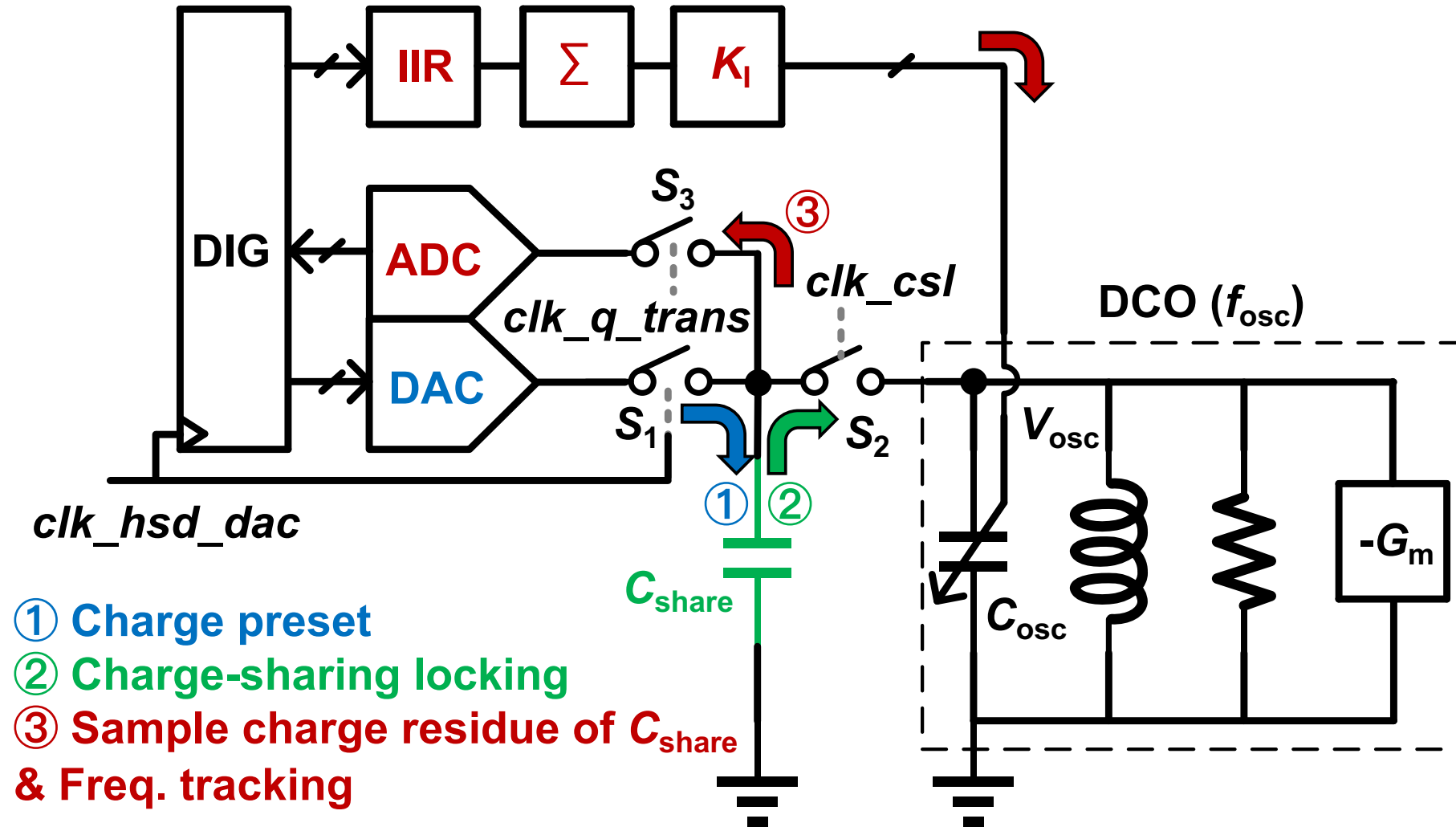
- Only one locking point (bottom), no ambiguity
- Wider locking range
 - $f_{osc} - Nf_{ref} \neq 0$, locking point close to sensitive ISF region of V_{osc}
- But, ΔV_{share} **CANNOT** distinguish the polarity of $f_{osc} - Nf_{ref}$

Comparison of Locking Points

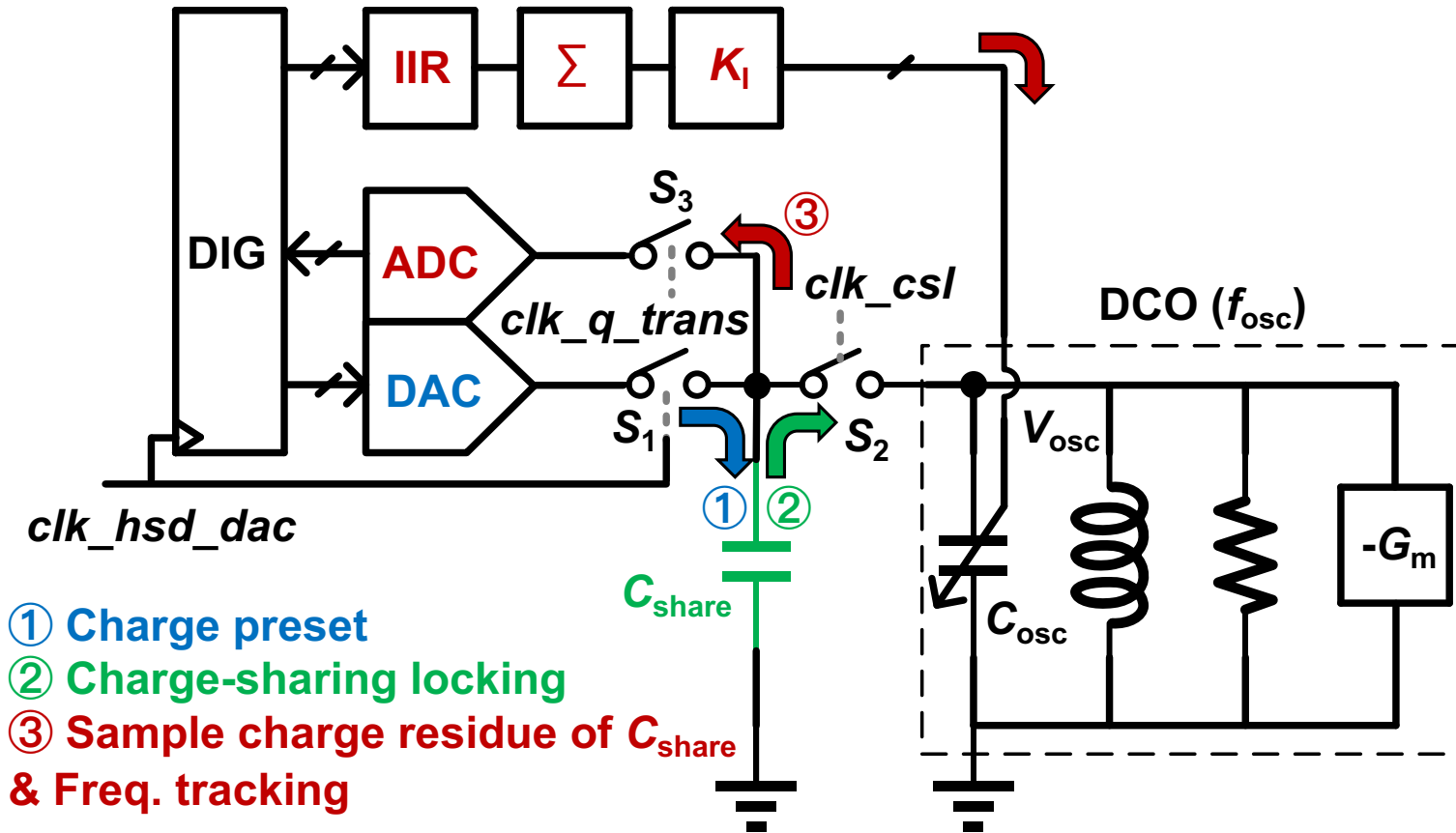
	Rising edge of zero-crossing	Falling edge of zero-crossing	Peak	Bottom
$f_{\text{osc}} - Nf_{\text{ref}} < 0$	$\Delta V_{\text{share}} < 0$	$\Delta V_{\text{share}} > 0$	$\Delta V_{\text{share}} < 0$	$\Delta V_{\text{share}} > 0$
$f_{\text{osc}} - Nf_{\text{ref}} \approx 0$	$\Delta V_{\text{share}} \approx 0$	$\Delta V_{\text{share}} \approx 0$	$\Delta V_{\text{share}} \approx 0$	$\Delta V_{\text{share}} \approx 0$
$f_{\text{osc}} - Nf_{\text{ref}} > 0$	$\Delta V_{\text{share}} > 0$	$\Delta V_{\text{share}} < 0$	$\Delta V_{\text{share}} < 0$	$\Delta V_{\text{share}} > 0$

- Locking point @ zero-crossing
 - Ambiguities in rising or falling edges 
 - ΔV_{share} represents $f_{\text{osc}} - Nf_{\text{ref}}$ if locking known @ rising/falling edges 
- Locking point @ peak/bottom
 - No ambiguities 
 - But, polarity of ΔV_{share} CANNOT distinguish polarity of $f_{\text{osc}} - Nf_{\text{ref}}$ 

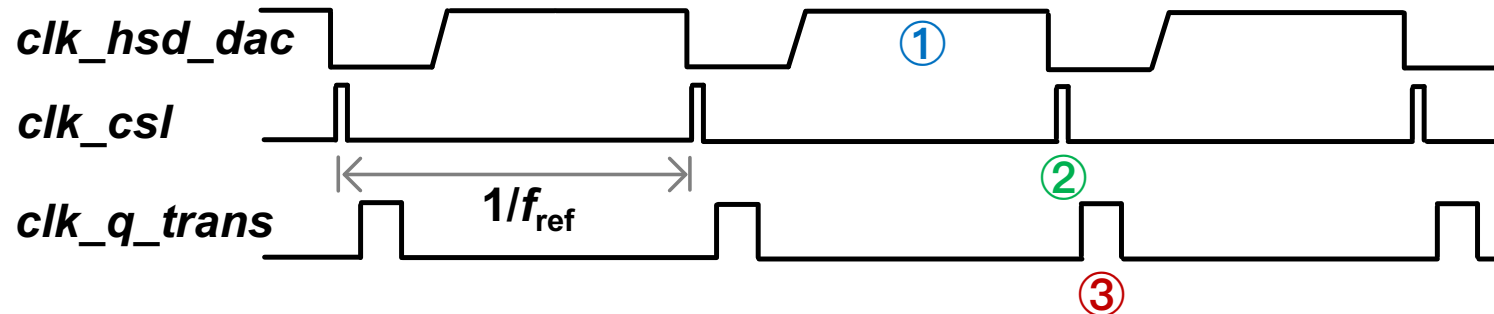
CSL w/ Digital Frequency-Tracking Loop (FTL)



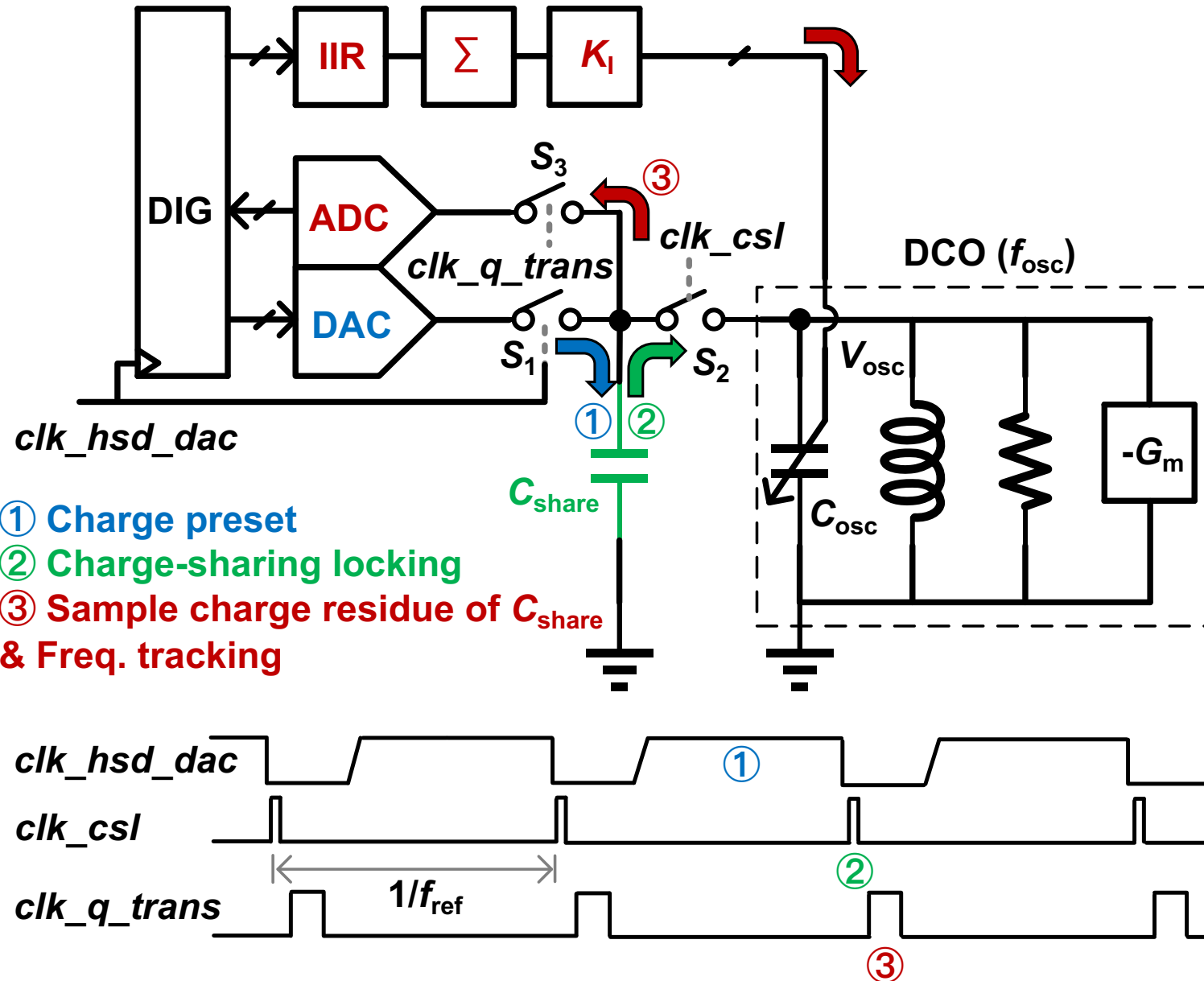
CSL w/ Digital Frequency-Tracking Loop (FTL)

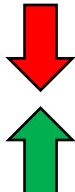


- Assume lock @ V_{osc} rising edge
- 1. Charge preset
 - S_1 ON, S_2 OFF, S_3 OFF
- 2. Charge-sharing locking
 - S_1 OFF, S_2 ON, S_3 OFF
- 3. Sample charge residue of C_{share}
 - S_1 OFF, S_2 OFF, S_3 ON

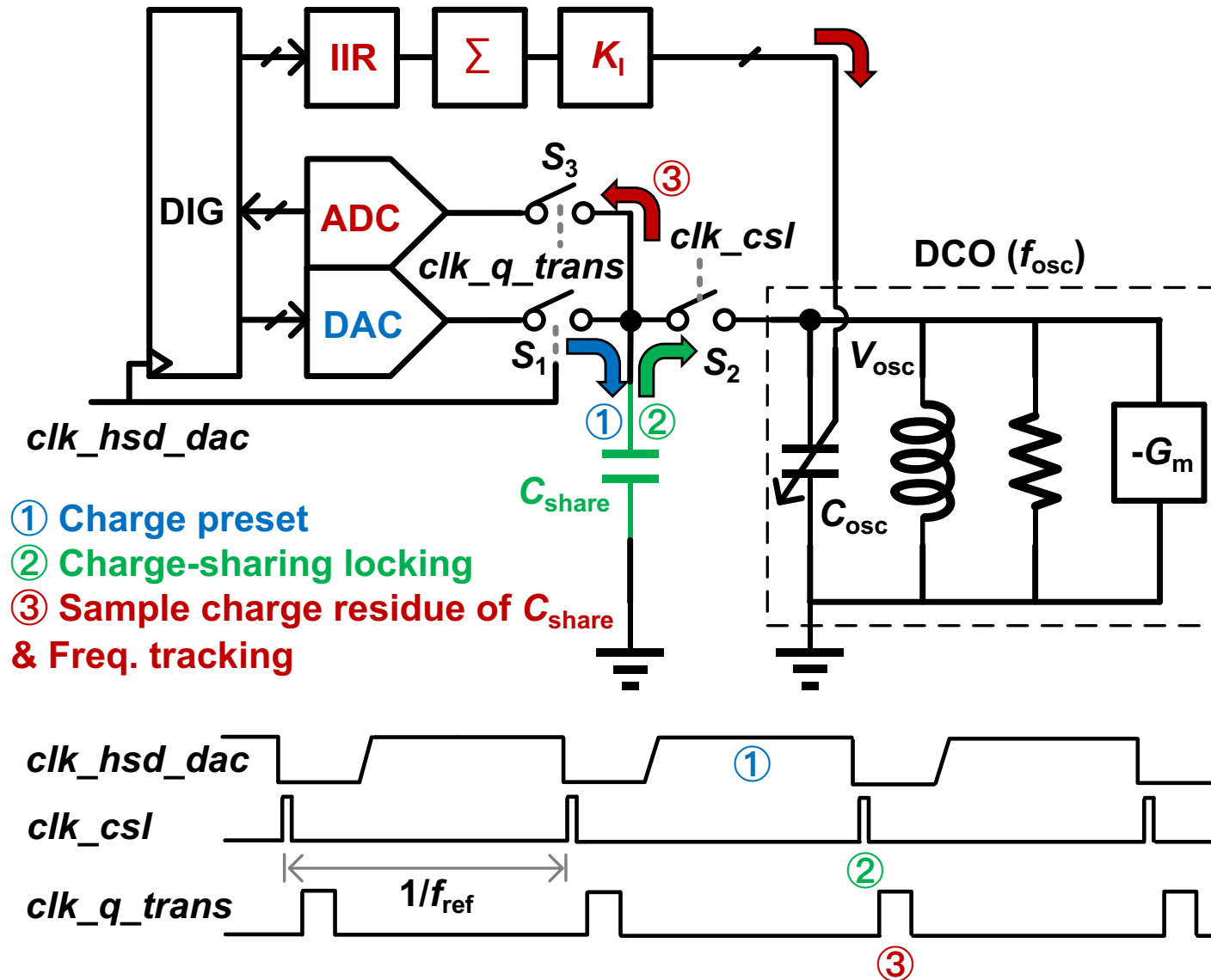


CSL w/ Digital Frequency-Tracking Loop (FTL)



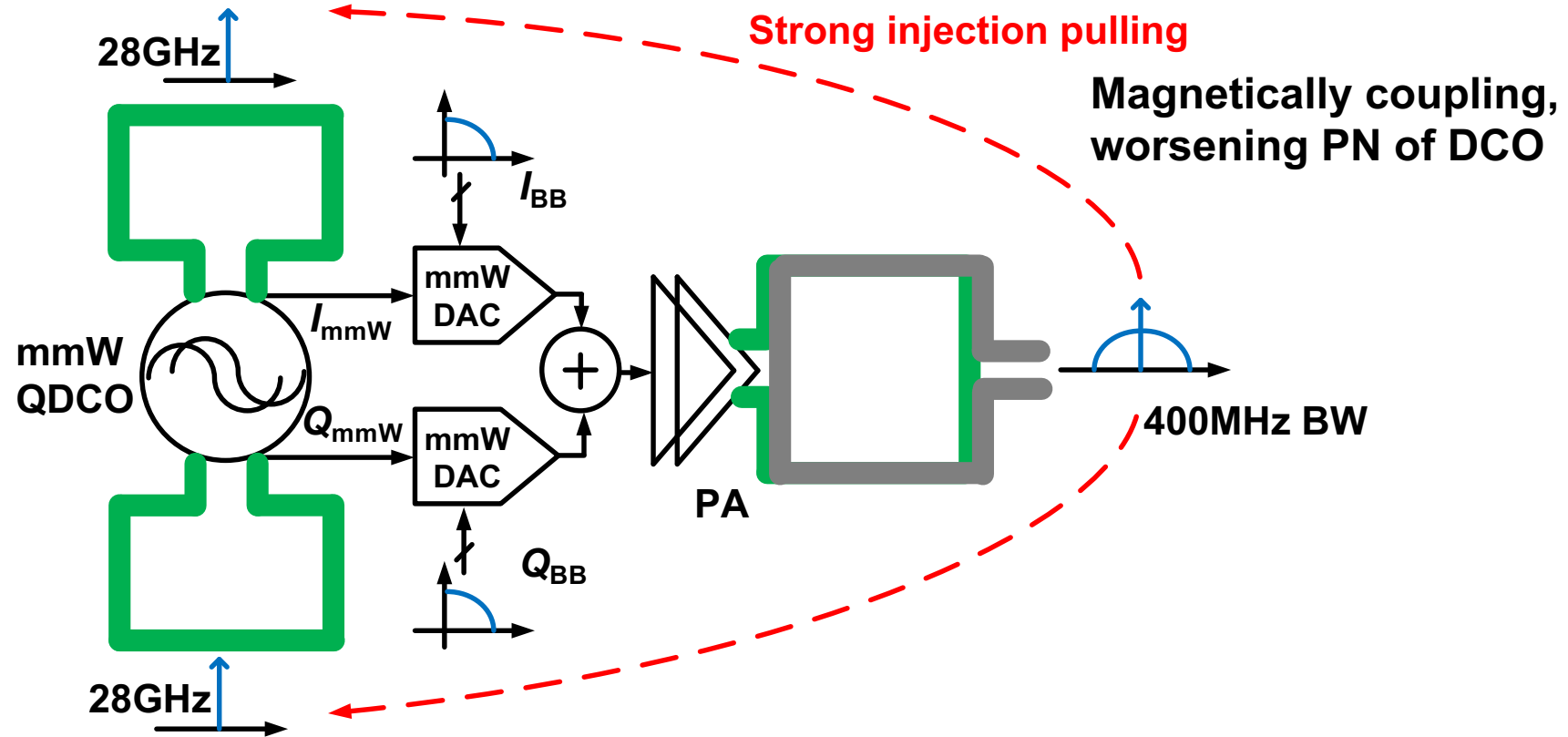
- Assume lock @ V_{osc} rising edge
 - 1. Charge preset
 - 2. Charge-sharing locking
 - Proportional path
 - Phase correction
 - 3. Sample charge residue of C_{share}
 - Integral path
 - Frequency tracking
 - $\Delta V_{share} < 0 \Rightarrow C_{osc}$
 - $\Delta V_{share} > 0 \Rightarrow C_{osc}$
- 

CSL w/ Digital Frequency-Tracking Loop (FTL)



- Freq. tracking after CSL
 - No timing-race problem 😊
- SAR-ADC-based Dig. FTL
 - Ultra-low power (0.1mW) 😊
 - Tiny area
- sw-cap. banks in DCO 😊
 - ONLY used to track freq. rather than phase correction
 - Relieve requirements on
 - Resolution, no need DSM
 - Linearity
- No isolating buf., low power 😊

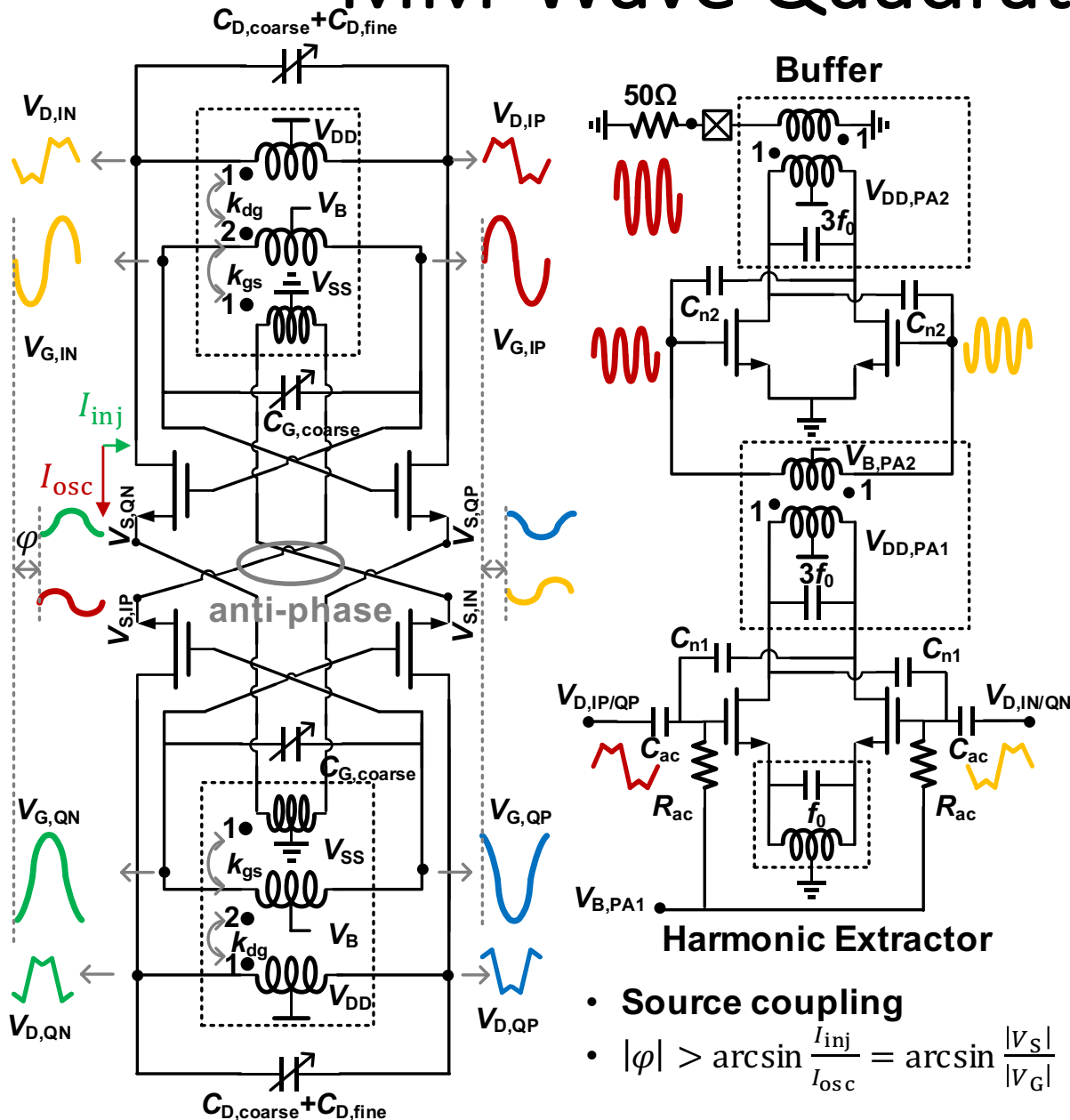
MM-Wave Quadrature Freq. Generation



- IQ mmW-DAC promising candidate for large BW of baseband 😊
- However, directly run QDCO @mmW
 - Low-Q cap. bank 😞
 - Strong inj. pulling due to large modulation BW ($\gg$ BW of PLL) 😞

[Mirzaei, JSSC-14]

MM-Wave Quadrature Freq. Generation



• Two class-F DCOs

[Hu, JSSC-18]

• Running @ f_0 (e.g. 28/3 GHz)

- Better Q of tank, less inj. pulling

• Square wave @drain

- 3rd harm. for extraction

• Sinusoidal wave @gate for CSL

• Source coupling to generate quadrature signal

[Ng, JSSC-07]

[Mirzaei, JSSC-07]

• Harmonic extractor

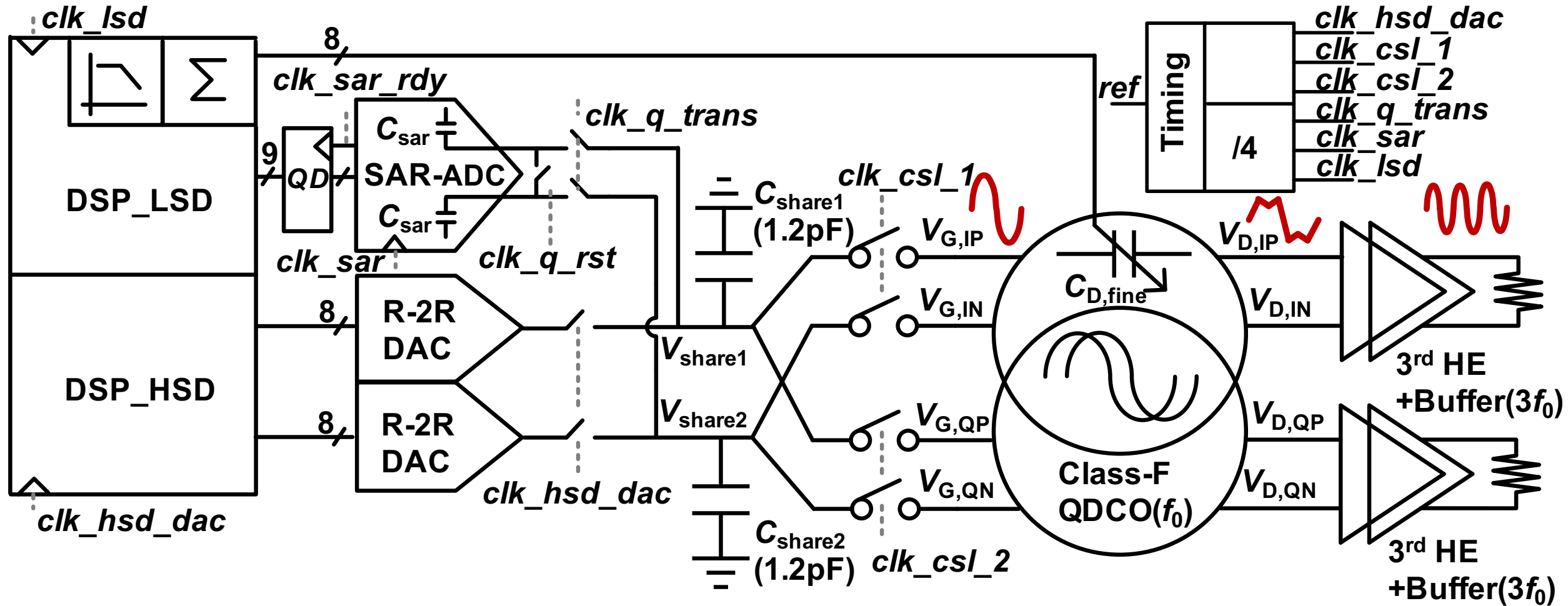
- Amplify $3f_0$, suppress f_0

[Zong, JSSC-19]

• Source coupling

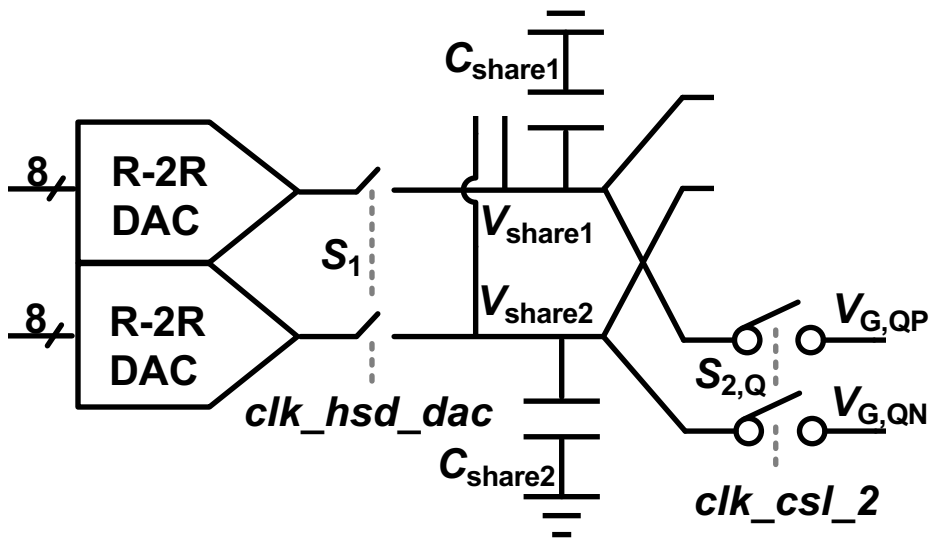
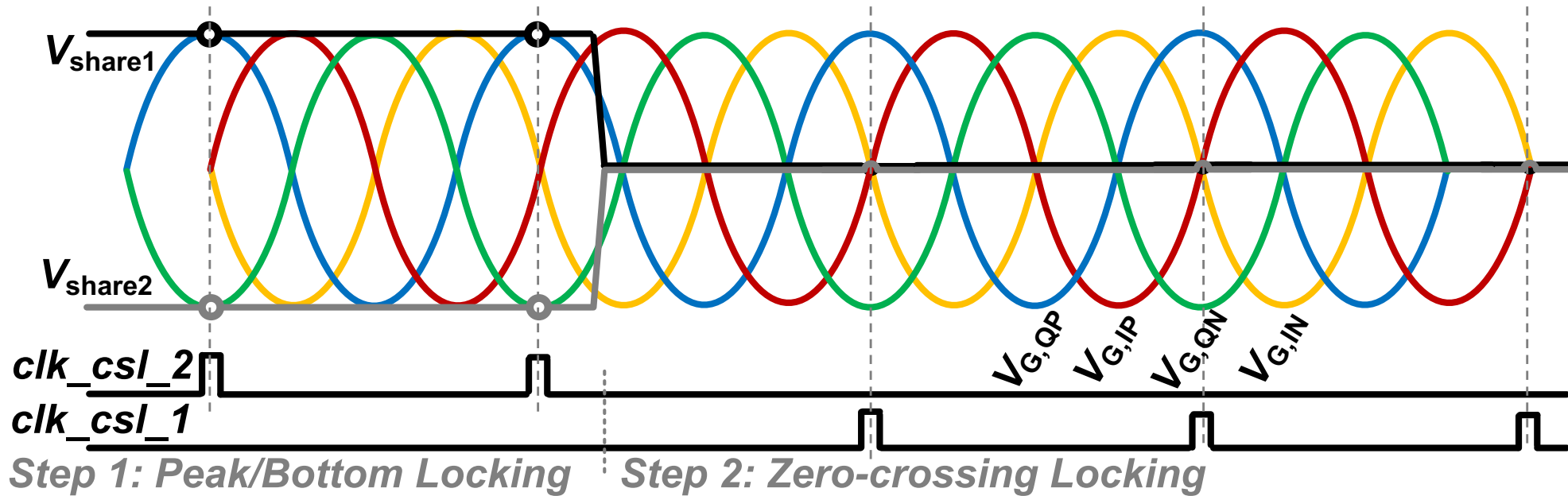
$$|\varphi| > \arcsin \frac{I_{inj}}{I_{osc}} = \arcsin \frac{|V_S|}{|V_G|}$$

Detailed Architecture of CSL Quadrature PLL



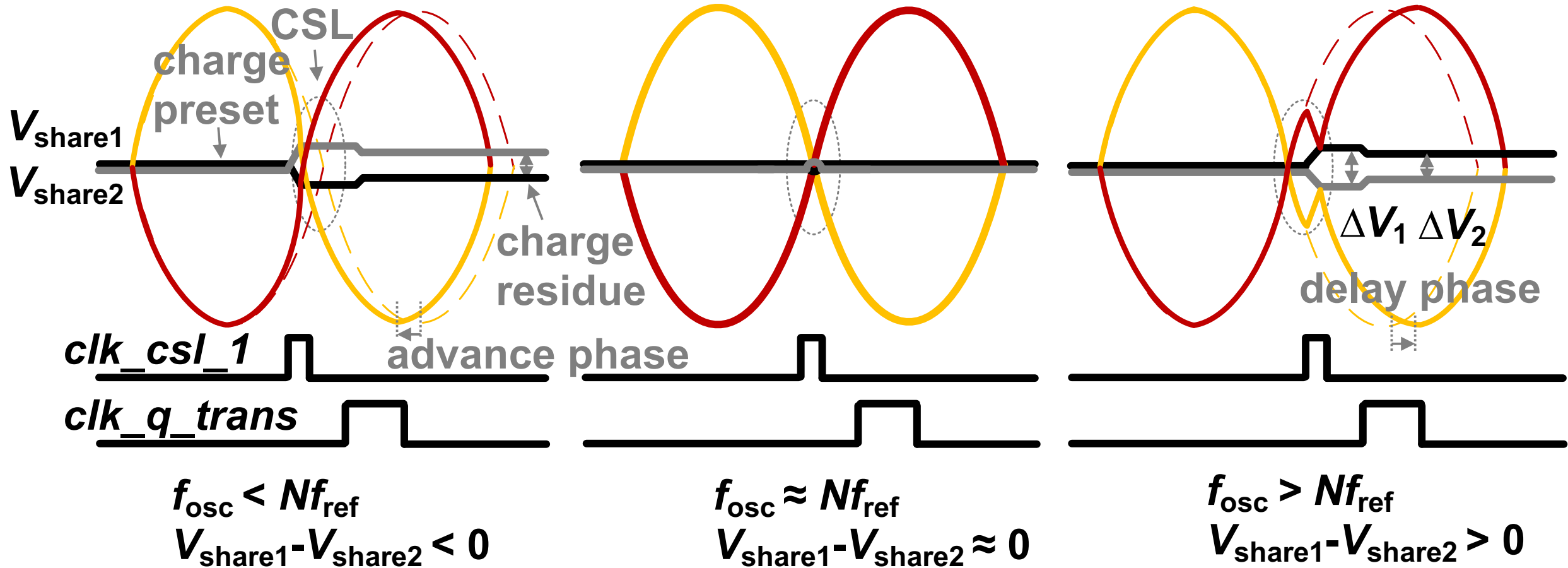
- Fully diff. operation mitigates all CM coupling and leakage
- FTL operates @ $f_{ref}/4$ to further save power

Two-Step Locking—Peak/Bottom Locking



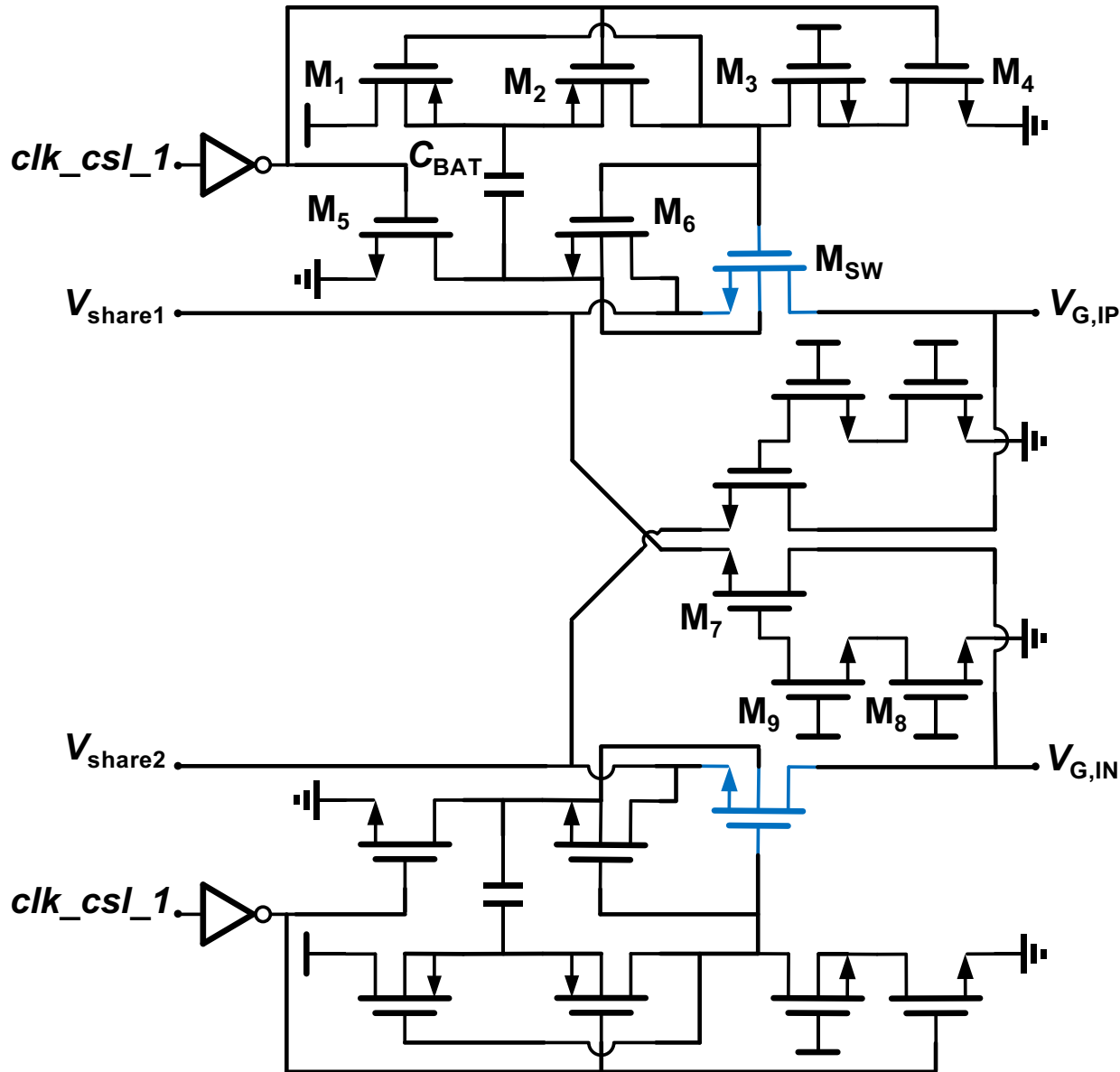
- Preset V_{share1} (V_{share2}) to V_{DD} (V_{SS})
 - Align $V_{\text{G,IP}}$ rising edge to $\text{clk_csl_1}/2$
- Preset V_{share1} (V_{share2}) to V_{cm}
 - CSL @ rising edge of $V_{\text{G,IP}}$
 - Avoid uncertainties of rising or falling edges

Differential Charge-Sharing Locking



- $\Delta V_2 = \Delta V_1 C_{share} / (C_{share} + C_{sar})$

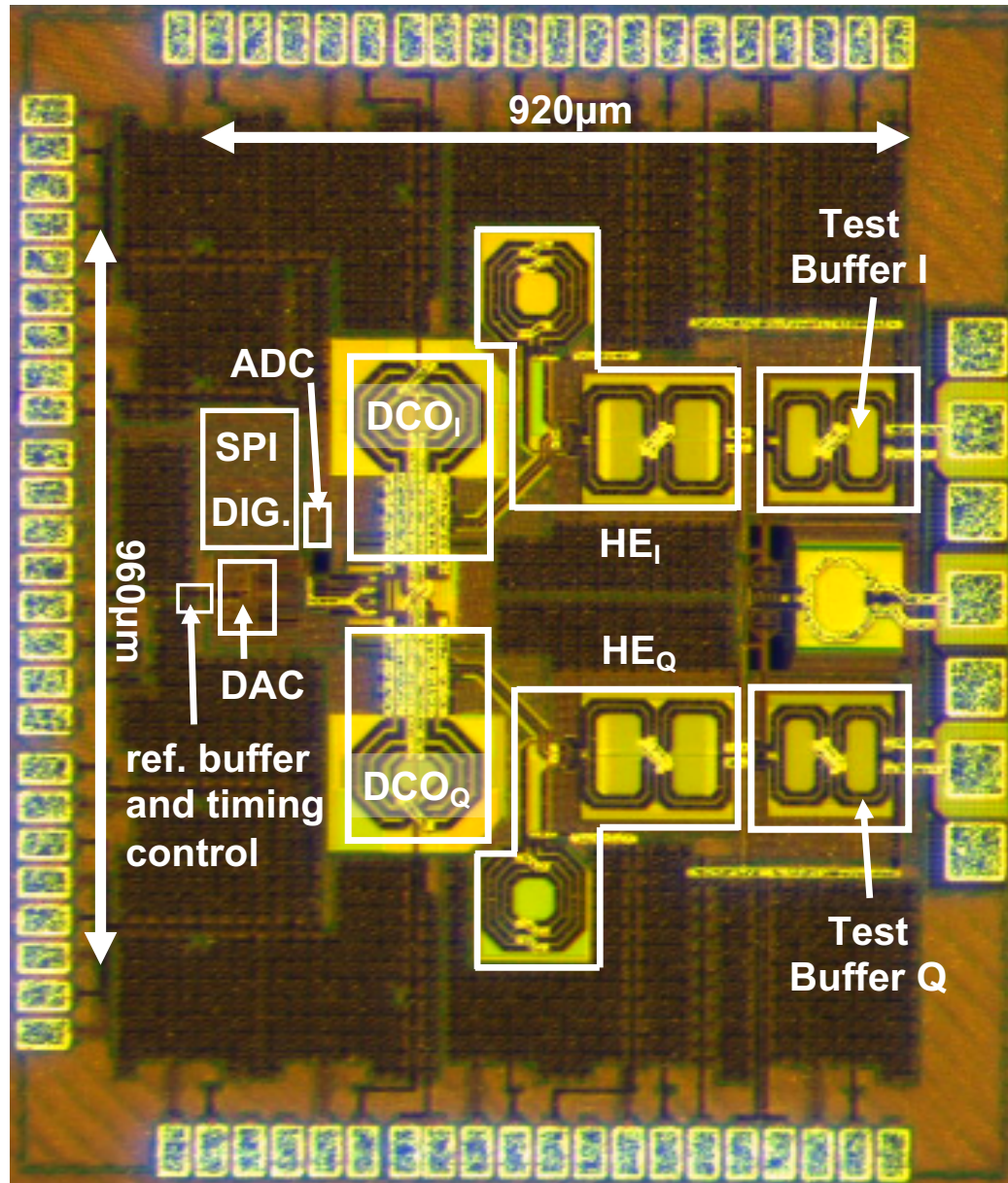
Charge-Sharing Switch



- Proposed bootstrap w/ adaptive body-biasing of M_{sw}
 - ON: body connects to source
 - Lower R_{on} , more linearity
 - OFF: body connects to GND
 - Higher R_{off} , more isolation
- $M_{7,8,9}$ as dummies of $M_{sw,3,4}$
 - Cancel “hold-mode feedthrough”
- Transmission gates for other switches

[Razavi, *MSSC-15*]

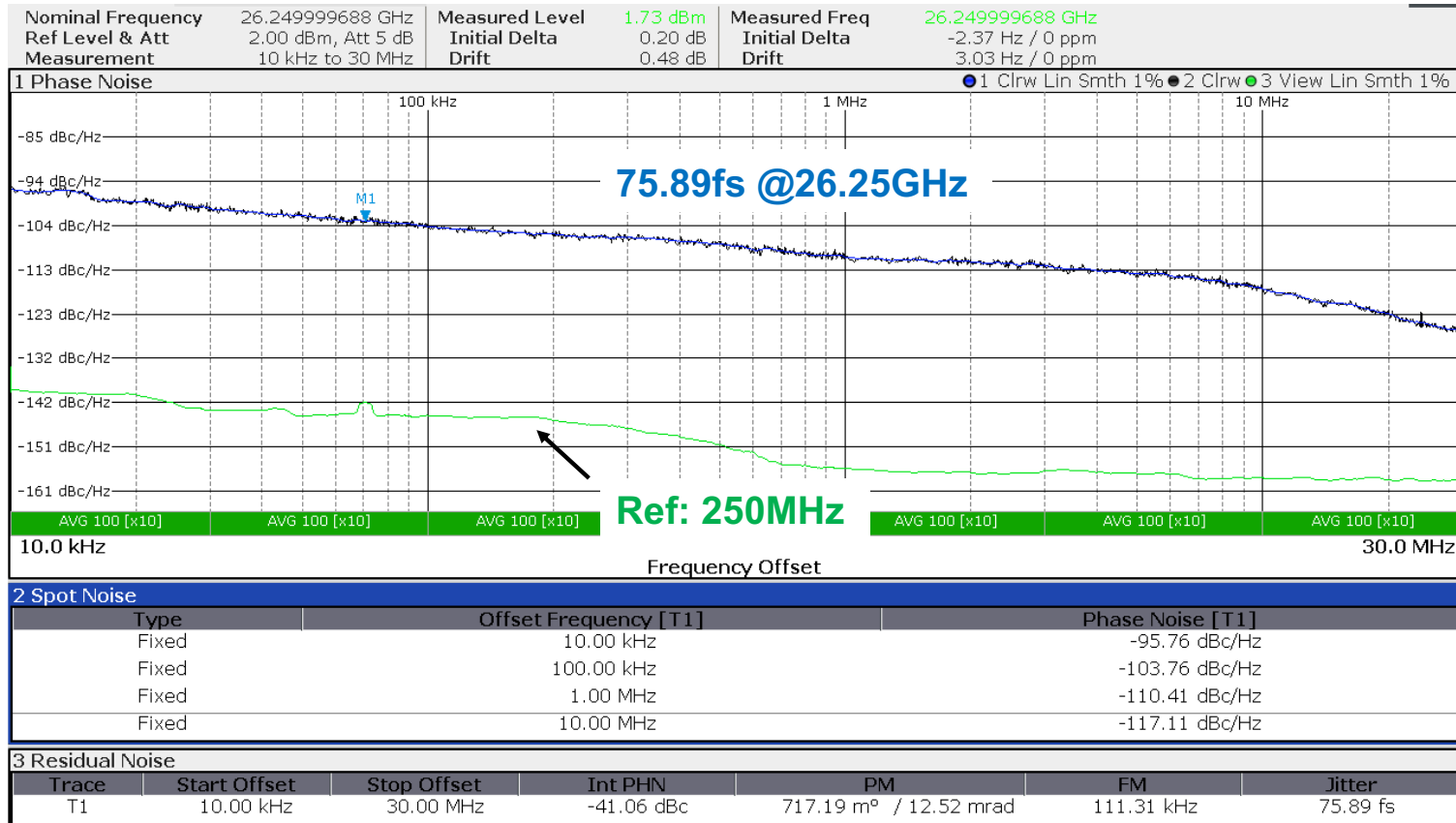
Chip Micrograph & Power Breakdown



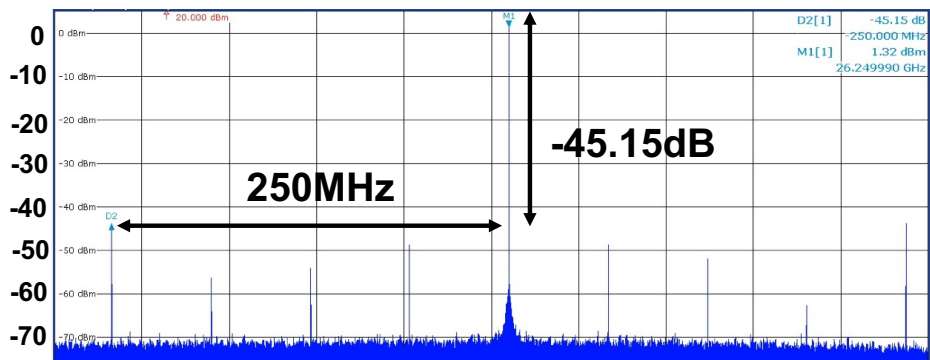
Building Blocks	P_{DC} (mW)
QDCO ($V_B=0.5V$, $V_{DD}=0.4V$)	3.92
Harmonic Extractors (HE) ($V_B=0.7V$, $V_{DD}=1V$)	2×5
R-2R DACs	2×0.75
SAR-ADC	0.1
Ref. buf. + Timing logics + Others	0.82
SPI + Digital	0.14
Total	16.5

- 28nm LP CMOS
- Ref. freq.
 - 250MHz
- Output freq.
 - 21.71–26.49GHz
- Active area
 - 0.5mm²

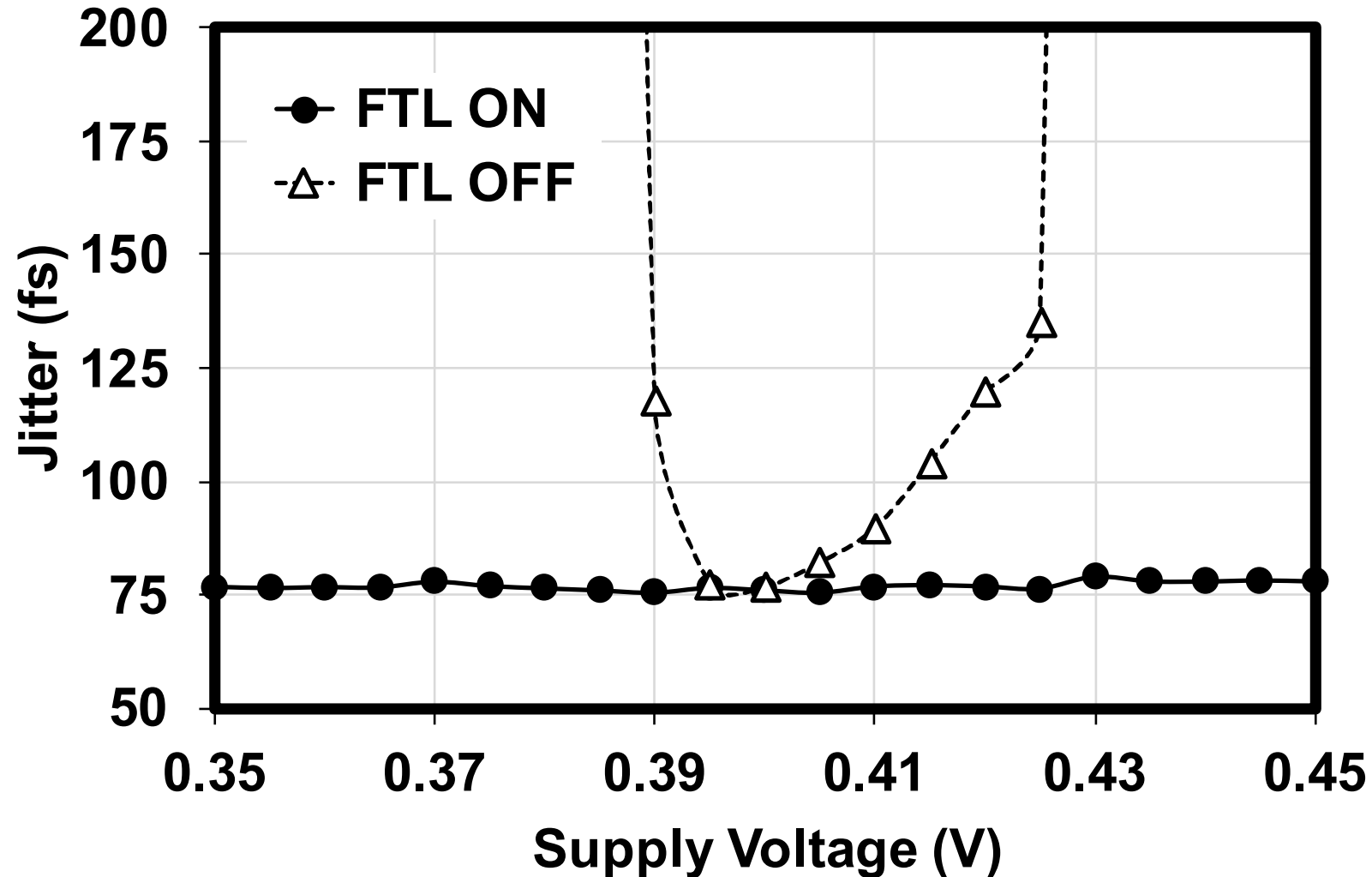
Measurements – Jitter & Spur



- @26.25GHz
- PN@1MHz: -110.41dBc/Hz
- IPN: -41.06dBc
- Jitter_{rms}: 75.89fs
- Ref. Spur: -45.15dBc
- Total Power: 16.5mW
- FoM@1MHz: -250dB

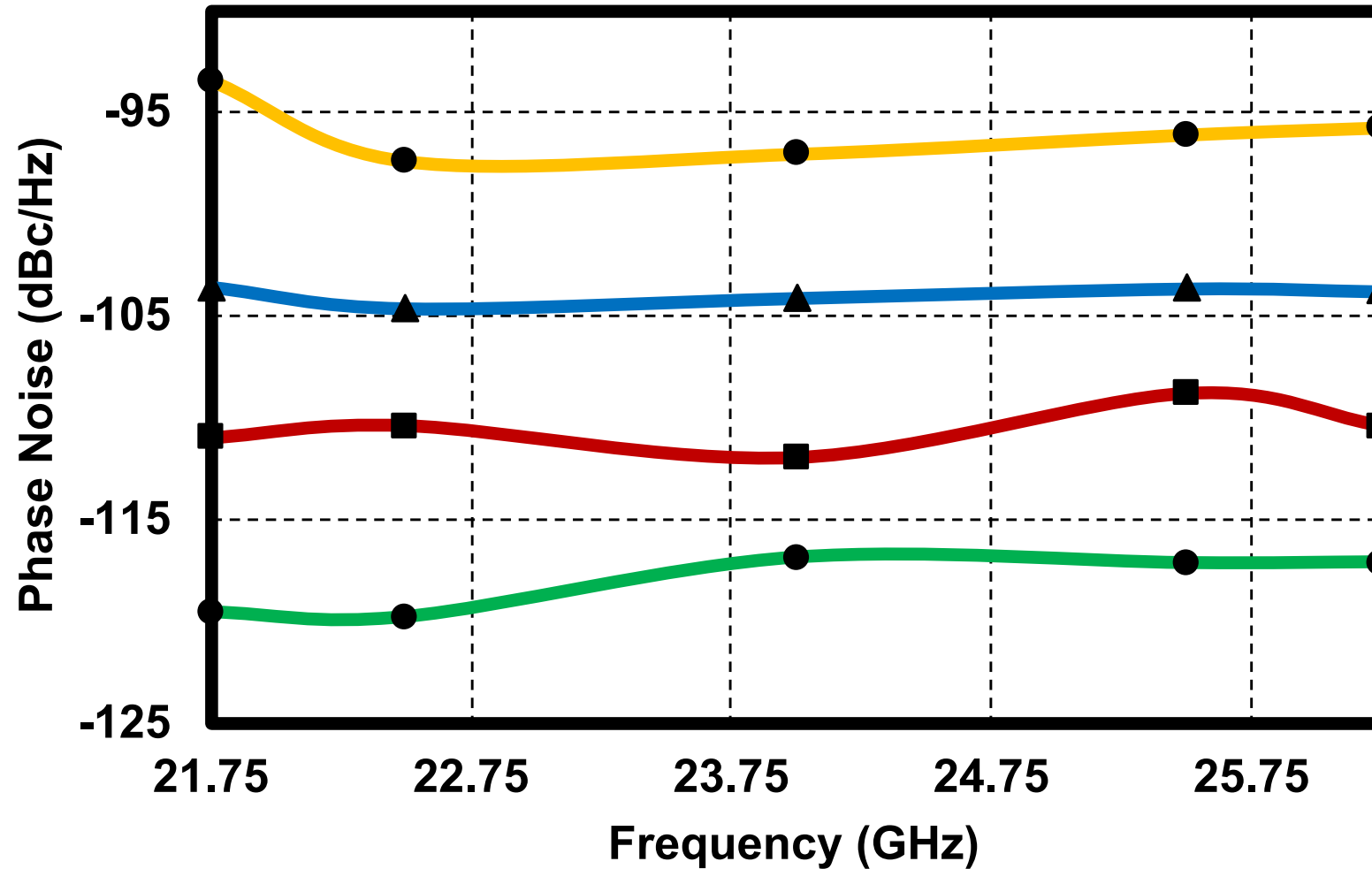


Measurements – Voltage Variation Robustness



- CSL w/ digital FTL maintains ~75fs jitter when V_{DD} of DCO changes from 0.35V to 0.45V

Measurements – PN Across Tuning Range



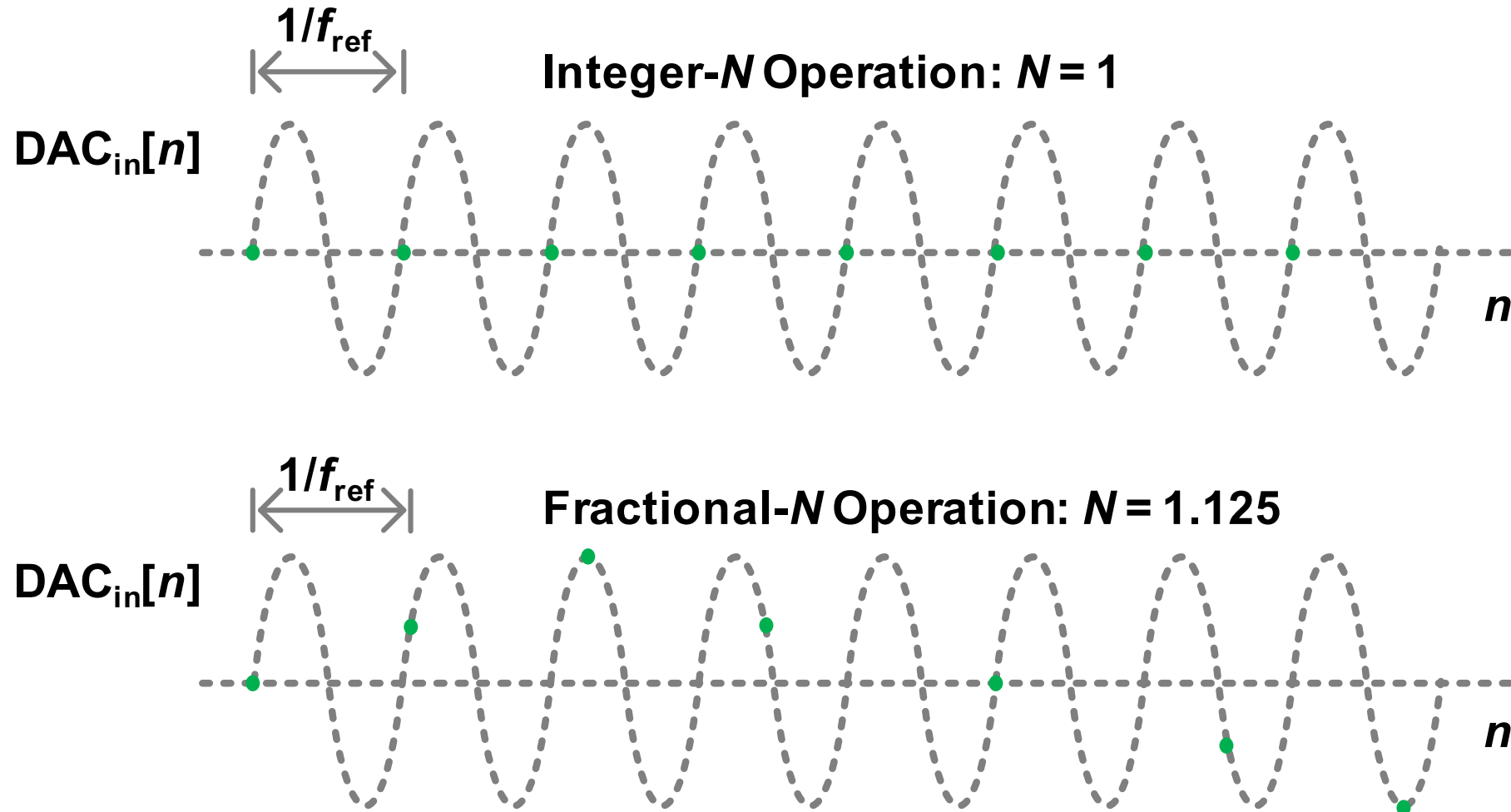
- PN @1MHz maintains around **-110dBc/Hz** from 21.75–26.25GHz

Performance Comparison

	This work	Kim, <i>ISSCC'19</i> [1]	Siriburanon, <i>JSSC'16</i> [4]	Yang, <i>ISSCC'19</i> [2]	Szortyka, <i>ISSCC'14</i> [3]	Elkholy, <i>JSSC'15</i>
Tech. (nm)	28 LP	65	65	65	40	65
Architecture	Charge-Sharing Locking	Cascading 4GHz SS-DPLL + QILO	Cascading 20GHz SSPLL + QILO	Analog SSPLL	Analog SSPLL	Injection Locking
Freq. Tracking/Locking Loop	Digital FTL (SAR-ADC)	Analog Phase-Avg. FTL	ILFD + PFD/CP	High-speed Divider + PFD/CP	CML+ PFD/CP	Gated Pulse + BBPD
Quadrature	Yes	Yes	Yes	No	Yes	No
Approach	Class-F QDCO + 3 rd Harm. Extraction	mmW QILO	mmW QILO	Direct mmW VCO	Direct mmW VCO	DCO
Output Freq. (GHz)	21.71 to 26.49 (19.3%)	28 to 31 (10.1%)	55.6 to 65.2 (15.8%)	25.4 to 29.5 (14.9%)	53.8 to 63.3 (16.2%)	6.75 to 8.25 (20%)
Ref. (MHz)	250	100	40	103	40	105 to 129
PN@1MHz (dBc/Hz)	-110.4	-105.84	-91	-112.8	-88.3	-112.3
Jitter (fs) (Integration range)	75.89 (10k to 30M)	76 (1k to 30M)	290 (NA)	71 (1k to 100M)	214 (1k to 100M)	179 (10k to 40M)
Ref. Spur (dB)	-45	-58@100MHz/ -40@2GHz	-73	-63	-40	-40
P_{DC} (mW)	16.5	41.8	32 / 35 (including FLL)	15.3 [#] / NA	42 / 75 (include FLL)	2.25
P_{DC,FTL} (mW)	0.1	0.9 ^{&}	NA	NA	33	NA
FoM/FoM* (dB)	-250.2/-253.2*	-246/-249*	-235.7/-238.7*	-251.1	-237/-240*	-251
Active Area (mm²)	0.5	0.32	N/A	0.24	0.16	0.25

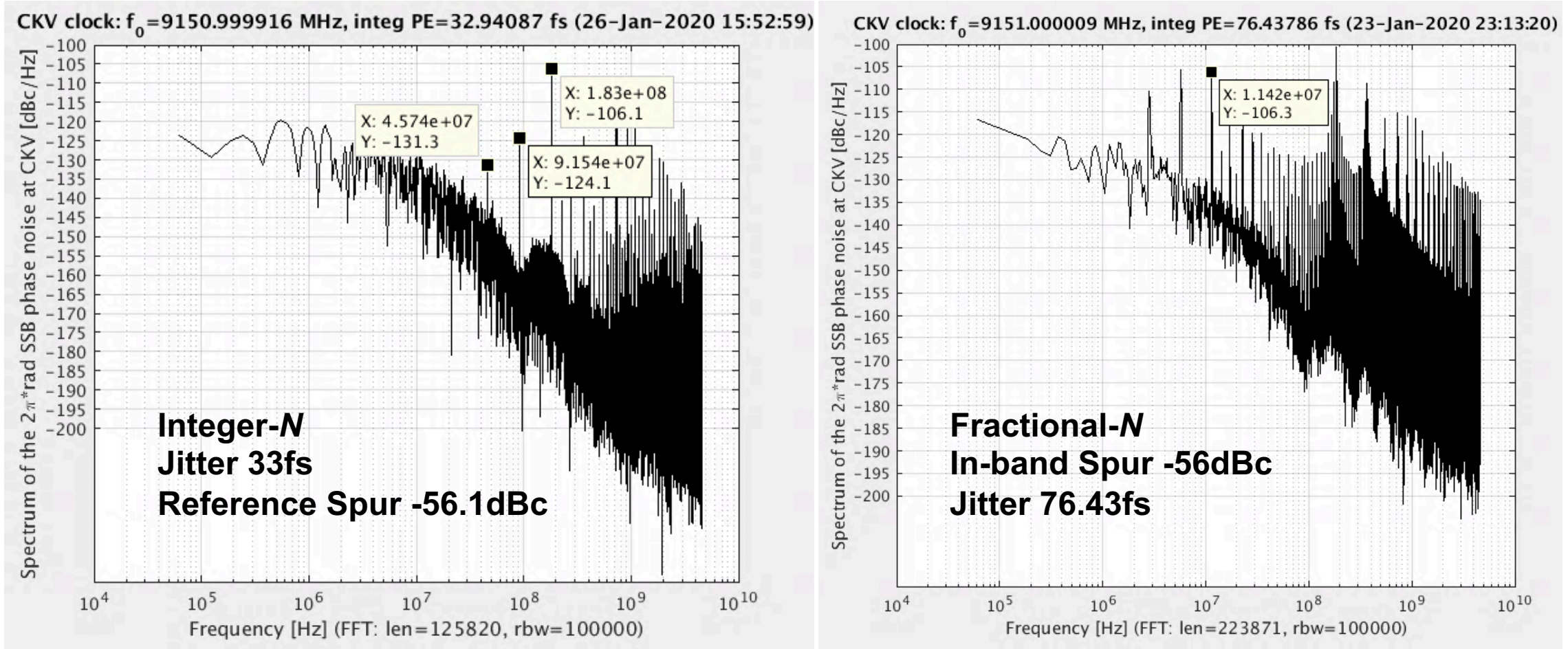
FoM = $20\log_{10}(\text{Jitter}/1\text{s}) + 10\log_{10}(\text{P}_{\text{DC}}/1\text{mW})$, FoM* = FoM+ $10\log_{10}(2/\#\text{phase})$, [#]Include the power of ref. buf., [&]From ISSCC'19 Visuals Session 16

Concept of Fractional- N Charge-Sharing Locking



- Charge-sharing locking enables fractional- N operation

System Simulation of 2nd Version of CSL



- 2nd version of CSL target sub-50fs for integer- N mode & sub-100fs for fractional- N mode

Conclusion – Charge-Sharing Locking

- Sub-100fs PLL is a must for 256QAM in 28/39 GHz bands
- Charge-sharing locking is proposed
 - A “big” capacitor w/ preset voltage briefly connects with the LC-tank for phase correction (proportional path)
 - Cap. charge residue for frequency tracking (integral path)
- Only DCO, DAC, ADC, switches, and digital circuits are needed
 - Scales very well with CMOS
- It achieves ultra-low jitter (75fs) & low power (16.5mW) for mmW quadrature frequency synthesis
- Extendible to fractional- N , phase modulation, FMCW...